

**Final Report
For
Initial Risk Assessment of Earthquake-Induced Landslides in
Jajarkot and Rukum Paschim Districts**



Submitted To:
Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority (NDRRMA)
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Executive Summary

This executive summary outlines key findings and recommendations for addressing landslide risks following the Jarkot Earthquake. Although the earthquake primarily caused damage to buildings, a limited number of landslides were reported. However, extensive cracking occurred on slopes and agricultural fields due to ground shaking. While the shaking wasn't intense enough to trigger significant landslides, it did reactivate existing ones and created numerous cracks.

A total of 206 landslides and 300 cracks were identified during field visits. The primary goal is to prioritize high-risk areas for settlement, considering both subsidence and landslide factors. These identified areas require further detailed assessment for resettlement and mitigation purposes. To prioritize the work, all visited sites were evaluated based on severity, risk to downslope settlements, and accessibility and classified into green (Low risk), yellow (moderate risk), red zones (high risk) zones. Among the red zones 27 critical locations earmarked for detail study, followed by evaluation of yellow and green zones in a second stage. Immediate action is imperative, especially in sealing cracks on slopes, agricultural fields, and existing landslides to prevent erosion and collapse. Promoting effective water management practices, like "bhal katne kaam," is essential to reduce water infiltration and mitigate landslide risks.

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List of Abbreviation

DDMCs : District Disaster Management Committees

GPS : Global Positioning System

MS: Surface Wave Magnitude Scale

MW: Moment Magnitude Scale

NDRRMA : National Disaster Risk Reduction and Management Authority

NEOC: National Emergency Operations Centre

NRA : National Reconstruction Authority

NSEG : Nepalese Society of Engineering Geologists

PGA : Peak Ground Acceleration

UNICEF : United Nations Children Fund

USGS : United State Geological Survey



1 Introduction

1.1 Background of the Jajarkot Earthquake

An earthquake of moment magnitude (MW) 5.7 (local magnitude, ML6.4) struck the Jajarkot district (Figure 1) of Karnali Province in western Nepal on November 3, 2023, at 23:47 local time. This is one of the strongest seismic occurrences to strike the region since 1505, as reported by Subedi et al. (2024). According to UNICEF (2023), this earthquake claims the lives of 154 people (84 females, 70 males) including 81 children (37 girls and 44 boys) and leaves 366 people injured (NEOC and DDMC report 2023). Approximately 200,000 more people require humanitarian aid. Epicenter of this earthquake was located in Ramidanda village, Barekot Rural Municipality, at a latitude of 28°50'24" N and a longitude of 82°11'24" E, with a focal depth of 10 km. The epicenter is 331 Km WNW of Kathmandu. Two aftershocks of magnitude M 3.5 and 3.8 is also occurred with the 10 Km radius of mainshock. The area is seismically very active associated with collisional tectonics where Indian plate subducts beneath the Eurasian Plate. This earthquake caused significant aftershocks in the vicinity and strong tremors were felt in Kathmandu and neighboring Indian cities including New Delhi. In the wake of the seismic events that unfolded in Jajarkot, the vulnerability of communities in the earthquake affected regions of Jajarkot and West Rukum has become a pressing concern. Most impacted area are the five local municipalities; Kushe, Nalgad and Bheri in Jajarkot district and Athbiskot and Sani Bheri in Rukum West.

The National Disaster Risk Reduction and Management Authority (NDRRMA) recognizes the imperative to address this issue proactively and has commissioned Geotech Solutions International Pvt. Ltd. to conduct an initial risk assessment. This report outlines the collaboration between NDRRMA and Geotech Solutions International Pvt. Ltd., emphasizing the need for a comprehensive evaluation of the earthquake-induced landslide risk in the Jajarkot and Rukum Paschim Districts. The earthquake on November 3, 2023, and the aftershock of November 6, 2023 not only left a trail of destruction but also underscored the critical importance of understanding and mitigating the secondary hazards, such as landslides, that often follow seismic activity. Geotech Solutions International Pvt. Ltd., renowned for its expertise in geotechnical engineering and disaster risk assessment, has been selected as the consulting firm to spearhead this crucial initiative. Their multidisciplinary approach and extensive experience in similar projects position them as the ideal partner to conduct a comprehensive risk assessment in these high-risk areas.

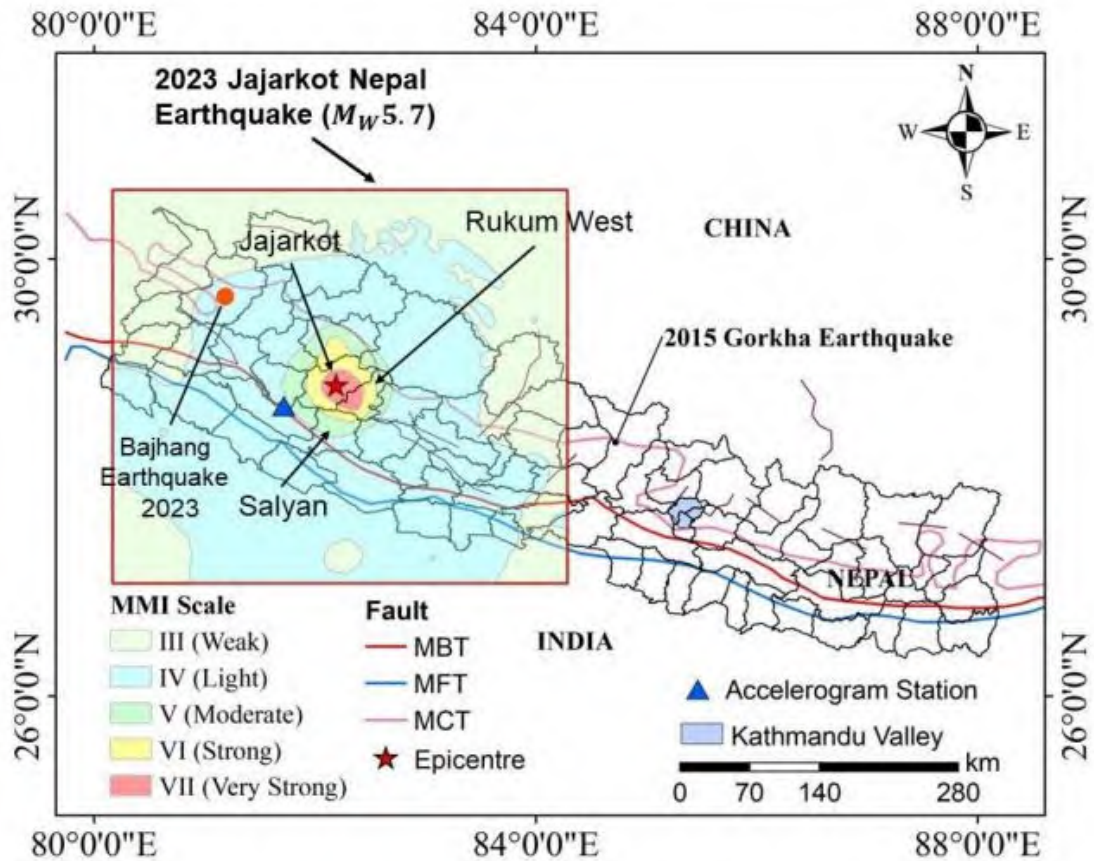


Figure 1: Map of Nepal showing 2023 Jajarkot Earthquake intensity with its epicenter - Jajarkot, Rukum West and Salyan are three mostly affected districts. (Source: USGS 2023).

This report provides an overview of the scope and objectives of the project, highlighting the significance of early detection and mitigation strategies in reducing the potential impact of earthquake-induced landslides. The collaboration between NDRRMA and Geotech Solutions International Pvt. Ltd. reflects a commitment to building resilient communities and safeguarding lives and infrastructure in the face of natural disasters. As we embark on this journey, it is with the shared understanding that the insights gained from this risk assessment will not only inform immediate disaster response efforts but will also lay the foundation for long-term strategies to enhance the resilience of these communities to future seismic events and their associated hazards.

1.2 Objective of the Assignment

The overall project aims to conduct initial risk assessment of earthquake-induced landslides in Jajarkot and Rukum Paschim Districts while prioritizing areas of risk for settlement considering subsidence and landslide factors. This study helps in developing a landslide

susceptibility map of the affected areas along with preparation of short term, mid-term and long-term risk reduction measures. Some of the major specific objectives of this study are:

- Conducting initial risk assessment of landslides prone area in Jajrkot and Rukum Paschim districts.
- Creating a comprehensive landslide inventory map for the identified landslide-prone areas.
- Identifying and prioritizing areas at risk for settlement, considering subsidence and landslide factors.
- Developing a landslide susceptibility map for the designated landslide-prone region.
- Formulating short-term/immediate investment plans, as well as mid-term and long-term plans for effective risk mitigation and management.

1.3 Scope of Work

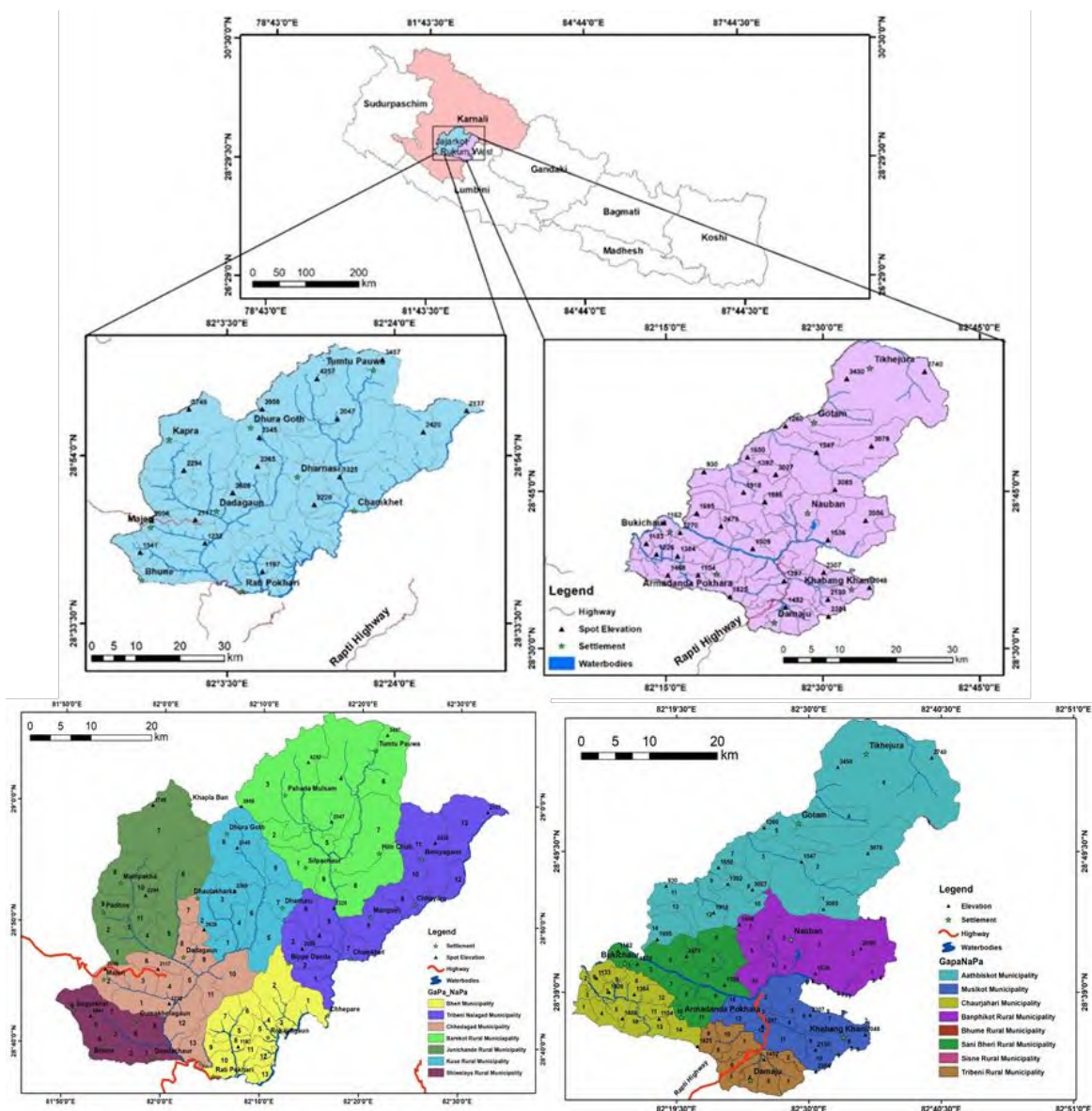
The scope of work of the present study is listed below.

- Analyze freely available satellite imagery for any traces of landslides and or subsidence and cracks.
- Collect copies and review landslide susceptibility assessment from available sources.
- Review available media report on landslides and cracks following the November earthquakes.
- Review the NRA guideline used during the 2072 earthquake and its application after modification.
- Consultation with Department of Mines and Geology (DMG), provincial governments, DAO offices and other relevant stakeholders.
- Communication with DAO office and dispatch plan with NDRRMA.
- Identify hotspot communities or settlements, structures like roads etc. that have been impacted by the earthquake induced landslides.
- Photographic documentation of the landslide.
- Site visit of affected settlements or stretches of road.
- Identify geological evidence of earthquake such as cracks, subsidence area, and landslide area.
- Identify and prioritize settlements that would require re-settlement and or risk reduction works.

2 Study Area

2.1. Location of the area

The study area lies in the earthquake affected Karnali Province which includes 10 districts which have been affected by the seismic events of 3rd November 2023. During our field study, we conducted initial risk assessment of all the affected municipalities and rural municipalities of Jajarkot and Rukum West districts.



2.1 Geology of the Study Area

In Jajarkot District, the geological composition is characterized by a variety of lithologies sourced from the Lesser Himalayan, Higher Himalayan crystalline, and granite formations as shown in Figure 3. Key geological formations in this area include the Dadeldhura Group, which consists of granites, gneisses, schists, carbonate rocks, and quartzites. Additionally, the Jaljala Group is present, comprising limestone, calcareous schist, and quartzite. The Mid-Land Group contributes to the geological diversity with phyllites, gneisses, limestone, shale, sandstone, and quartzite. Lastly, the Surkhet Group adds fossiliferous limestone, sandstone, and quartzites to the region's geological profile.

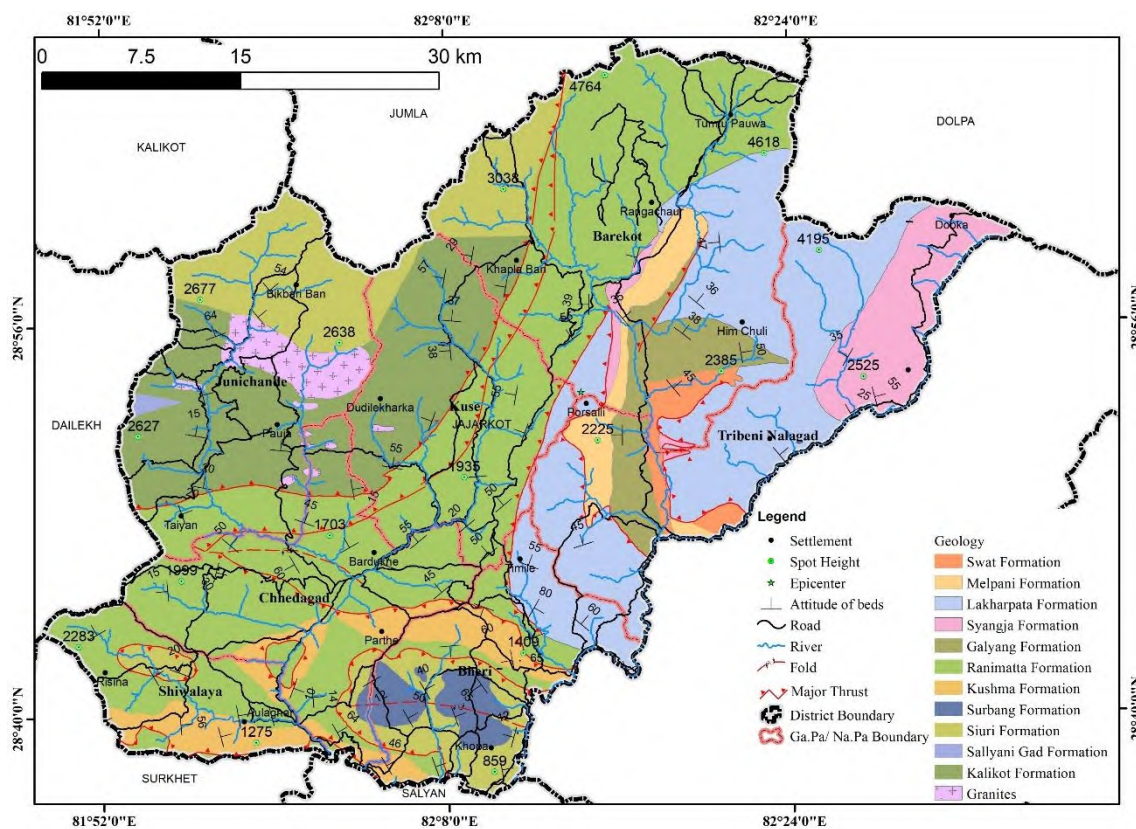


Figure 3: Regional geological map of the Jajarkot District (Source: DMG)

Regarding Rukum West, the geological composition mirrors that of Jajarkot, featuring lithologies from the Lesser Himalayan, Higher Himalayan crystalline, and granite formations as shown in Figure 4. The Dadeldhura Group is prominent, with its mix of granites, gneisses, schists, carbonate rocks, and quartzites. Similarly, the Jaljala Group is present with its limestone, calcareous schist, and quartzite formations. The Mid-Land Group contributes to the region's geological makeup. Additionally, the Surkhet Group adds to this diversity with fossiliferous limestone,

sandstone, and quartzites. These geological details are sourced from the Department of Mines and Geology.

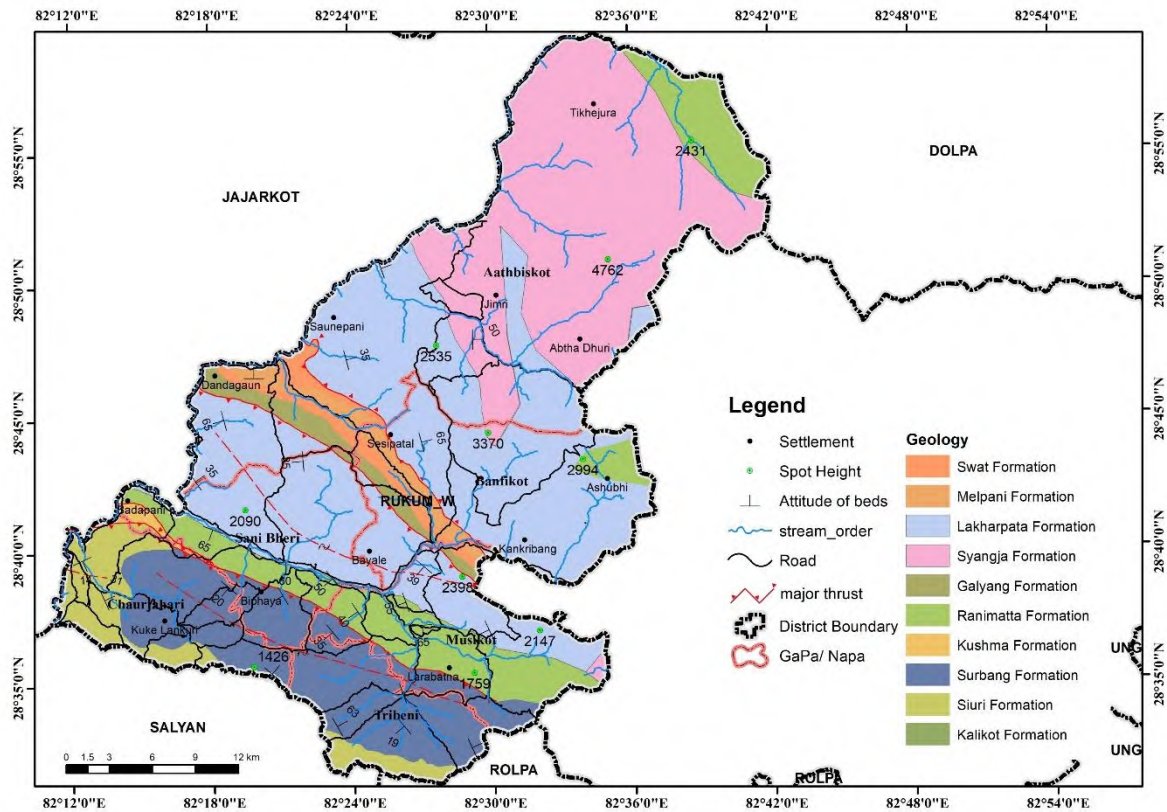


Figure 4: Geological map of the Jajarkot District (Source: DMG)

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2.2 Seismic Gap in Western Nepal



In spite of the fact that there have been a lot of minor earthquakes in western Nepal in the last few years, this one was among the strongest since 1505 (MS~8.2) (Figure 5). There may have been a large buildup of tectonic stress and energy in the area because there hasn't been any noteworthy seismic activity in western Nepal for the previous 518 years. Several studies have

warned that an enormous build-up of strain in the region is likely to result in at least one earthquake of MW8.5 or more in the western part of Nepal (Ghazoui et al., 2019, Srivastava et al., 2013). Scientists have warned of a large seismic gap in western Nepal between Pokhara and Dadeldhura. There has not been a major earthquake since 1505, and an earthquake of more than eight magnitudes is anticipated in the area. This serves as a reminder to Nepal to improve its readiness and resilience.

In 1505, western Nepal experienced an MW8.2 earthquakes west of the 2015 rupture zone. The accumulated strain since then, without release, suggests a likelihood of the occurrence of stronger earthquakes in the area in the future (Chiaro et al., 2015). The seismic gap in the region can be seen in Figure 5 and Figure 6. The seismic activity in the western Nepal Himalayan region showed variations between high and low seismic phases from 1963 to 2006 (Paudyal et al., 2010). The existence of a weak Main Himalayan Thrust beneath Tibet with the initiation of the Main Central Thrust can be explained by the South Tibetan Detachment and corresponding stress field in Western Nepal (Paudyal et al., 2010).

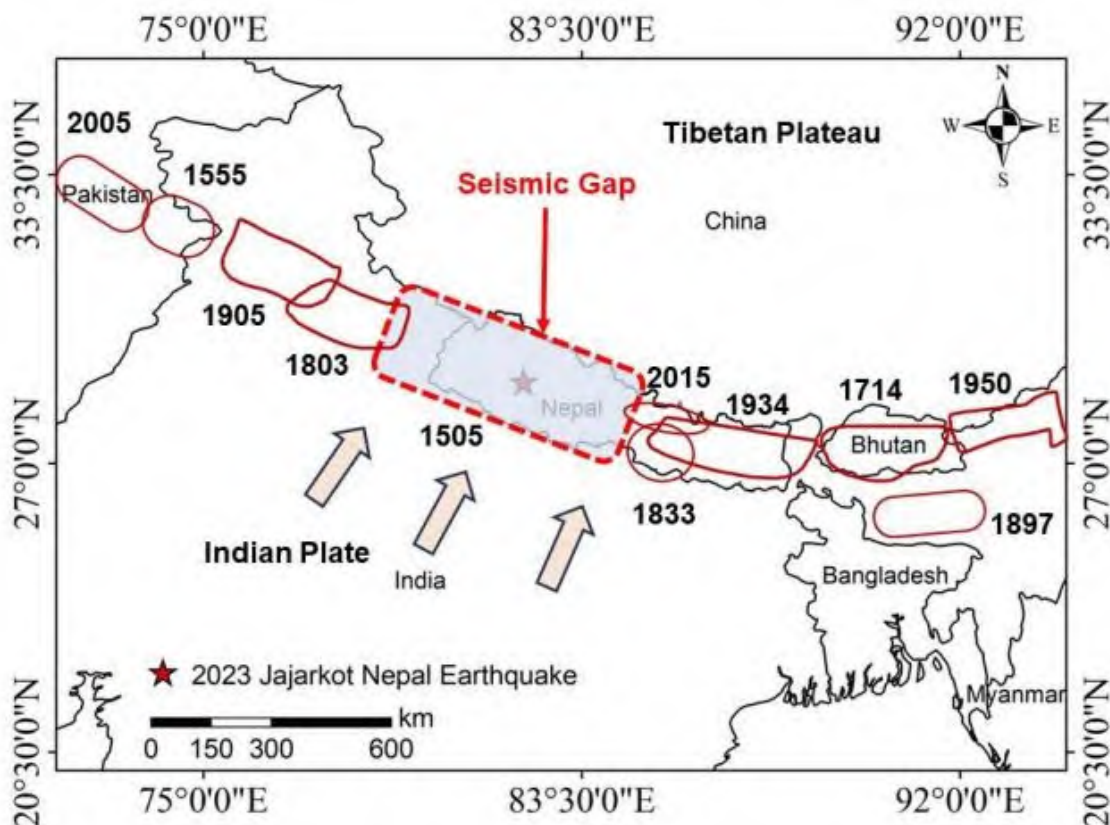


Figure 5: Map showing historical earthquakes with year occurred and approximate rupture area (solid redlines), and seismic gap (dotted red line) in the western Nepal region (Subedi et al., 2024).

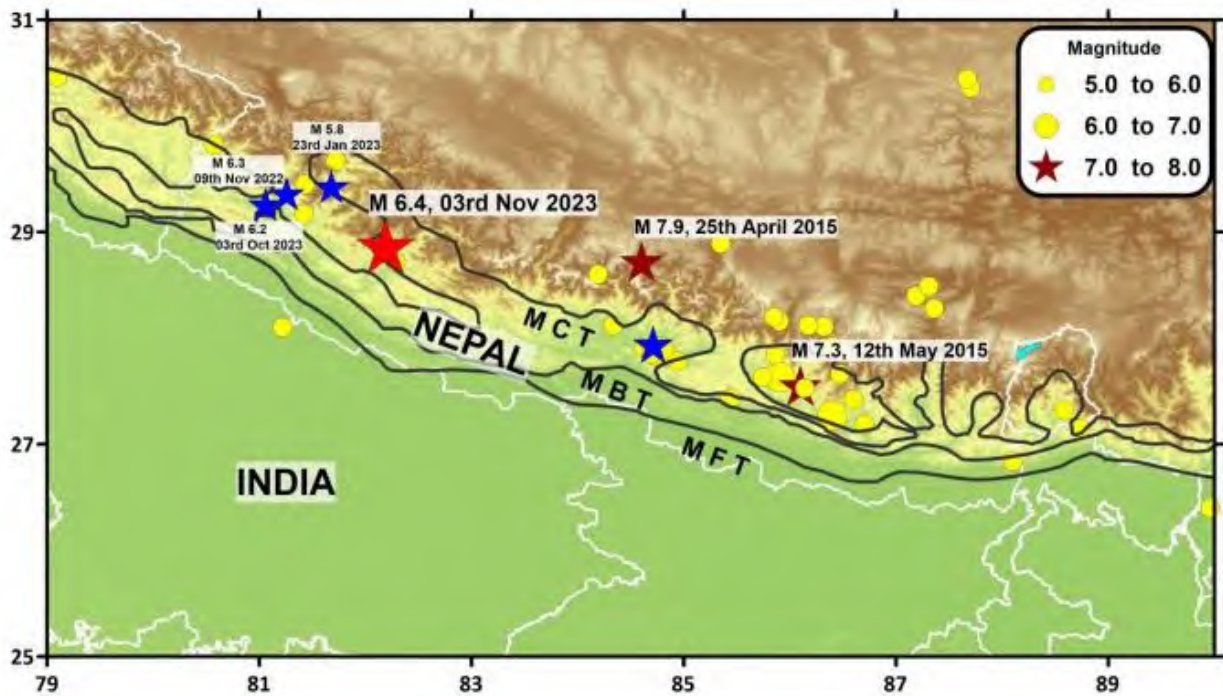


Figure 6: The present earthquake (M 6.4) of 3rd November 2023 and seismicity ($M > 5.0$) in the epicentral and surrounding region (yellow star), (source: www.seismo.gov.in; NCS-MoES).

2.3 Seismo-Tectonics Aspects of Jajarkot Earthquake

The three-component recorded accelerogram that was taken at Bhimchula station during the MW 5.7 Jajarkot, Nepal, and Earthquake of 2023 mainshock are shown in Figure 7. The closest accelerogram station to the epicenter is Bhimchula, and Nepal has limited seismological and accelerogram stations scattered throughout the nation. It is clear that during the mainshock, the peak ground acceleration (PGA) for the ground motions that were recorded reached up to 70 cm/s^2 . The PGA recorded is lower than the projected PGA ($295\text{--}340 \text{ cm/s}^2$) obtained using probabilistic seismic hazard analysis (PSHA) with a 10% likelihood of exceedance over a 50-years' time period by NBC (NBC 105:2020). PGA records (Figure 8) from EEW network sensors show the potential of earthquake early warning. Time since the onset of earthquake and arrival of strong ground motion to Kathmandu (topmost record) is nearly 80 seconds.

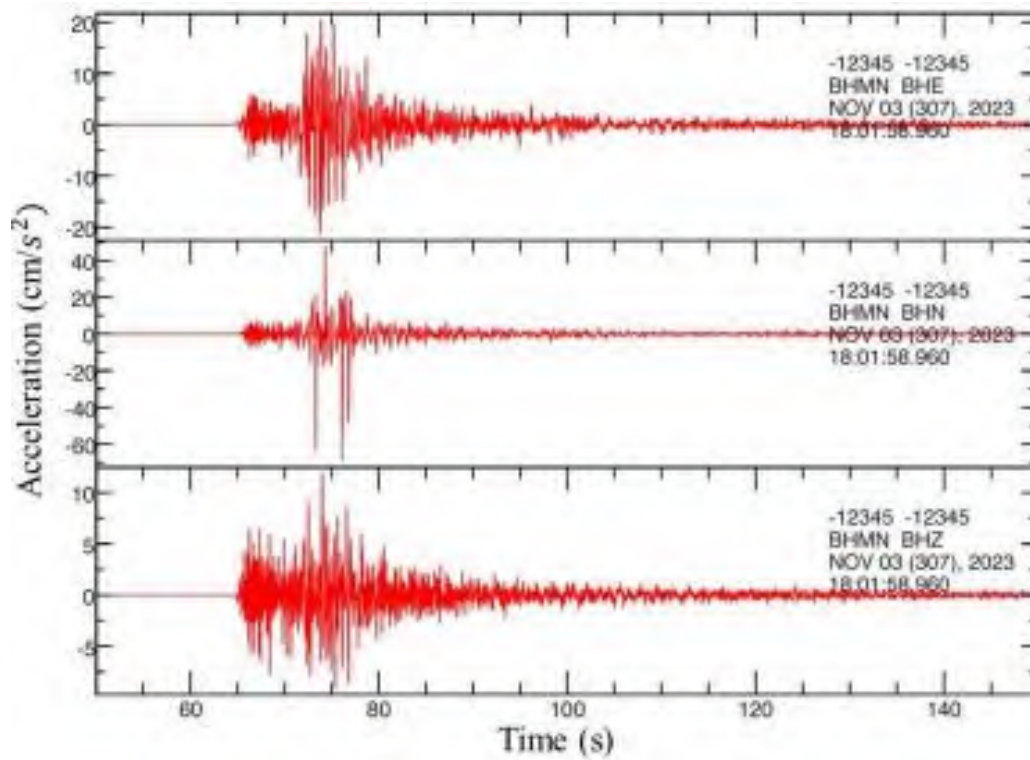


Figure 7: Recorded accelerogram at Bhimchula station (28°39'20.88" N, 81°42'51.84" E) for the mainshock of 2023 Jajarkot Earthquake. (Source: Seismicity and Seismotectonic related to Jajarkot Earthquake 2023).

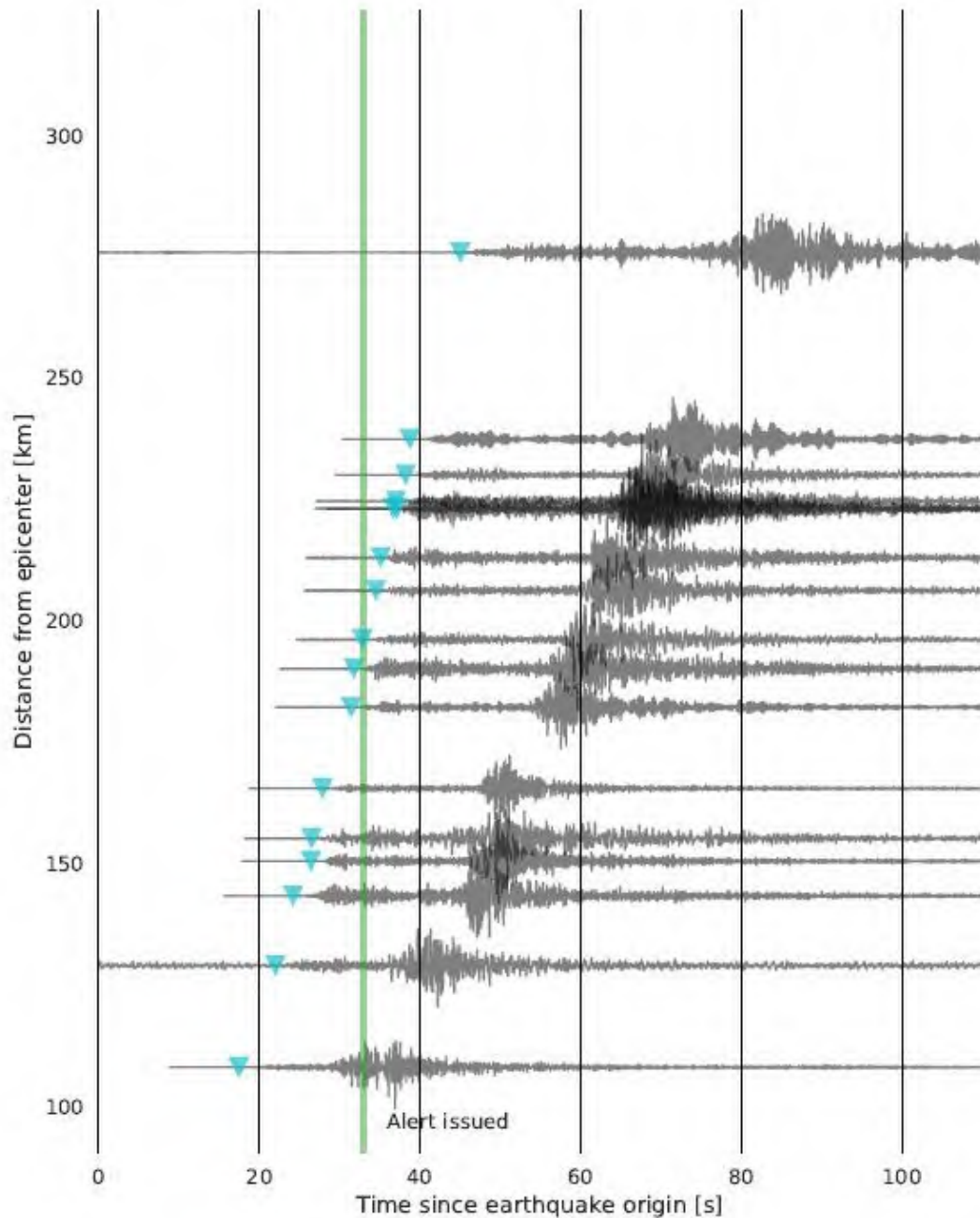


Figure 8: Peak Ground Acceleration (PGA) records from EEW Network Sensors (Source: National Society for Earthquake Technology- NSET)

2.4 Socio-economic aspect of Study Area



Jajarkot and Rukum West districts, both located in Karnali province - Western Nepal, share some similarities in their socio-economic aspects due to their geographical proximity and similar development challenges. However, there are also some distinct characteristics that differentiate the two districts. Here's an overview of the socio-economic aspects of Rukum West and Jajarkot districts:

1. **Economy:** Both districts heavily rely on agriculture as the primary economic activity. Subsistence farming is common, with crops like rice, maize, millet, wheat, and vegetables being grown. Livestock rearing, including cattle, goats, and poultry, also contributes to the local economy. However, limited market access and inadequate infrastructure hinder agricultural productivity and economic growth in both districts.
2. **Income and Poverty:** Both Rukum West and Jajarkot districts face challenges related to poverty and income disparity. Poverty rates are relatively high compared to national averages. Marginalized communities, such as ethnic minorities and remote villages, are particularly affected. Limited employment opportunities, lack of access to education and healthcare, and underdeveloped infrastructure contribute to the persistent poverty in both districts.
3. **Education:** Access to quality education is a common challenge in both districts. Educational infrastructure, teacher shortages, and low literacy rates are prevalent, especially among marginalized groups. Access to schools, particularly in remote areas, is limited. Efforts are being made to improve education through the construction of schools, scholarships, and initiatives to enhance enrollment and retention rates in both districts.
4. **Health and Healthcare:** Both Rukum West and Jajarkot districts face challenges in terms of healthcare services and infrastructure. Access to quality healthcare facilities, trained medical professionals, and essential medicines is limited, especially in remote areas. Health indicators, such as infant and maternal mortality rates, are relatively higher compared to national averages. Efforts are underway to improve healthcare services through the establishment of health posts, deployment of health workers, and health awareness programs.
5. **Infrastructure:** The socio-economic development of both districts is hindered by the lack of proper infrastructure. Challenges include road connectivity, electricity supply, clean water, and sanitation facilities. Limited access to transportation and basic amenities hampers economic activities, trade, and overall development in both districts.

While Jajarkot and Rukum West districts share these common socio-economic challenges, it's important to note that there may be specific variations in the severity of these issues and the specific context of each district. Efforts by the government, non-governmental organizations, and development partners are focused on addressing these challenges through infrastructure

development, education and healthcare initiatives, poverty reduction programs, and livelihood improvement schemes in both districts.

Jajarkot district a part of Karnali Province, is one of the seventy-seven districts of Nepal. The district, with Khalanga as its district headquarters, covers an area of 2,230 square kilometers (860 sq.m) and has a population of 171,304 in 2011 Nepal census. There are 3 municipalities and 4 rural municipalities in this district. Jajarkot district consists of three municipalities:

Bheri Municipality: Bheri municipality was established in 2017 and is the largest municipality in Jajarkot, bordering the Dolpa district to the north. The municipality is divided into total 13 wards. It has a population of around 37,892 and is known for its scenic beauty and religious significance. The municipality encompasses the historical town of Khalanga, the district headquarters, which is famous for the ancient Daha Deurali temple. Its area is approximately 265.77 sq km.

Chhedagad Municipality: This municipality is formed in 2017 and is situated in the eastern part of Jajarkot, bordering the Rukum East district. The municipality is divided into total 13 wards and its headquarter is located at Karkigaun. It has a population of approximately 37,877. The municipality is known for its natural resources and agricultural potential. Chhedagad boasts scenic landscapes, including the popular Chapakot Daha lake, a popular destination for locals and tourists. Its area is approximately 214.19 sq km.

Nalgad Municipality: This municipality is established in 2017 and is located in the southern part of Jajarkot, bordering the Salyan district. The municipality is divided into total 13 wards and its headquarter is located at Khagenkot. It has a population of about 28,922 and is known for its historical and cultural significance. The municipality houses the Triveni Sangam, the confluence of three rivers: The Tila, the Lundi, and the Nalgad. This confluence is considered a holy site by Hindus and attracts pilgrims throughout the year. Its area is approximately 187.44 sq km.

Jajarkot district consists of following four rural municipalities:

Barekot Rural Municipality: Barekot is a rural municipality located in Jajarkot District of Karnali Province of Nepal. The rural municipality is divided into total 9 wards and its headquarter is Nayakwada. According to Ministry of Federal Affairs and General Administration Barekot has an area of 577.5 square kilometres and the total population of the rural municipality is 18,083 as of 2011 Nepal census.

Junichande Rural Municipality: Junichande is a rural municipality located in Jajarkot District of Karnali Province of Nepal. The rural municipality is divided into total 11 wards and

its headquarter is Majhkot. According to Ministry of Federal Affairs and General Administration Junichande has an area of 346.21 square kilometers and the total population of the rural municipality is 21,733 as of 2011 Nepal census.

Kushe Rural Municipality: Kushe is a rural municipality located in Jajarkot District of Karnali Province of Nepal. The rural municipality is divided into total 9 wards and its headquarter is Tharpur. According to Ministry of Federal Affairs and General Administration Kushe has an area of 274 square kilometers and the total population of the rural municipality is 20,621 as of 2011 Nepal census.

Shivalaya Rural Municipality: Shivalaya is a rural municipality located in Jajarkot district and is divided into total 9 wards and its headquarter is Sima. According to Ministry of Federal Affairs and General Administration Shivalaya has an area of 134.36 square kilometers and the total population of the rural municipality is 15,269 as of 2011 Nepal census. The rural municipality is divided into total 9 wards and the headquarters of this newly formed rural

Rukum West:

Rukum west district is a part of Karnali Province, is one of the seventy-seven districts of Nepal. The district, with Musikot as its district headquarters, covers an area of 1213.49 square kilometers (468.53 sq.m) and has a population of 154,272 in 2011 Nepal census. There are 3 municipalities and 3 rural municipalities in this district. Rukum West district consists of three municipalities:

Aathbiskot Municipality: Aathbiskot is a Municipality in West Rukum District in Karnali Province of Nepal that was established in 2015. The municipality is divided into total 14 wards. It has a population of around 33,601. Its area is 560.34 square kilometers.

Chaurjahari Municipality: Chaurjahari is established in 2015 through the merging the two former Village development committees Bijayaswori and Kotjahari. It lies on the bank of Sani Bheri River. It has a population of around 27,438(2011). Its area is 107.4 square kilometers.

Musikot Municipality: Musikot is a municipality and the district headquarter of Western Rukum District in Karnali Province of Nepal that was established as Musikot Khalanga on 2 December 2014 by merging the two former Village development committees Musikot and Khalanga. It has a population of around 32,939 (2011). Its area is 136.1 square kilometers.

Rukum district consists of three rural municipalities:

Sanibheri Rural Municipality: Sani Bheri has total 11 wards, which are scattered across 134 square kilometers of geographical area. Its headquarter is at Simli. According to 2011 Census

conducted by Central Bureau of Statistics (CBS), Sani Bheri Rural Municipality had total population of 22,194.

Banphikot Rural Municipality: Banphikot is a rural municipality located in Western Rukum District . According to 2011 Census conducted by Central Bureau of Statistics (CBS), Banphikot Rural Municipality had total population of 18,696. Total area of this rural municipality is 190.43 square kilometers and divided into 10 wards. Banphikot Rural Municipality was established in 2015 through the merging five the former Village development committees of Aathbiskot, Pipal, Duli (ward no. 4, 5 &6) and Magma (ward no. 1-6 & 9). This Municipality shares boarder with Sisne Rural Municipality (East Rukum district) in the East, Sani Bheri Rural Municipality in the West, Aathbiskot Municipality in the North, and Musikot Municipality in the South.

Triveni Rural Municipality: Triveni is a rural municipality located in Western Rukum District of Karnali Province of Nepal. Tribeni Rural Municipality had a population of 19,404. Of those, 99.8% spoke Nepali, 0.1% Urdu as their first language and 0.1% other languages as their first language. It was established in 2017 through the Local Government. Total area of this rural municipality is 175.3 square kilometers and divided into 10 wards.



3 Methodology

With the knowledge of the project background and objectives, the following approaches were adopted by GSI to execute the services outlined in the TOR, and to achieve the intended objectives. The proposed methodology presents the approach and procedure to be adhered to by the GSI team for the project output.

3.1 Inception and Work Planning

Upon offer of the task, GSI coordinated and conduct an initial consultation with project team, team members and experts for developing a detailed work plan and shared with the NDRRMA for comments and feedback.

3.2 Literature Review and Desk Study

Relevant project documents, progress reports, available reports regarding landslide study area will be reviewed. Various reports from the government has be used to gather necessary information regarding the affected areas. Topographical maps, satellite images and geological maps of the study area has been collected for the general understanding of the affected area. The list of landslides provided by the NDRRMA and local government were reviewed before visiting the site. Also a landslide survey form has been developed taking reference from the guidelines prepared by NDRRMA.

3.3 Communication and Meeting with NDRRMA and relevant stakeholders

During the preliminary stage of the study all the relevant stakeholders were contacted to gather general information about the affected areas. All the municipalities and rural municipalities were contacted and affected places were noted down to further plan on how to execute the field work. Concerned authorities including the mayors, deputy mayors, chairperson and vice-chairperson of concerned rural municipalities, ward chairperson and vice-chairperson and members were contacted in order to gather information about the areas affected by landslides and cracks that were induced during the earthquake. A complete list was compiled and generated which includes all the affected regions of the area.

GSI team presented the Inception Report to NDRRMA team and describe the approach that will be adopted in the field. Based on the suggestions provided by the NDRRMA team the landslide survey form has been updated and deployed in the field. Some glimpses of the inception report sharing is shown in Figure 9. Before mobilizing to field,



Figure 9: Inception report sharing at NDRRMA office.

During field visit our team met with the relevant stakeholders of the affected area along with the government agencies to collect the existing scenario of the area. After that the team visited affected municipalities to collect data and to assess the damage caused by the earthquake. Required information regarding historical landslide events and impacts has also been discussed with the local residents and concerned authorities.

3.4 Field Work and Data Collection

After literature review and desk study detailed field work was carried out in all the municipality and rural municipality of the Jajrkot and Rukum west district where landslide were detected. Figure 10 and Figure 11 shows some glimpses of field visit and interaction with local people and local government. Figure 12 shows photographs expert interaction during field work with CDO and Municipality. During Field work was carried out for the risk assessment of earthquake-induced landslides and earthquake triggered old landslides which involves a systematic approach that combines geological, geotechnical, and seismological information. The technical guidelines and landslide inventory survey sheets published by NDRRMA were used in the field investigation in the affected areas.





Figure 10: Field visit photographs, interacting with affected municipality and ward office.



Figure 11: Field visit photographs, interacting with affected municipality, local people and ward office.



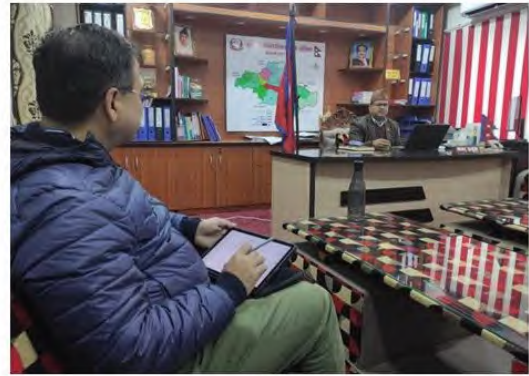
a. Expert Discussion with CDO at Jajarkot



b. Discussion with CDO at Rukum West



c. Discussion with CDO and attending DDMC meeting at Jajarkot District



d. Discussion at Musikot Municipality

Figure 12: Photographs of field interaction with District CDO and municipalities.

Preliminary research before field visit includes the gathering of existing geological and geotechnical maps, hydrogeological map, seismic hazard maps, and historical landslide data. Identification of the seismic history of the area, including the magnitude and frequency of past earthquakes is also very important for the seismic risk assessment and seismically induced geotechnical effects.

The team selected sites for field investigation based on a combination of factors such as seismic activity, slope steepness, soil type, and land use. The team also collected information regarding the geological formations, faults, and other relevant features in the landslide affected areas.

The team also measured slope angles, height, and other relevant topographical features. And Identify signs of previous landslide activity, such as scarps, debris, or displaced vegetation. All the condition of existing infrastructure (buildings, roads, etc.) and their vulnerability to landslides are also documented well.

4 Landslide Susceptibility Assessment

In context of earthquake induced landslides susceptibility mapping, there are various approaches to evaluate the susceptibility. For susceptibility mapping, landslide caused by earthquake shall only be used for the training dataset, however in our case very few location has been identified as landslide triggered by the earthquake and most of them are reactivated landslide by the existing landslides caused by rainfall. In total, 211 no of landslides which are mainly induced by rainfall and reinduced by earthquake are located and about 300 crack locations were collected from both districts. In such scenario the susceptibility mapping cannot be performed by the identified landslide location. Therefore, in this study we have taken reference from the F value map. To understand this phenomenon, an analysis has been done using two geomorphological parameters, mainly slope and curvature. PGA value provided by USGS is also used to evaluate the terrain for occurrence of shallow landslide after the earthquake. Uchida et al.2004 have used similar method in Japan to evaluate shallow landslides in granitic terrain of Hyogo Prefecture. Dahal 2018 has also used the same methodology to evaluate the landslide susceptibility after the Gorkha earthquake 2015. Using similar concept and considering landslides as grouping variables and slope, PGA and curvature as independent variables, following Discriminant Function (Eq 1) has been developed to understand earthquake-induced shallow landslide susceptibility in Jajrkot and Rukum West area.

$$F = 4.536x (\text{Curvature}) + 0.004x (\text{Slope}) + 0.005x (\text{PGA}) - 2.766 \text{----- Eq 1}$$

The above equation determines F-value or landslide susceptibility index value. The landslide susceptibility zonation of the study area is accomplished in ILWIS. For the zonation of landslide susceptibility map, natural break method is used to classify the map into various zones. The natural breaks classification method (also known as Jenks' optimization) is based on natural groupings based on the inherent in the data. The resulting susceptibility map has been overlaid with the existing landslide/cracks location to determine the success rate of the susceptibility map. The equation can be used to predict future landslide scenario in central Nepal and its success rate is 72 % with 28 % of false negative value.

In total, 211 no of landslides which are mainly induced by rainfall and reinduced by earthquake are located and about 300 crack locations were collected from both districts. Therefore, in summary all the landslides are mainly induced by rainfall.



4.1 Curvature map

Curvature map represents the rate of change of slope and is the second derivative of elevation. It shows the convexity or concavity of the slope surface. The curvature of slope of the area is classified to convex, planar and concave as shown in Figure 13, where A represent the Convex, B represent the concave and C represent the planar curvature. During earthquake the topographic effect and resonance can be seen in the mountain ridges indicating convex slopes will have higher tendency to fail. The curvature map of the area is represented in Figure 14.

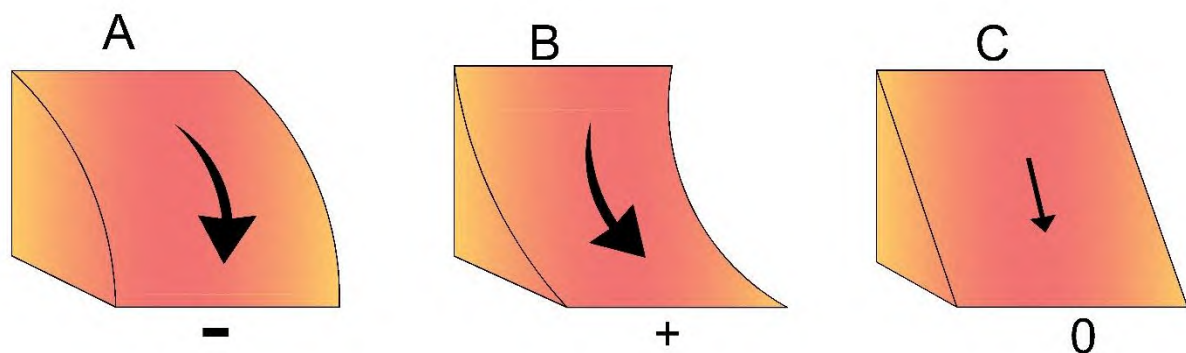


Figure 13: Schematic representation of the Curvature of Slope.

4.2 PGA value map

PGA map represents the ground acceleration that has been recorded on the ground during the earthquake. In our case we have used USGS PGA value map that is free available is the USGS website. The PGA map of the area is shown in Figure 15.

4.3 Slope map

A slope map indicates the terrain steepness. The slope within the area varies from 0 to 60° degrees which was classified into different categories as <15°, 15-30°, 30-45°, 45-60° and 60-90° as shown in Figure 16.

4.4 Landslide inventory map

Landslide inventory map refers to the location, its distribution, and characteristics of the past landslides. The landslide inventory map is created from the landslide occurred after earthquake 3 November 2023. The landslide inventory map of the area for analysis of landslide hazard map is shown in Figure 17.

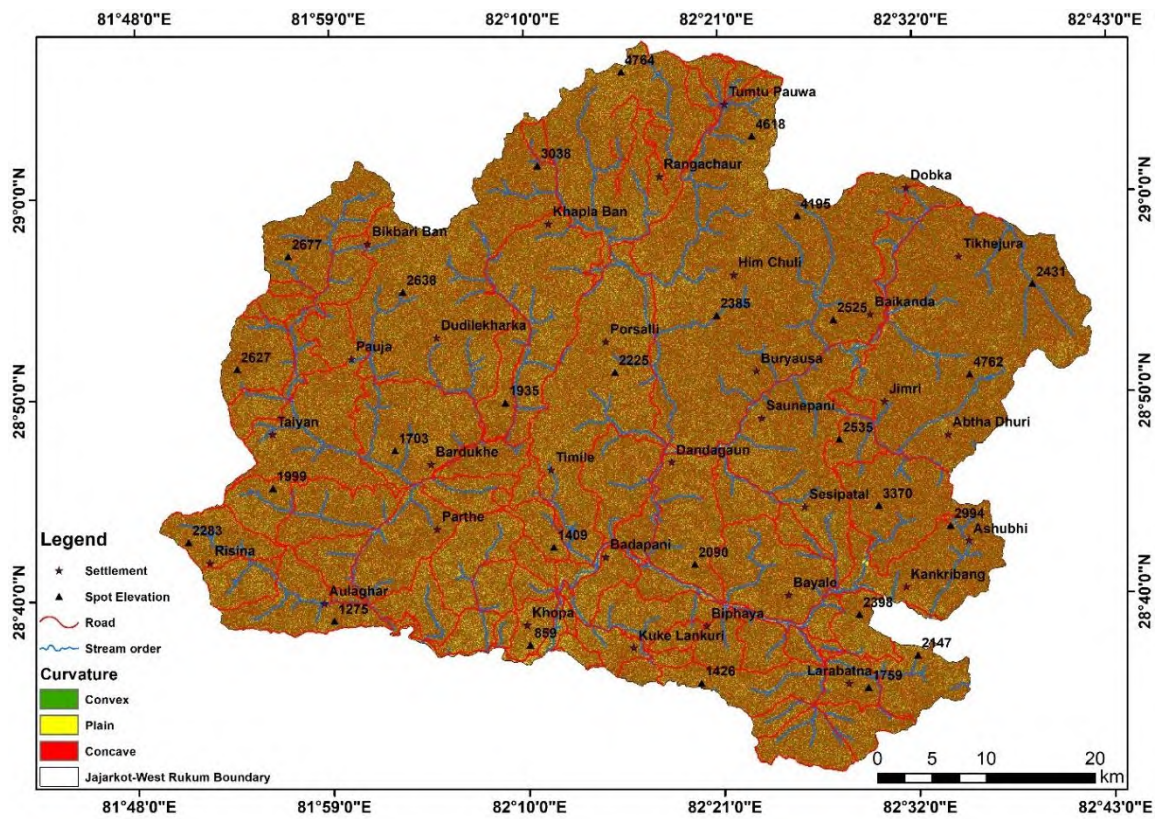


Figure 14: Curvature map of Jajarkot-West Rukum District.

Figure 15: PGA map of Jajarkot Earthquake 2023.

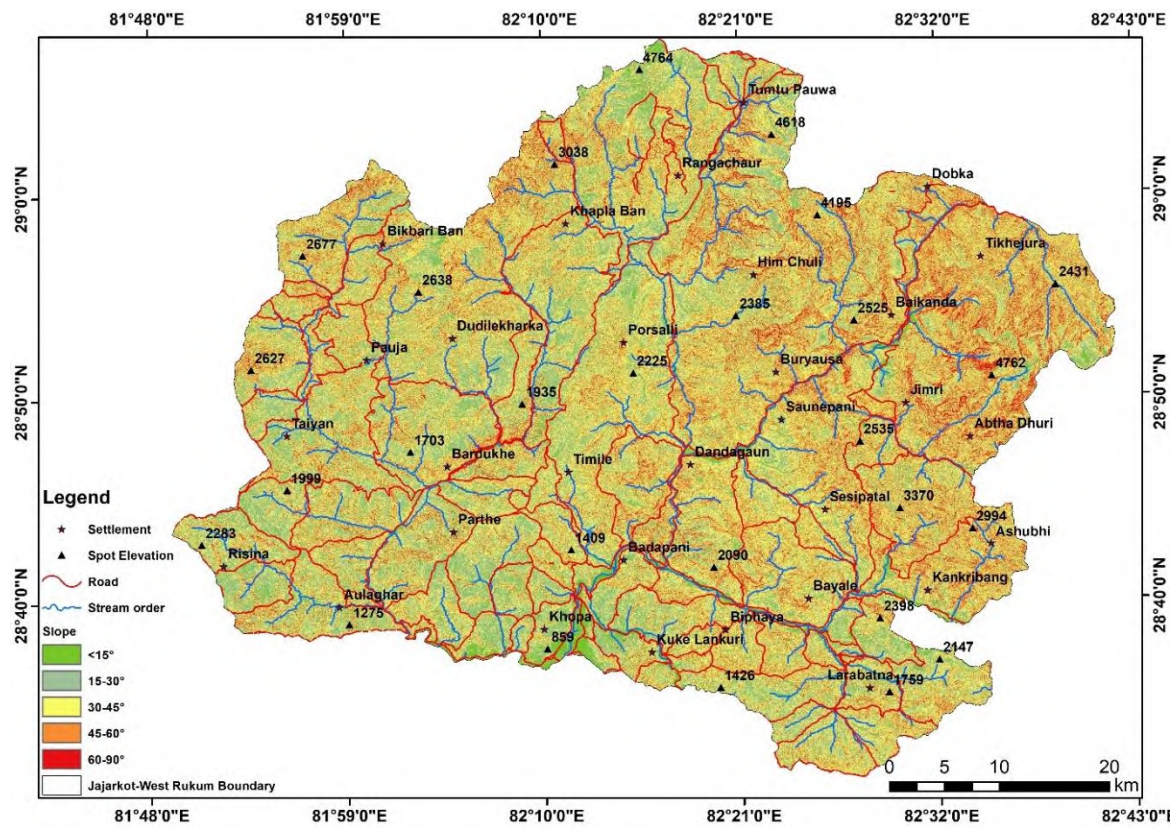


Figure 16: Slope map of Jajarkot-West Rukum District.

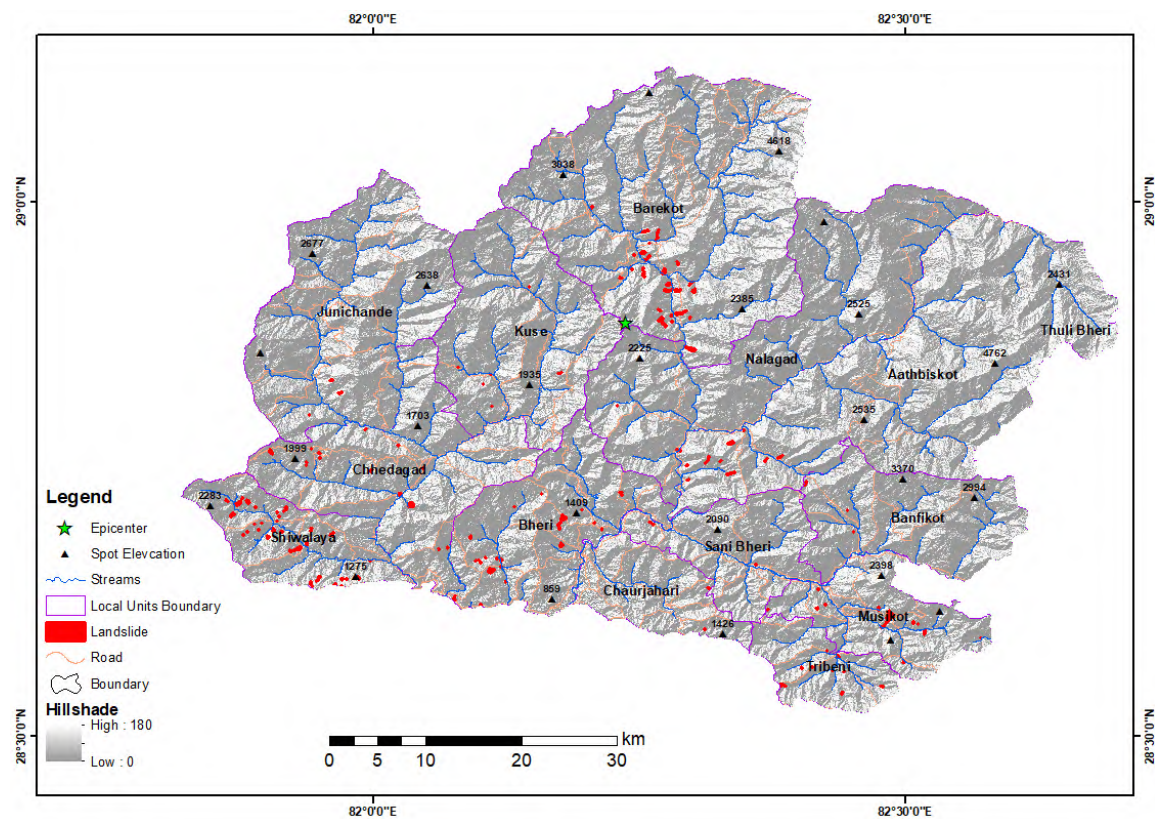


Figure 17: Landslide inventory map of Jajarkot-West Rukum District.

The landslide and crack inventory reveals significant impact in areas like Shiwalaya, Bheri, Barekot, Aaathbiskot, Musikot, Chedagad, and Junichande. Based on this data, a landslide susceptibility assessment has been conducted, categorizing the entire region into five classes according to susceptibility. The color scheme ranges from green indicating very low to low susceptibility, to red representing very high susceptibility to landslides. The susceptibility map highlights Shiwalaya, Barekot, Chedagad are highly susceptible, while areas like Sani Bheri, Nalgad, Kuse Chaujhari, and Kuse exhibit moderate to low susceptibility. Figure 18 shows the susceptibility map of the Jajarkot and Rukum district based on the 2023 earthquake event.

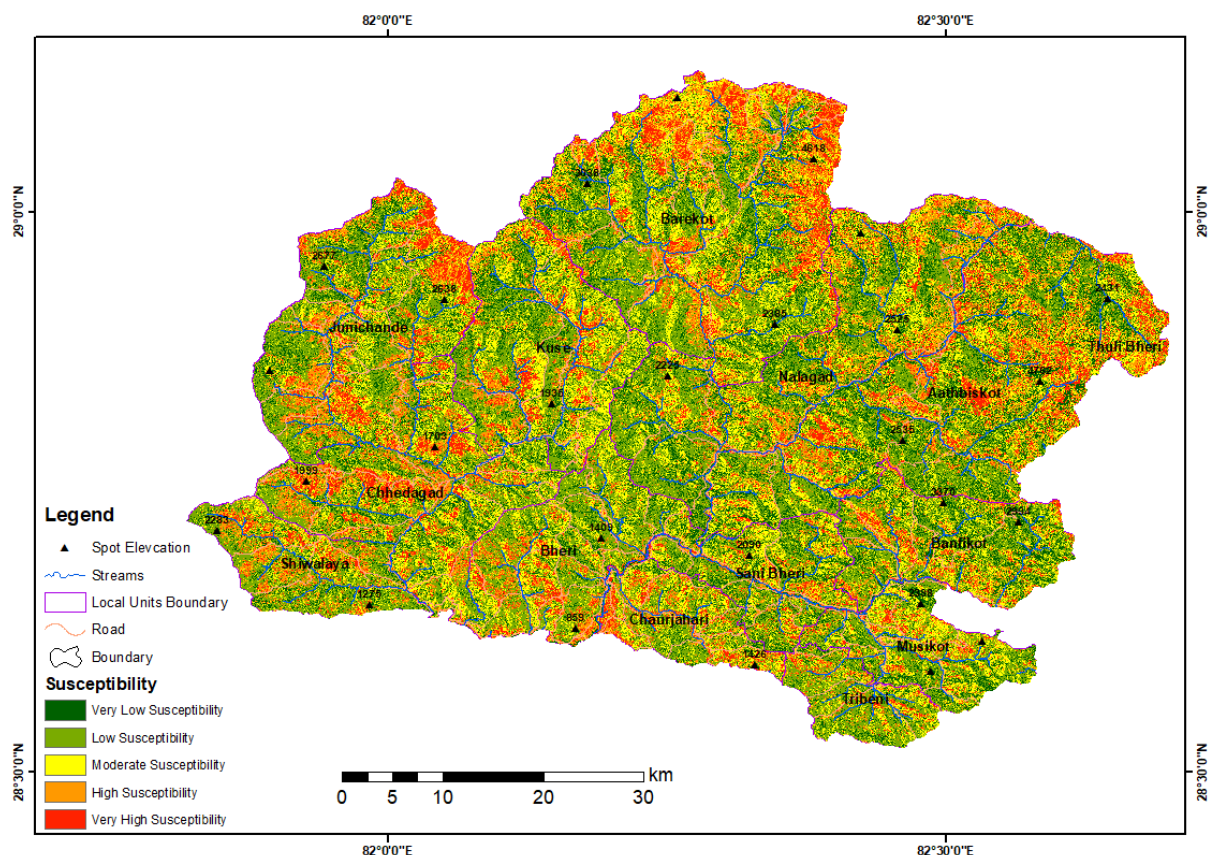


Figure 18: Susceptibility map (based on F map) of the Jajarkot and West Rukum Districts after Jajarkot Earthquake.

5 Landslide Risk Assessment of Jajarkot District

5.1. Nalgad Municipality

A significant portion of Nalgad Municipality has been impacted by landslides, cracks and ground subsidence, particularly Wards 1, 2, 3, and 4. These Wards have extensive cracks, some as deep as 1.5 meters and wide as 30 centimeters, posing a major threat to settlements. Landslides, both earthquakes induced and reactivated by the earthquake, have affected Wards 1, and 4, 5 while Ward 4 has experienced rockfall. In summary, Wards 1 to 7 in Nalgad Municipality have sustained various degrees of damage, while Wards 8 to 13 appear to be unaffected. The damage details of Nalgad Municipality are on Annex III.

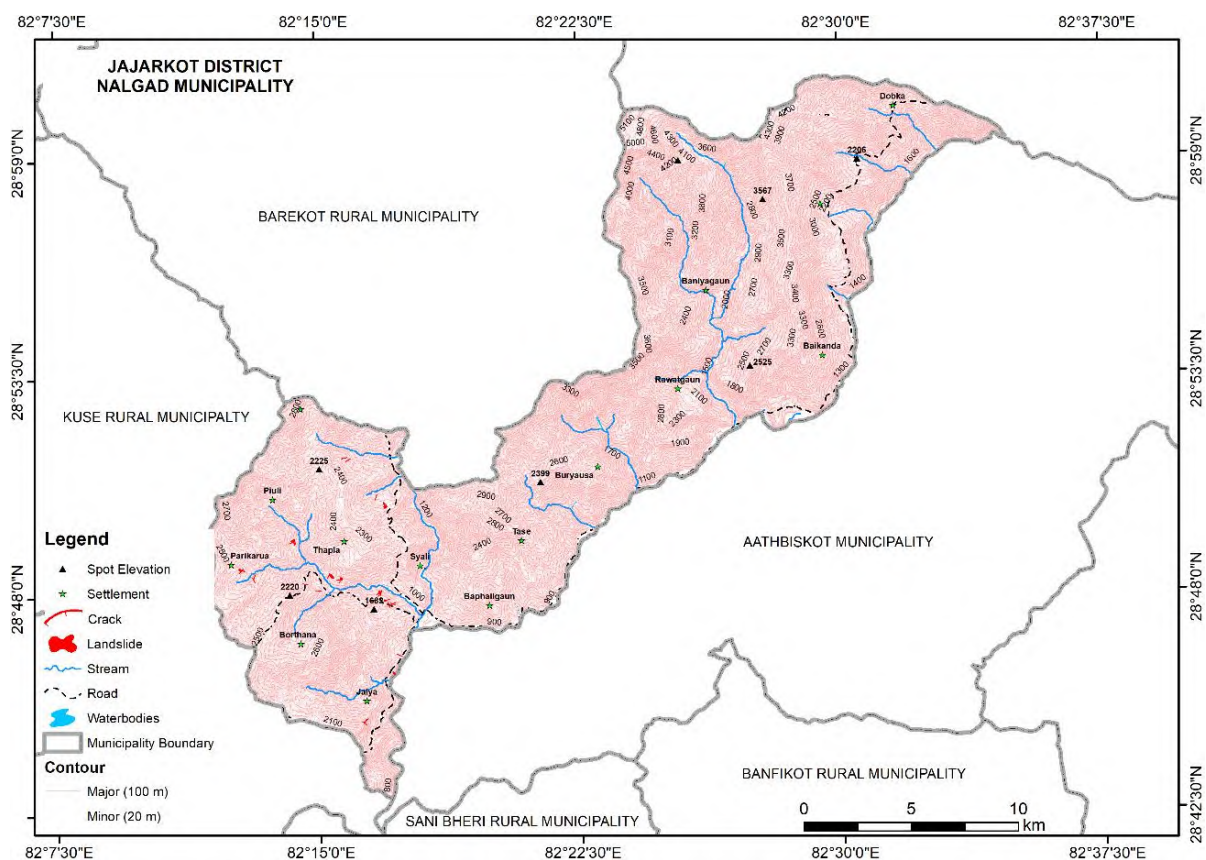


Figure 19: Cracks and Landslides observed in Nalgad Municipality

In Nalgad Municipality, Kalpat Tole, Rumlai Tole, Bakoti, Rani Gau, Darpakha Tole, Garagli, Farsa Gau, Aul Tole, Pari Karuwa, Karuwa Gau, Chaurila, Pali, Chiure and Lalta Chaur area have some significant damages.

Darpakha Tole: In ward 4, there lies an area known as Dharpakha Tole, characterized by long cracks along its slope. Situated below this crack-ridden terrain is Gynodaya Primary School,

nestled amidst the slopes. Figure 20 shows the location of Daharpakah area and cracks around it and how it can affect the downstream community.

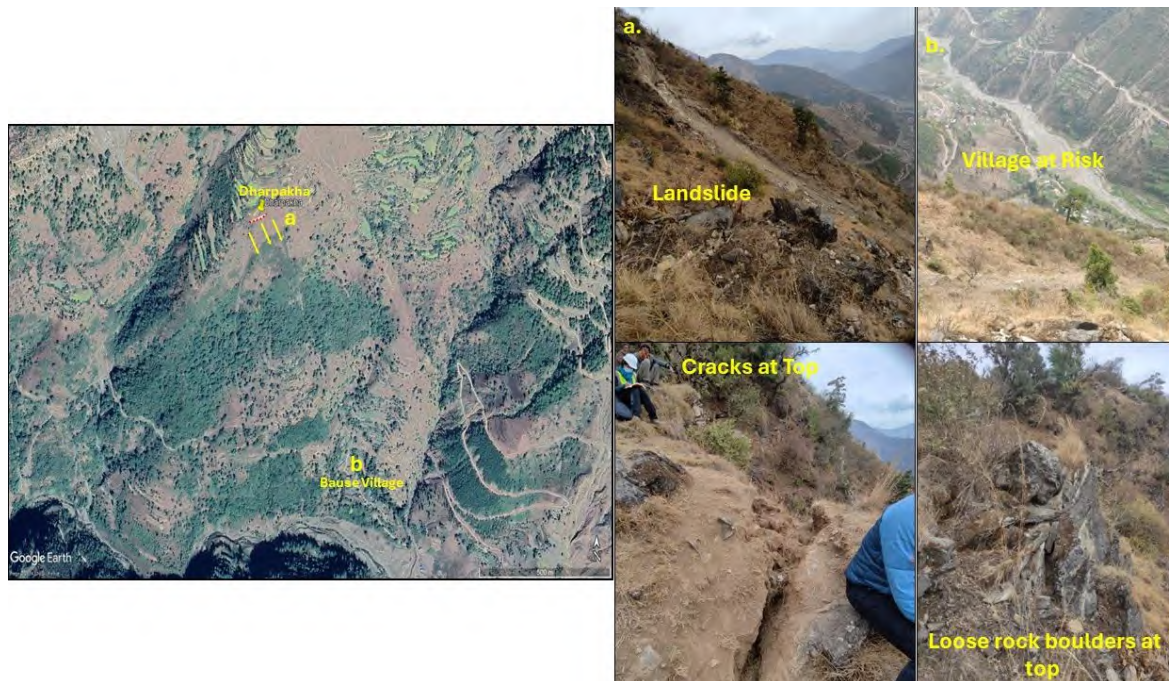


Figure 20: Cracks observed at Daharpakah area and downslope settlement at risk.

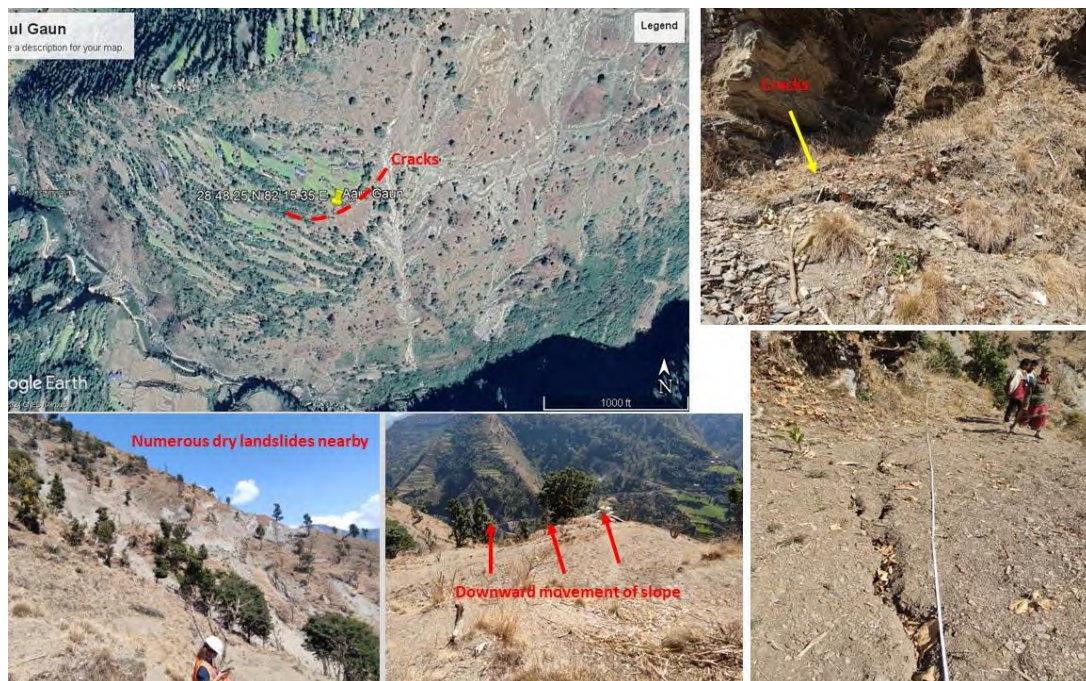


Figure 21: Cracks observed at Aul village area and downslope settlement at risk.



Figure 22: Cracks observed at Garagli village area and downslope settlement at risk.



Figure 23: Cracks observed at Latachaur village area and downslope settlement at risk.

5.2. Chedagada Municipality

Several Wards in Chhedagad Municipality have been affected by landslides triggered by the earthquake. These Wards include Ward 2, 4, 5, 12, and 13. Ward 3, on the other hand, has been primarily affected by cracks in the ground. Land subsidence, impacting both the ground itself and roads, is also observed in Wards 3, 5, and 12. Some villages of Ward 3, 4 and 13 are the

most affected from the earthquake and some places within these Wards are to be migrated. Some major locations such as Bata Chaur, Lewal Bazar and Saga Chaur area have some significant landslide risk. The damage details of Chhedagad are on Annex III.

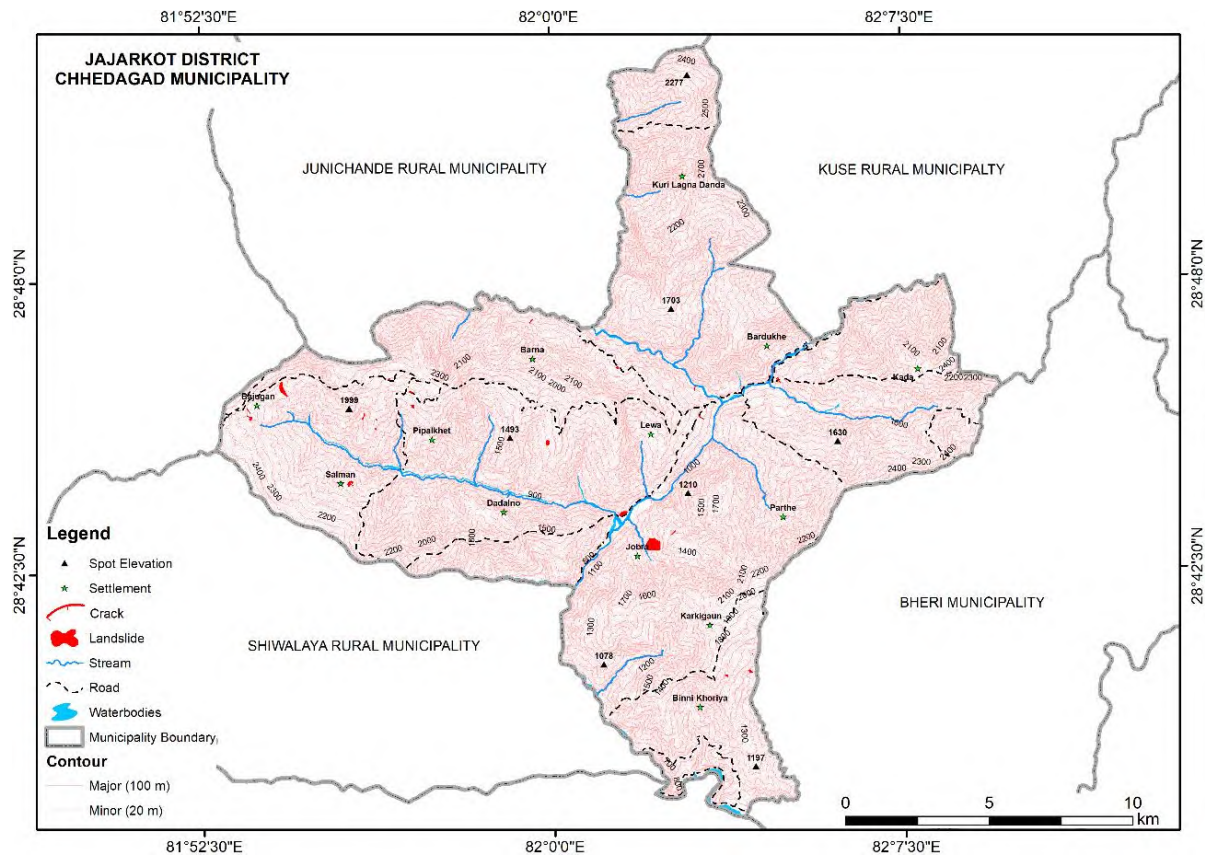


Figure 24: Cracks and Landslides observed in Chhedagad Municipality

5.3. Kushe Municipality

Kushe Rural Municipality has suffered the most damage in Ward 2, where earthquake-induced landslides, cracks, and subsidence are widespread. Dogri village in particular has numerous cracks measuring around 20 cm wide and 25 cm deep. Landslides have also been observed in Wards 4 and 5. Road and land subsidence can be observed in Ward 2 and Ward 3.

Some major locations such as Sagri gau, Talegau, Manedhunga Ma Vi, Bagge Khola, Dogri, Bhera, Khurpa and Damdala have some significant landslide issues.

The damage details of Kushe Rural Municipality are on Annex III.

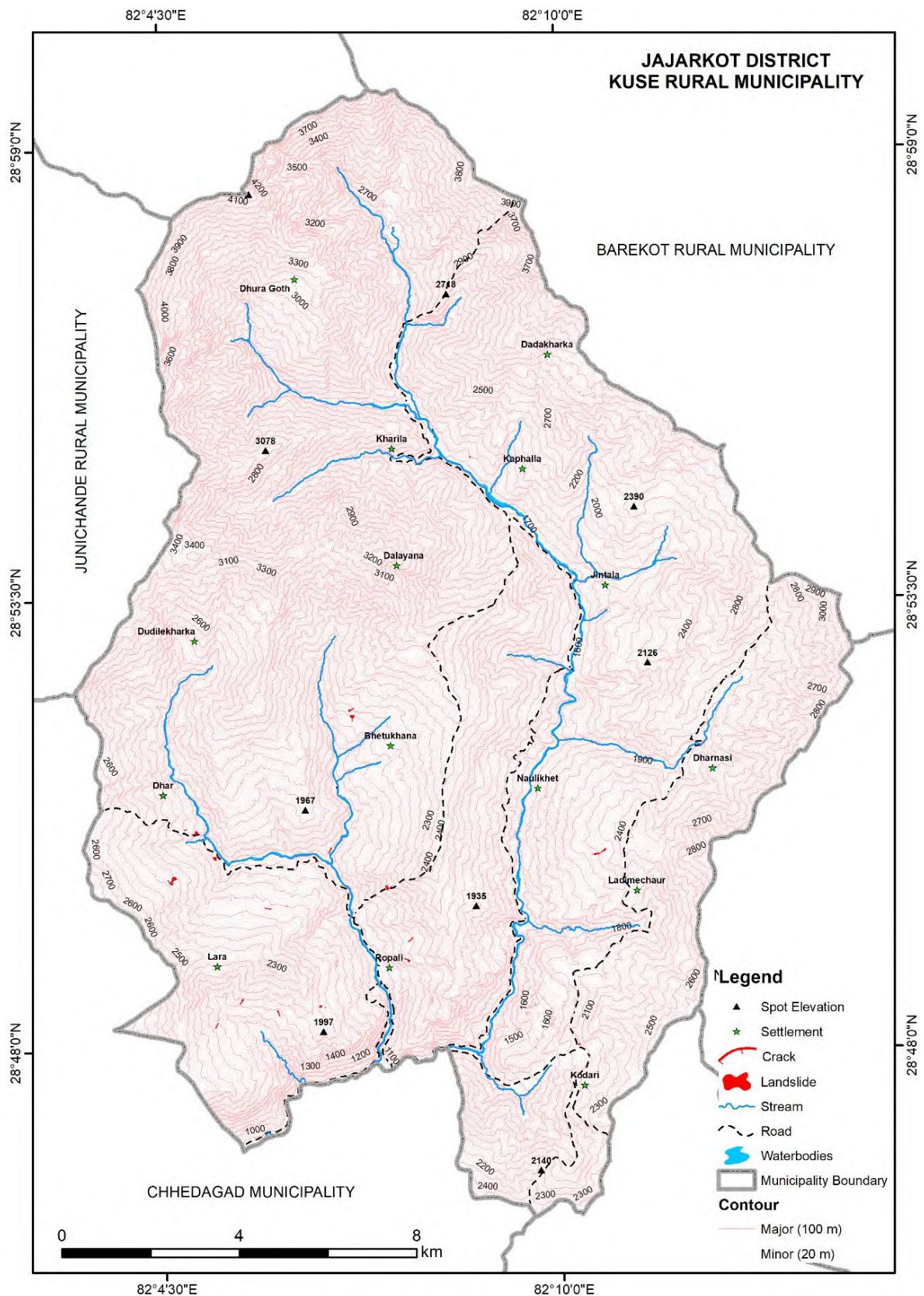


Figure 25: Cracks and Landslides observed in Kuse Rural Municipality

5.4. Shivalaya Municipality

The most affected Wards in Shivalaya Rural Municipality are Ward 2, 3, 6 and 7. All these Wards are affected by landslides which are old and reactivated by earthquake. Some of the villages from this Ward need to be migrated as soon as possible. Dry spring problems can also be seen in Ward 1 and 2. Subsidence of land can be observed in Ward 1, 2, 4, and 5 of Shivalaya Rural Municipality.

Some major locations such as Dingaghar tol (Sisne), Biurekuna Raha, Purano Tip, Kolpari Tol, Lothpakha, Pokhare, Syala gau, Viusa, Gosai Tol (Viusa), Thanbada Tol (Viusa), Malubela Chaur, Nigale Kholi, Talla Mela Odar area have some significant risk. The damage details observed in Shivalaya Rural Municipality are on Annex III.

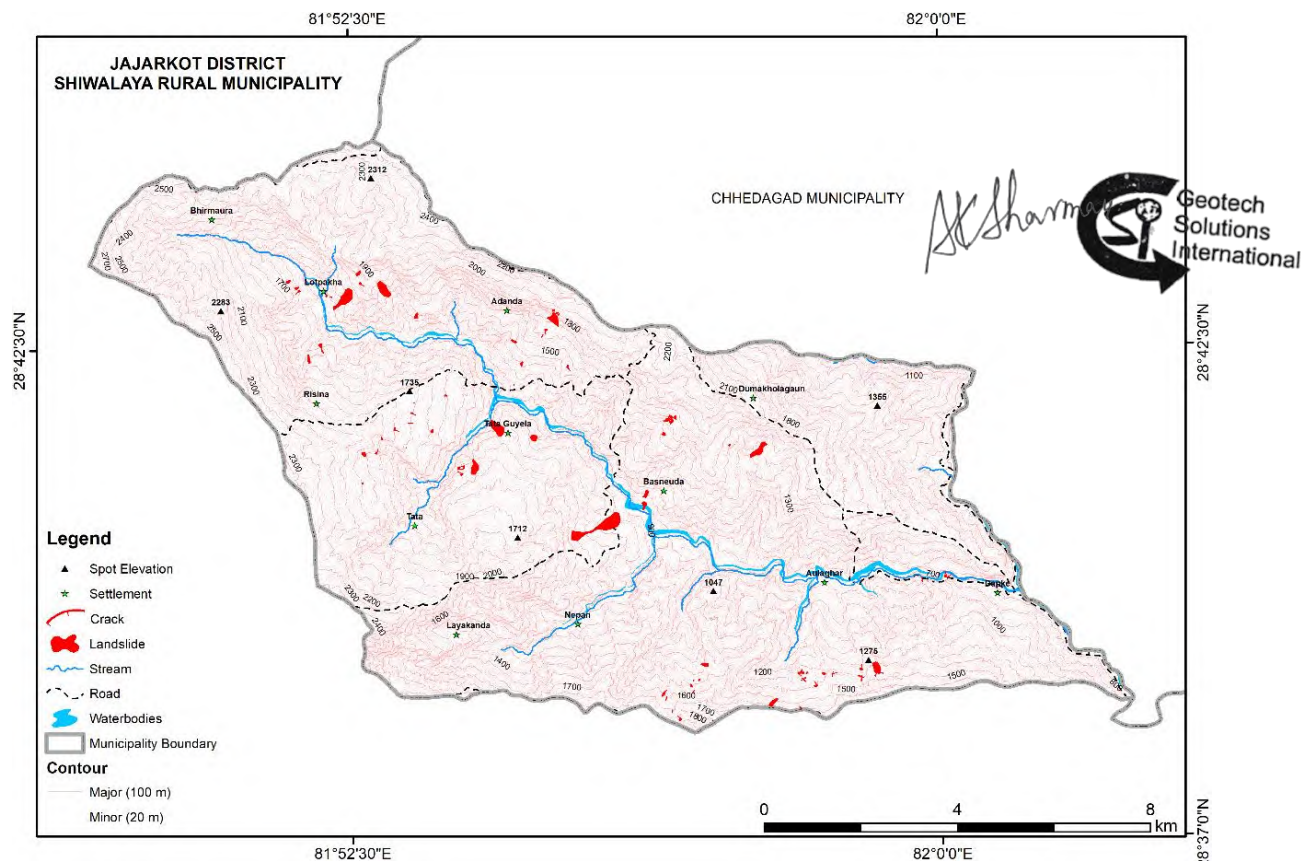


Figure 26: Cracks and Landslides observed in Shiwalaya Rural Municipality

5.5. Junichande Rural Municipality

Major earthquake induced and rainfall induced landslides in this Rural Municipality are in Karkigau (Ward 1), Kortan, Mujkot (Ward 11), Patalpani, Khattekhol (Ward 4). Major rock fall problem was observed in the Badibati tol of Ward 4. Some major locations such as Karki Gau, Saraswati Adharbhut School, Batkule, Mujkot, Sanokhol (Gau), Badibati Tole and Kortan

Tol has some significant landslide risk. The damage details in Junichande Rural Municipality are on Annex III.

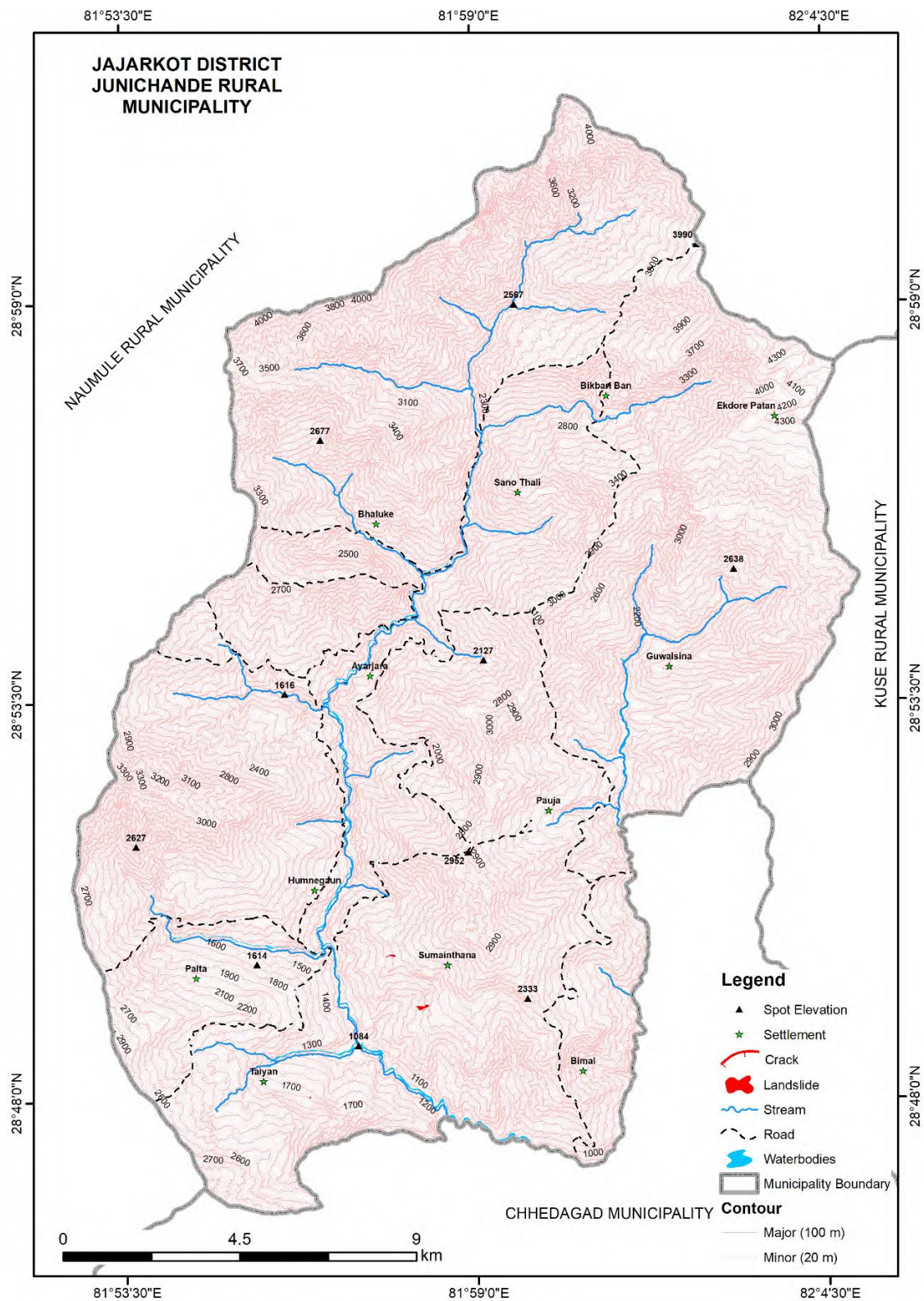


Figure 27: Cracks and Landslides observed in Junichande Rural Municipality

5.6. Barekot Rural Municipality

Ward 1, 5, 6, and 8, are prone to landslides in Barekot Rural Municipality, which can be dangerous for the people living there. Specific areas Sarkijuila in Ward 4, Tolkhan and Karuka in Ward 5, Nauli and Dimyara village in Ward 7, Gapichaur in Ward 8, Laikham, Bandala, Rawaldhara in Ward 9, are particularly affected by reactivated landslides and earthquake induced landslide. In Laikham village, Ward 9, there are cracks on the top of the hill that may cause rocks to fall during the coming monsoon. Koteli Village and Okhaldhunga are also at a high risk of stream erosion landslides. The information was collected through field surveys and is represented in Annex III.

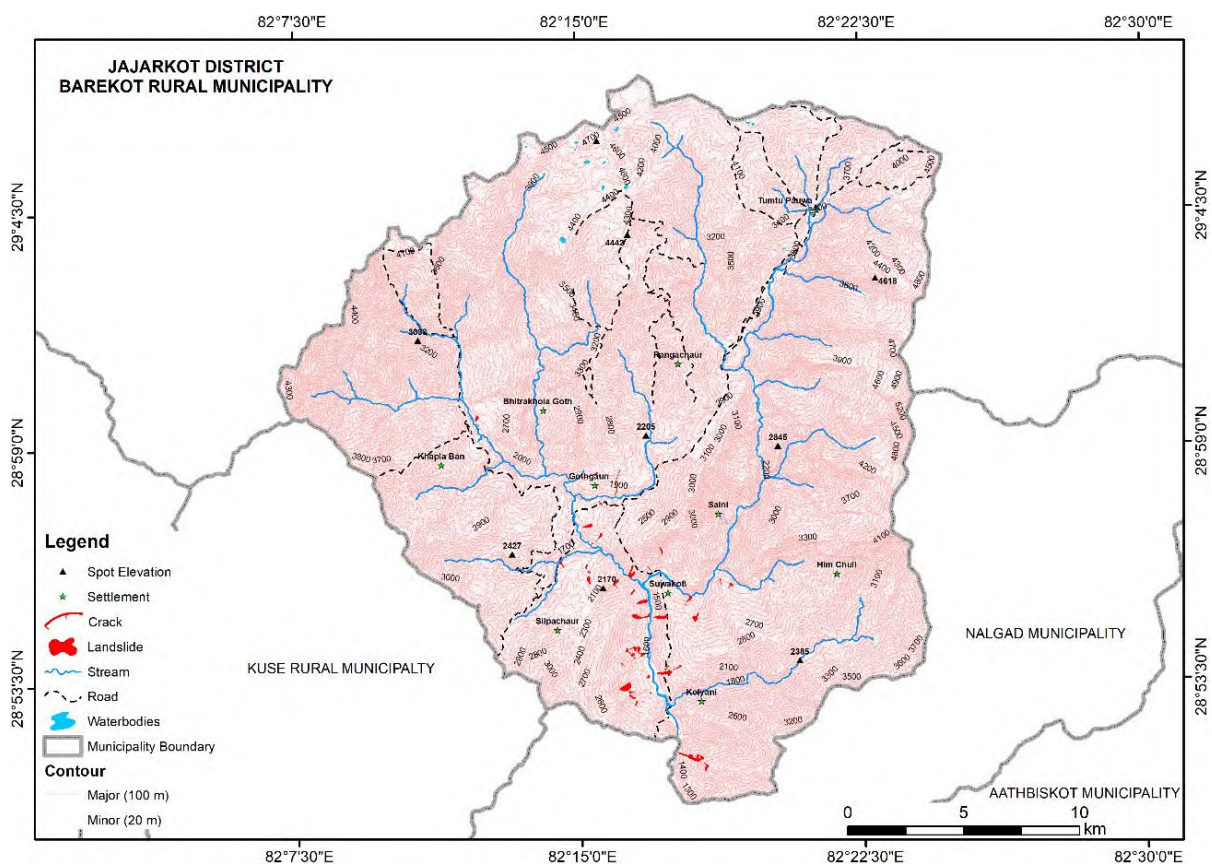


Figure 28: Cracks and Landslides observed in Barekot Rural Municipality.

5.7. Bheri Municipality

The most affected Wards of Bheri Municipality are Ward 1 and Ward 2. These Wards have the highest number of cracks within the settlement area induced by earthquake. The most vulnerable villages are Maide, Chaupani and Chaur of Ward 1 and Dulley of Ward 2. The areas potential to further rainfall induced slides are Ward 3 and Ward 4. These Wards have numerous medium scale slides around the settlement area. Rokayagaun of Ward 5 and Gangatiya of Ward

6 have terrain which are susceptible to land subsidence and cracks developed by creeping. As per Ward Chief, the ground water at about 4-5m is maximum present at Punma and the water level rises during monsoon. Some major locations such as Maide, area between Maide and Chaupani, Chaupani area, Thati Bazar, Harla Tol, Samlayana, Playa, Kopacha Ghosa Khola, Lamkatiya, Bhoor Chaur area, Balyatar, Damahabagar, Dhulpahiro and Aale Gau area have some significant landslide risk.

Land subsidence and creeping is also a major problem in this area. Dokana tol of Ward 7 has cracks and creeping of land is at Ward 7 due to stream erosion. The damage details are noted in Annex III.

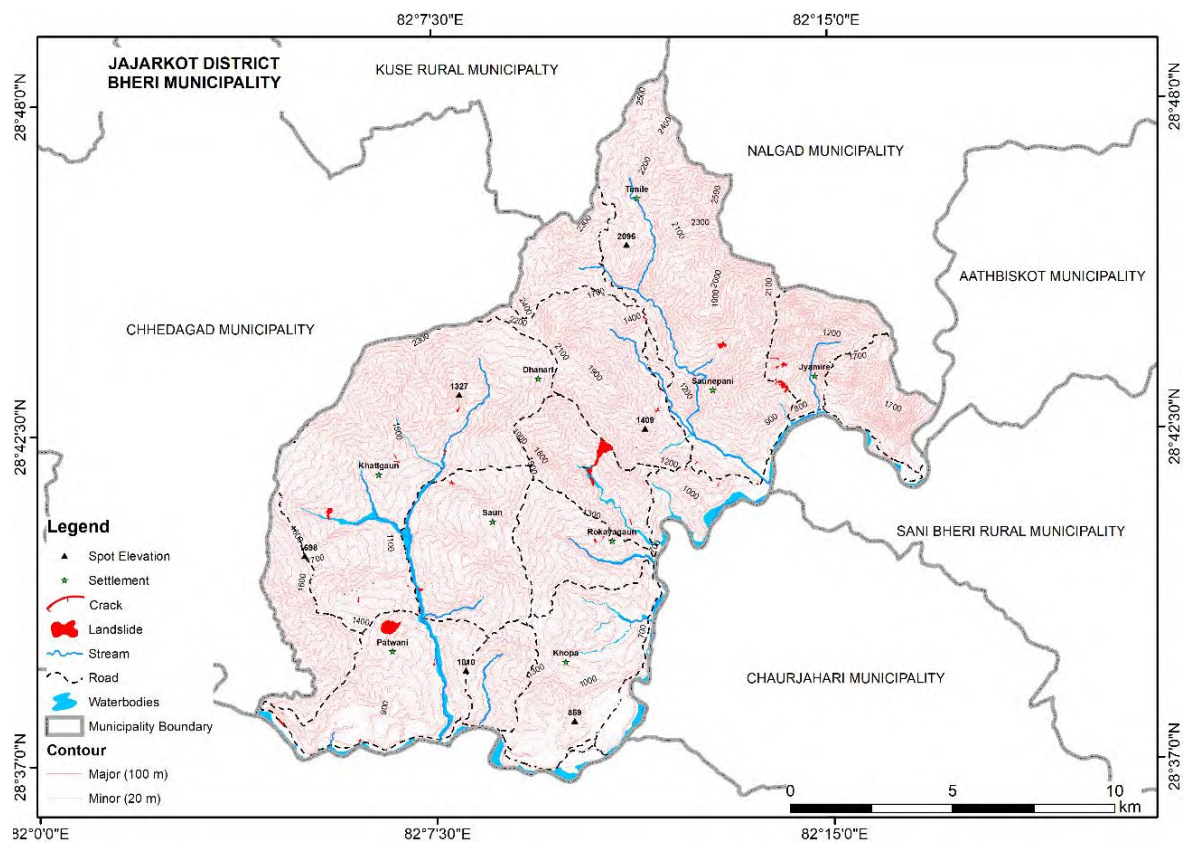


Figure 29: Cracks and Landslides observed in Bheri Municipality

6 Risk Assessment of Western Rukum District

6.1 Aathbiskot Rural Municipality

Aathbiskot is a Municipality in West Rukum District in Karnali Province of Nepal. Some parts of the municipality of Ward 11, 12, 13 and 14 are prone to landslides, which can be susceptible to people living there. Specific areas like Tatapani in Ward 9, Simpakha, Mungraha, of Ward 11, Kotdanda of Ward 12, Maurighari, Hampa, Bhalma Ghartipakha of Ward 13 and Bankegaira of Ward 14 are particularly affected by rainfall induced landslides reactivated by earthquake. In Mathilo Kuridhara in Ward 13 is high risk to the rock fall and is situated at the top of the boulders of the debris consisting of a gap of more than 3 m and depth of more than 5 m. The damage details of Aathbiskot Municipality are on Annex IV.

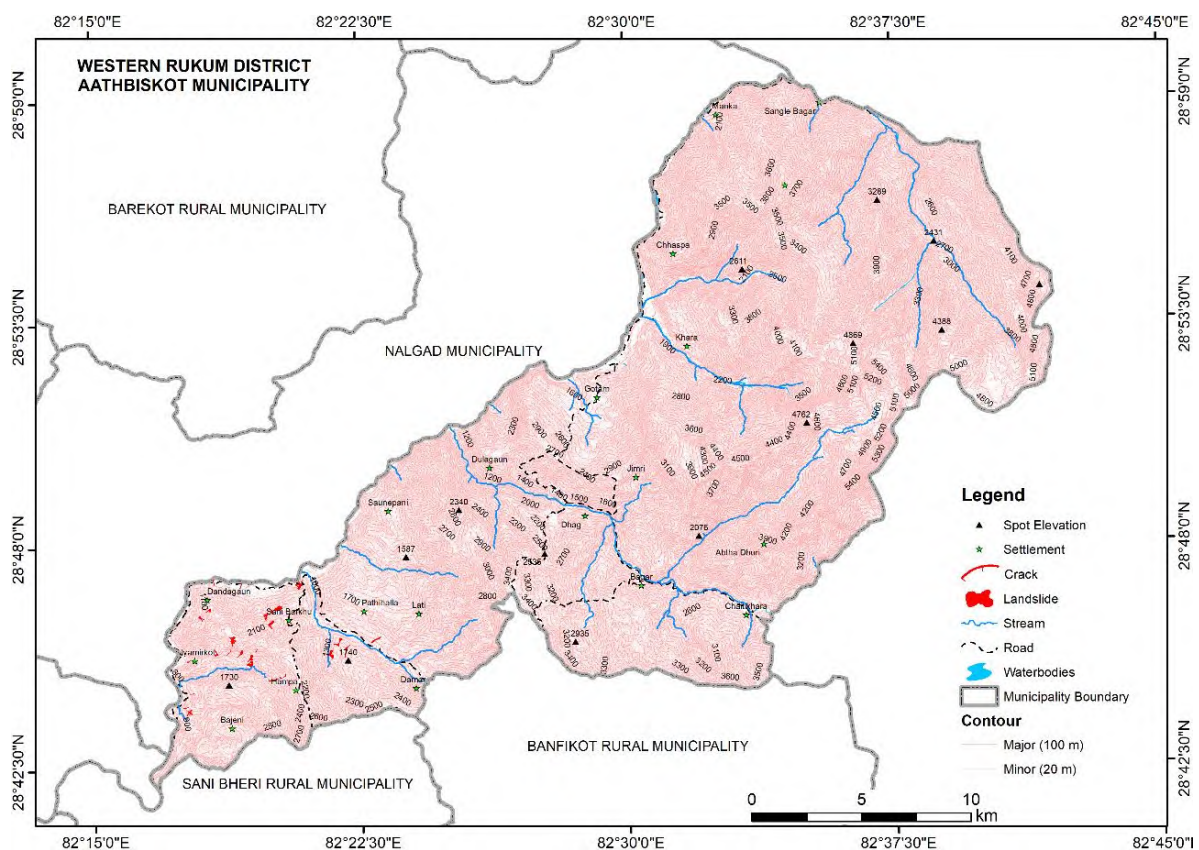


Figure 30: Cracks and Landslides observed in Aathbiskot Municipality

5.8. Triveni Rural municipality

Triveni Rural Municipality in West Rukum District in Karnali Province of Nepal. Some parts of the municipality Ward 1, 3, 4, 5, 6, and 7 are susceptible to landslides, which can pose a risk to people living here. Specific areas like Waski Thala, Barkhe Takura, Hampal, Jyalan, Chauki

Dhunga, Haiban, Thulo Gaira and Lalit are particularly affected by reactivated and earthquake induced landslide. The damage details in Triveni Rural Municipality are on Annex IV.

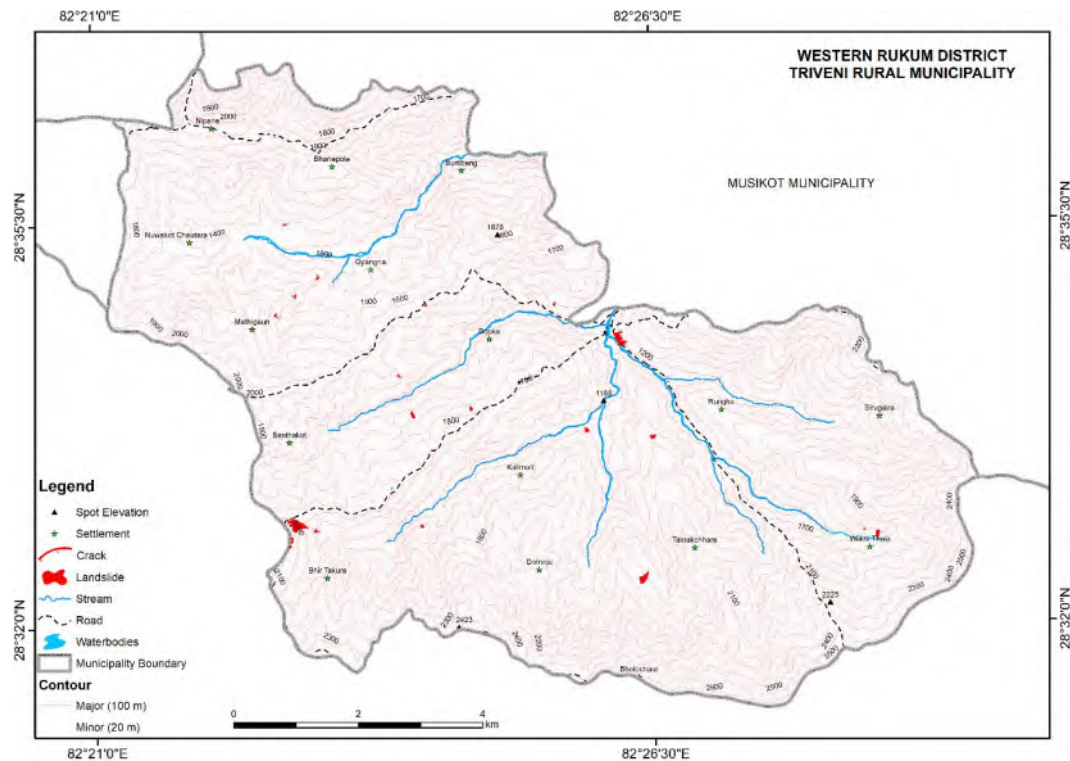


Figure 31: Cracks and Landslides observed in Triveni Rural Municipality

6.2 Bafikot Rural Municipality

Banphikot Rural Municipality in West Rukum, Nepal, faces potential dangers from landslides, particularly in Wards 5, 6, 7, 9, and 10. These areas, including specific locations like Kurlengjur, Kalabang Jhari Gau, Dabang, Mathi Gaira, Bhandari Kanda, Pandey Gau, Kauleni Gau, Gyamma Khola, Dhaule Pahren, Thulo Pahren, Jhyang Gau, kasera pakha, Ukaedadha area pose some significant risk of landslides that are reactivated by earthquake. Additionally, Ward 5 faces the unique risk of rockfalls and cracks due to its dolomitic topography. This combination of factors poses a significant threat to the residents of these areas. The damage details observed in Banphikot Rural Municipality are on Annex IV.

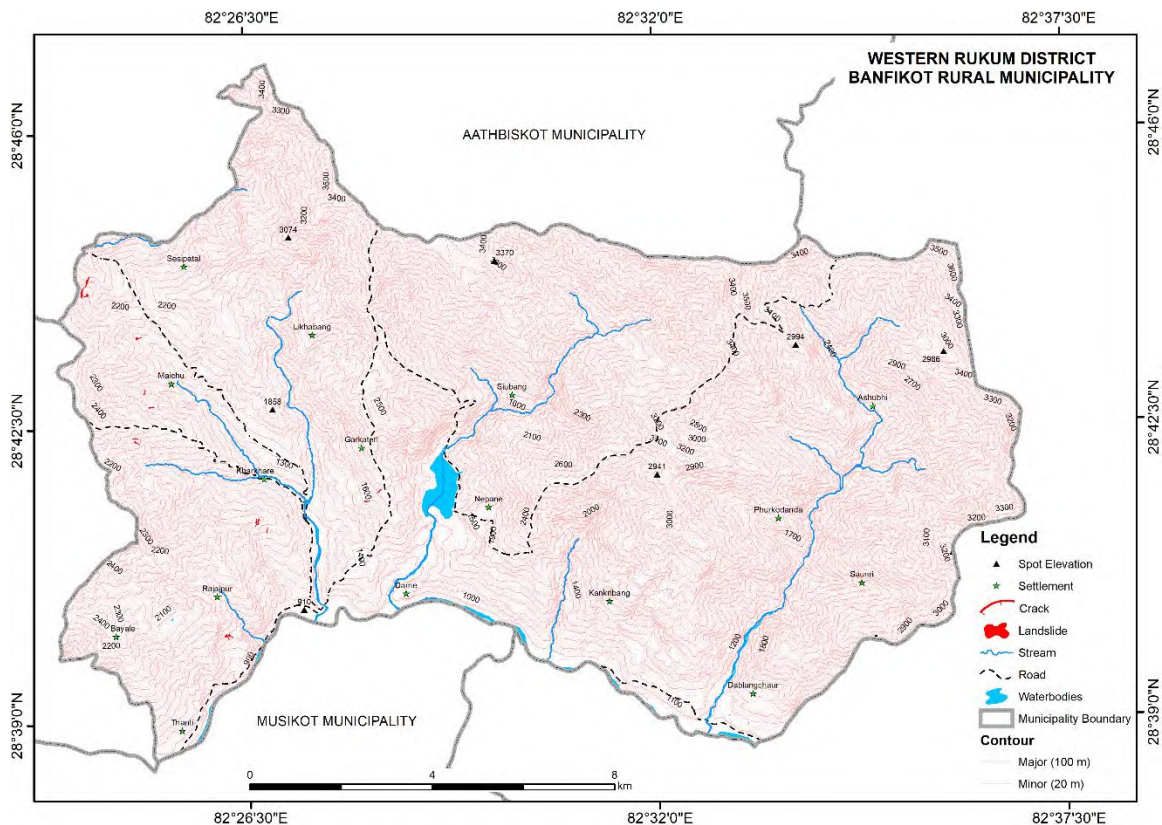


Figure 32: Cracks and Landslides observed in Banfikot Rural Municipality

6.3 Musikot Municipality



Major earthquake reactivated landslide in Musikot Municipality is observed in the Ghualpani (Dangdunge gau) of Ward 6, in Jharigau Dabang of Ward 7, in Mathigau Gyammakhola khandanda road (Ward 7 and 8), Roti Gyammakhola Road section in Ward 7, Kyamma area in Ward 8, Dhulepahiro area in Ward 9, Thulopahiro in Ward 10 and Rithabot landslide in Ward 11. Multiple series of cracks are observed in Jharigau in Dabang village of Ward 7. Land subsidence problems is observed in Sakh village of Ward 7, Kanshpokharai of Ward 12. Rock fall problem was observed in Ghorkhani village Ward 8. The damage details of Musikot Municipality are on Annex IV.

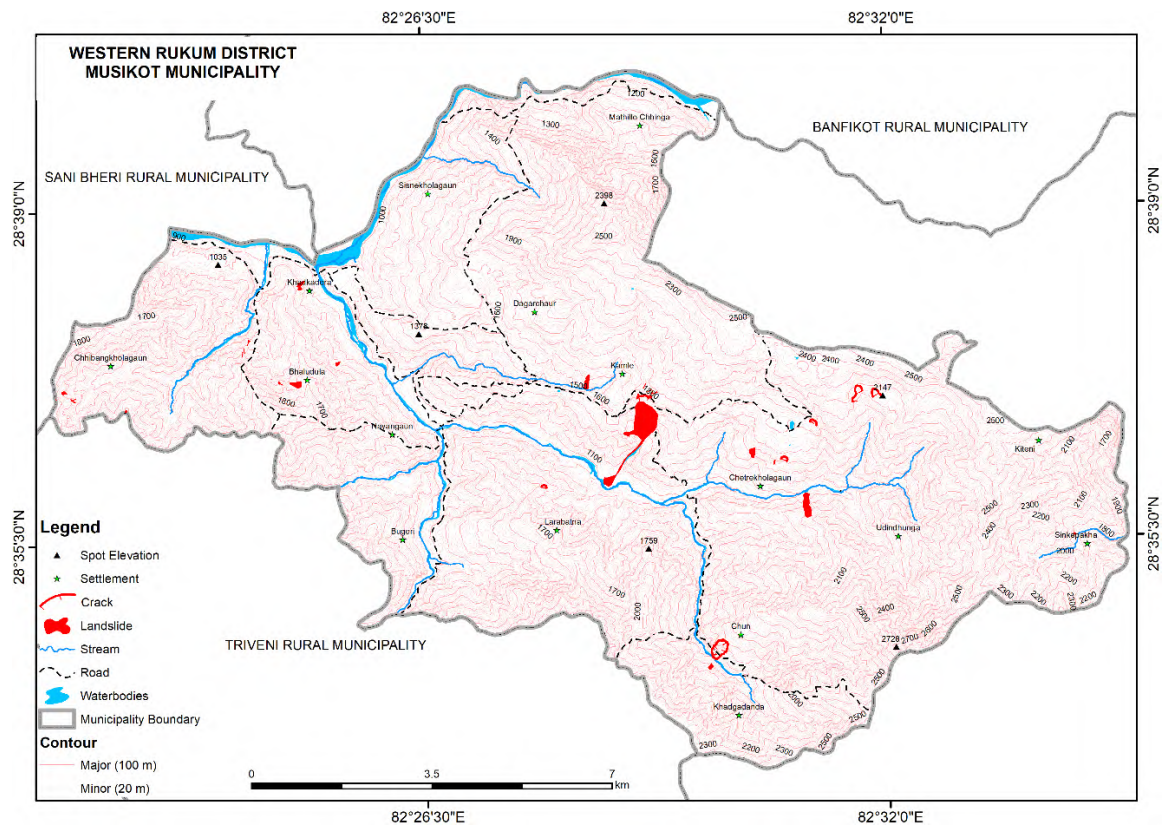


Figure 33: Cracks and Landslides observed in Musikot Municipality

6.4 Sanibheri Rural Municipality

Most of the affected areas in this rural municipality are Chinabagar (Ward 1) and Githakot (Ward 2) and Gaje Takuri (ward 3) Kotgau Bijaypr (ward 6), Tallo aamdanda (ward 8). Series of cracks and land subsidence are observed in the Chinabagar and Gajetakuri of Ward 3. Rock fall topple is observed in the Tarike of Ward 1. Water level fluctuation problems is observed in the Balebang (Ward 4) and Sarkemare of Ward 8. Prominent cracks near settlement and multiple crack lines are observed in the Balebang Ward 4. The damage details of Sani Bheri Rural Municipality are on Annex IV.

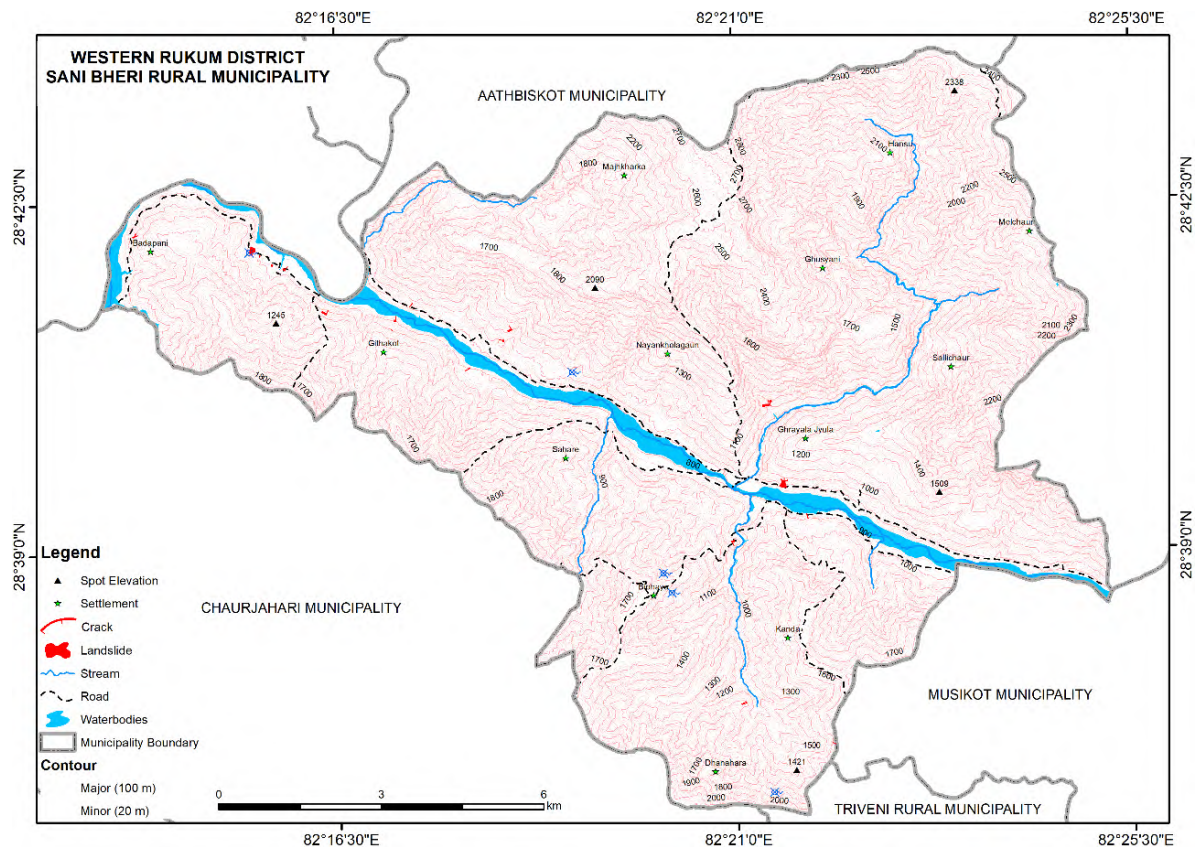


Figure 34: Cracks and Landslides observed in Sani Bheri Rural Municipality

6.5 Chaurjahari Municipality

Prominent landslides in this municipality are located in the Ratamata and Jhapekhola Gau of Ward 13 and Majhmela Area of ward no 12. Some major cracks in this municipality are located in the Mulpani of Ward 6 and Jumlipokhari of Ward 12, Sharkena of Ward 11. Similarly, crack and land subsidence are located in the Jhapekhola Ward 13. The damage details of Chaurjahari Municipality is on Annex IV.

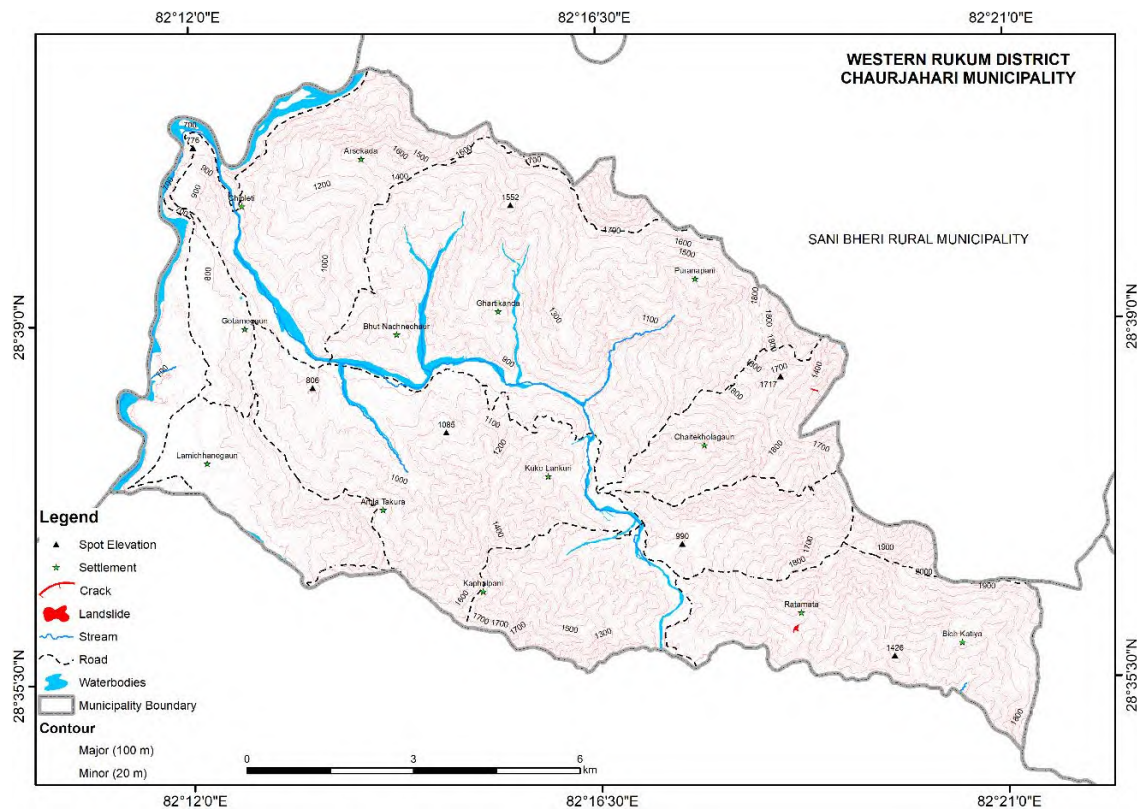


Figure 35: Cracks and Landslides observed in Chaurjahari Municipality

7 Risk Rating of Jajarkot and West Rukum Landslides

The field investigation revealed that the significant landslide triggered by the Jajarkot Earthquake itself is restricted to a small extent. Reactivation of landslides previously triggered by rainfall is evident in numerous locations. Throughout the majority of the area, the earthquake has resulted in cracks appearing at multiple sites, suggesting an acceleration insufficient to instigate a landslide. However, these cracks may act as potential triggers for landslides during the impending monsoon season due to infiltration and erosion. Therefore, ranking of these landslide and cracks are deem necessary to plan our immediate and long terms action. Risk ranking involves various approaches such and, in this study, we use the linear Risk ranking Scale This is probably the easiest and most used practice to rank risks: One has to simply give a number in a linear scale of 1 to 3, 5 to both the likelihood and severity (causes) of an event and multiply the two. The product is the risk level. Sometimes you find ratings in fractions, or percentage, but the idea remains the same. This is very similar to RPN (Risk Priority Number), as can be found in Failure Modes and Effects Analysis (FMEA).

In our case all the landslides are cracks and are extremely high risk zone if untreated and we need to categorized based severity case. To prioritize the landslides, they are rated across five parameters: Severity, Proximity to Settlement, Accessibility, Risk for Further Damages, and Potential for Mitigation, Community Vulnerability each rated from 1 (low) to 5 (maximum).

Severity Factors	1	2	3	4	5	Rating (%)
Severity (how big, area to be taken for reference)	Very Small	Small	Moderate	Big	Very big	
Proximity to Settlement (how far the settlement)	Very Far	Far	Moderate	Close	Very Close	
Accessibility (How easily the	Inaccessibl e	Difficult to access	Moderately Accessible	Accessible	Easily accessible	

site can be accessed for mitigation action)						
Risk for Further Damages (cracks are	Sliding and movement ceased	Slight Sliding and movement observed	Moderate Sliding and movement observed	Significant Sliding and movement observed	Continuous movement has been observed	
Potential for Mitigation (easy or difficult to mitigate)	Very Less intervention needed	Less invention needed	Moderate intervention needed	Robust intervention needed	Not possible for mitigation	

The total score for each landslide determines its percentage rating by summation of all scores divided by the number of parameters. Landslides are then categorized into three classes: ≤ 20 for low risk (Green), 21-60 for medium risk (Yellow), and >60 for high risk (Red). This classification guides planning for immediate and long-term mitigation measures.

Figure 36 indicates the locations of landslide with high medium and low risk rating in Jajarkot District. Figure 37 indicates the locations of landslide with high medium and low risk rating in Western Rukum District.



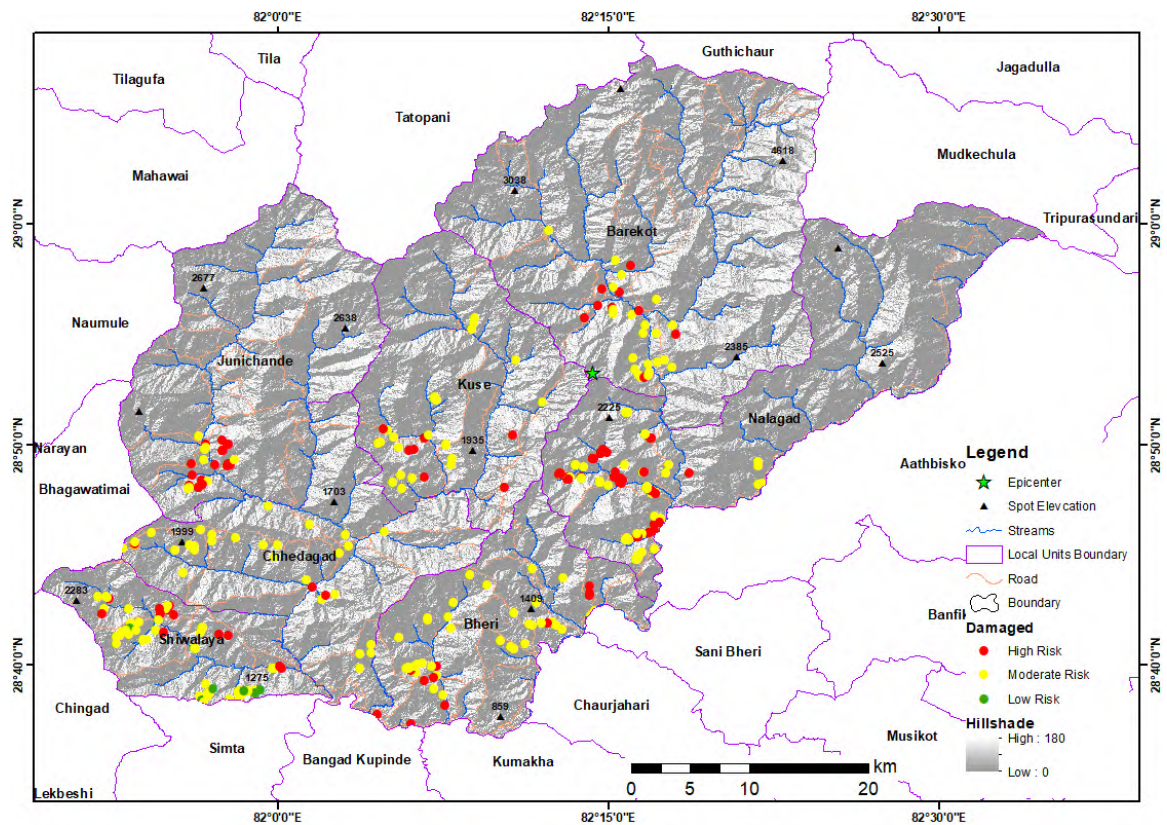


Figure 36: Landslide risk in Jajarkot District

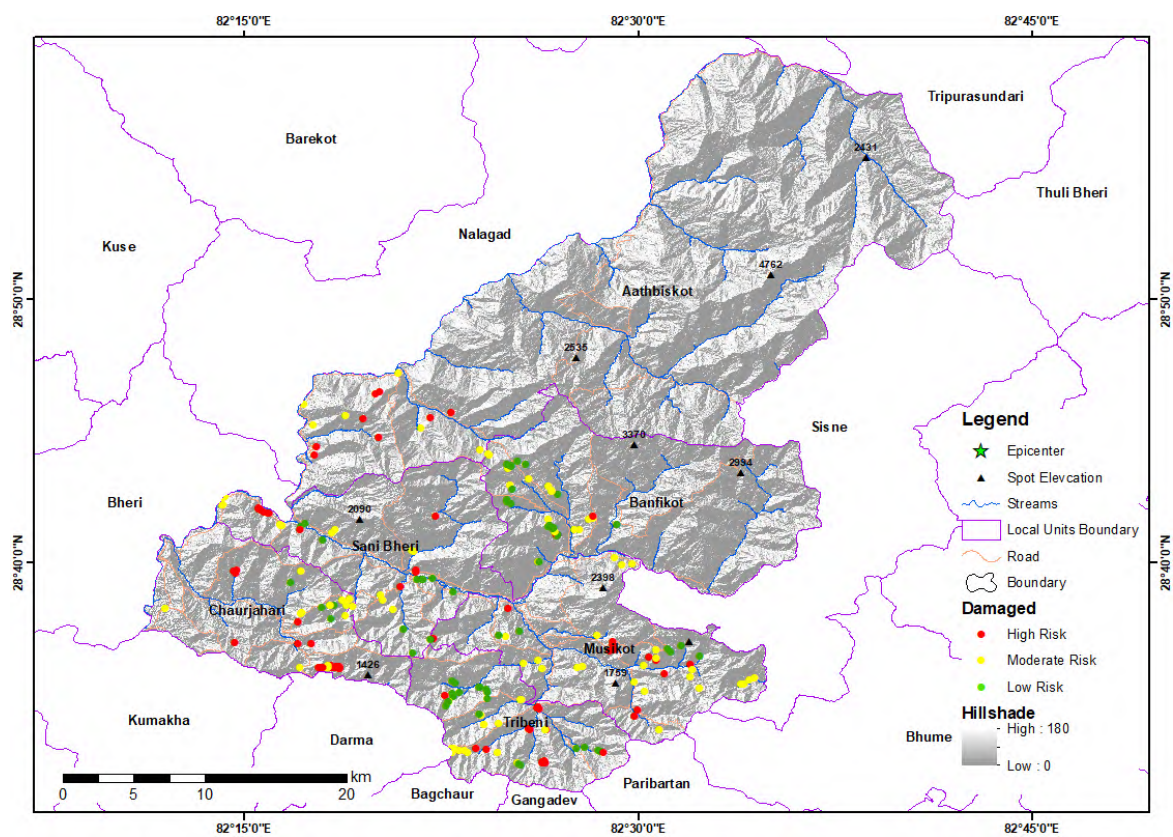


Figure 37: Landslide risk in Rukum District

The region's most severely impacted by the Jajarkot Earthquake 2023 include Nalgad Municipality, Barekot Rural Municipality, Bheri Municipality, and Siwalaya Municipality. Conversely, areas with comparatively lower effects are Junichande Municipality, Kushe Rural Municipality, and Chedagad Municipality.

Likewise, in Rukum district, Musikot, Chaujhahari, Tribeni Municipality, and Sani Bheri have experienced higher impacts, while Aaathbiskot and Bamfikot have seen lesser impact.

It's imperative to address the landslides and cracks in the red and yellow regions before the upcoming monsoon season. Prioritization should be given to the red zones for intervention, followed by the yellow zones, to streamline the mobilization of intervention teams. These areas in the red and yellow zones require ongoing monitoring for a couple of years, followed by further planning.

Among the red location, certain locations are particularly critical and necessitate detailed studies for potential resettlement. The specific locations requiring resettlement are detailed in

Table 1.



Table 1: List of villages that need relocation.

S.N	Village	Ward	Municipality	District	Image	Crack and Landslide Dimension
1	Maide	1	Bheri	Jajarkot		Crack: Length=111 m; width=4-12 cm; depth=15-35 cm
2	Chaupani	1	Bheri	Jajarkot		Crack: Length=7-8 m; width=10-15 cm; depth=15-20 cm
3	Chaur	1	Bheri	Jajarkot		Crack: Length=50 cm-3.5 m; width=10-25 cm; depth=20-55 cm
4	Tolkhan	5	Barekot RM	Jajarkot		Landslide: Length=300 m; breadth=42 m; depth=8 m
5	Nauli, Dimyari	6	Barekot RM	Jajarkot		Landslide: Length=560 m; breadth=230 m; depth=1 m
6	Laikham	9	Barekot RM	Jajarkot		Crack: Length=310 cm; width=2-13 cm; depth=60 cm (multiple cracks present within an area).
7	Chaurila	1	Nalgad M	Jajarkot		Crack: Length=5.8 m; width=10 cm; depth=1 m (multiple cracks present within an area).
8	Puwadadha	1	Nalgad M	Jajarkot		Crack: Length=2 m; width=7 cm; depth=30 cm (multiple cracks present within an area).
9	Chaitekafal	2	Nalgad M	Jajarkot		Crack: Length= 16 cm; width= 17 cm; depth=37 cm

S.N	Village	Ward	Municipality	District	Image	Crack and Landslide Dimension
11	Rumali	2	Nalgad M	Jajarkot		Crack : Length=45 m; width= 21 cm; depth= 1.5 m
12	Ayardadha	2	Nalgad M	Jajarkot		Crack and subsidence : Length=23 m; width= 7 cm; depth= 47 cm and subsidence=20 cm
13	Aule	4	Nalgad M	Jajarkot		Crack : Length= 30 m; width=25 cm; depth= 15 cm (multiple cracks present within an area).
14	Dedhkhulo	2	Chhedagad	Jajarkot		Landslide : Length= 100 m; breadth=50 m; depth= 7 m
15	Dalitbasti, Lewalbasti,	3	Chhedagad	Jajarkot		Lewalbasti Landslide : Length=40 m; breadth=30 m; depth= 3 m Dalitbasti Landslide : Length= 40 m; breadth= 20 m; depth= 3 m
16	Sagachaurbasti	12	Chhedagad	Jajarkot		Landslide : Length=100 m above; breadth= 90 m; depth=3 m
17	Bagaha	13	Chhedagad	Jajarkot		Landslide : Length= 30 m; breadth=20 m; depth=1 m
18	Kortang village	11	Junichande	Jajarkot		Landslide : Length= 80 m; breadth=700 m; depth=2.5 m
19	Dungra	2	Shivalaya	Jajarkot		Landslide : Length= 500 m; breadth=100; depth=4-5m
20	Sima	3	Shivalaya	Jajarkot		Landslide : Length=200 m ;breadth=60m ;depth=10 m

S.N	Village	Ward	Municipality	District	Image	Crack and Landslide Dimension
21	Lotpakha	6	Shivalaya	Jajarkot		Landslide: Length=200 m; breadth=40-50 m; depth=20 m
22	Viusa	7	Shivalaya	Jajarkot		Landslide: Length=35 m; breadth=40 m; depth=1.5 m
23	Kotgau bijaypur	6	Sani Bheri	Rukum		Crack: Length=100 m; width=0.5 m; depth=10-15 cm
24	Middle hill highway Chinabagar	1	Sani Bheri	Rukum		Crack: Length=10 m; width= 25 cm; depth=1 m
25	Gajetakuri	3	Sani Bheri	Rukum		Crack: Length=200 m; width=8 cm; depth=50 cm
26	Kalapahira	13	Aathbiskot	Rukum		Crack: Length= 2-8 m; width= 2-30 cm; depth= 50-60 cm
27	Mathillo kuridhara	13	Aathbiskot	Rukum		Crack: Length= 5-10 m; width= 2-5 cm; depth=44-58 cm (multiple cracks present within an area).

7.1 Coseismic Landslide

Understanding the impact of earthquakes on altering the likelihood of rainfall-induced landslides in the months and years following such seismic events has been a persistent challenge. The Gorkha Earthquake of 2015, for instance, instigated over 20,000 landslides across 14 districts of Central and Western Nepal, showcasing its immediate consequences. However, comprehending its longer-term implications has been difficult to predict. Subsequent years witnessed a surge in landslides and unforeseen incidents in the region, including the 2016 Bhotekoshi Glacier Lake Outburst Flood (GLOF), the 2020 cloud burst event triggering numerous landslides in Sindupalchowk, and the 2021 flooding of the Melamchi River. This trend is not unique to Nepal, as large earthquakes globally have been followed by widespread landslides, contributing significantly to the overall casualty toll. Such coseismic landslides bring about extensive disruption to infrastructure and facilitate the transport of substantial sediment volumes into river networks. A study by Rosser et al. 2021 highlighted that the landslide hazard increased notably by November 2019 compared to immediately post the 2015 Gorkha earthquake, with a significant portion of the landscape affected by landsliding.

In the case of the Jajarkot earthquake of 2023, while the immediate impact of landslides may not have been remarkable, the potential for rainfall-triggered landslides in the earthquake-affected terrain remains uncertain and daunting. Hence, it is recommended to expedite immediate actions to mitigate the exacerbation of this situation during future monsoon seasons.

7.2 Crack Sealing Methodology

In landslide prevention, it's crucial to keep surface water from seeping into the ground near potential landslide areas. Good drainage on the surface is needed for this. This is especially important where landslides have already happened or tension cracks have been developed in slope. If runoff water gets into cracks and gaps in the soil, especially at the top of the slope, it can cause some negative effects:

- Increase water levels in the unstable area, reducing the strength of the soil and making it more likely to slide.
- Add extra force to the slide because of the pressure from the water in the cracks at the top of the slope.



The most common method for sealing cracks consists of filling them with puddle clay or other impervious materials. It is often sufficient to excavate a trench along the tension crack and to backfill it with the excavated impervious material, possibly adding small quantities of bentonite, red clay or other natural material to reduce permeability further, and shaping the ground so that surface water does not pond in the area. If necessary, an impervious membrane may be added at or near the surface.

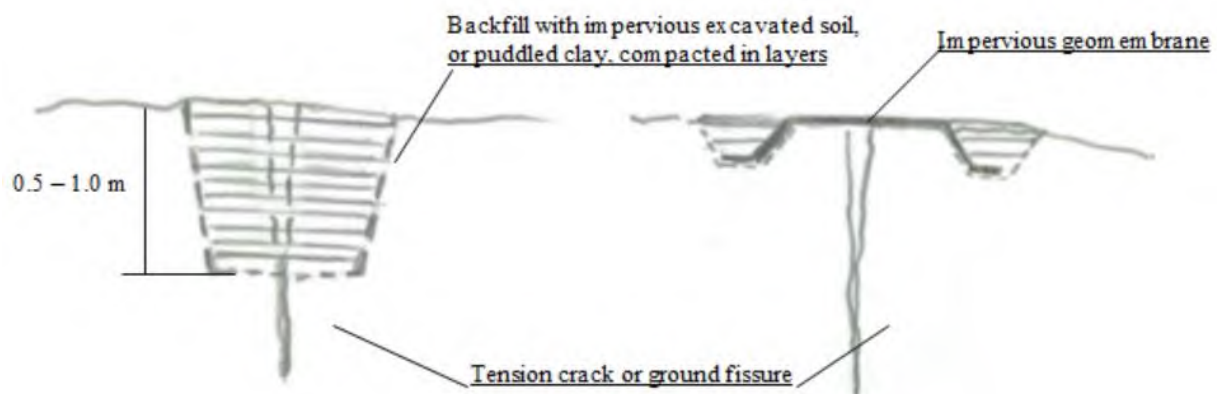


Figure 38: sealing tension cracks- typical example (Source: NGI, 2023)

Impervious membranes/ plastic sheets may be used as an emergency or temporary measure, while arrangements are being made for the works to be carried out. In addition to the crack sealing, regular inspection and maintenance is required in case of continued movement as it may cause previously sealed cracks to reopen.



Figure 39: A typical example of covering and sealing of cracks using impervious materials and plastic sheets.

8 Conclusion and Recommendation

In summary, the earthquake appears to have had minimal direct impact in triggering landslides. Most of the landslides and cracks appeared or identified by the study is located in already existing landslides which is mainly activated by rainfall. However, the primary cause of damage in the affected area caused by the reactivation of pre-existing landslides due to heavy rainfall. Additionally, the seismic activity has led to the formation of significant fissures, posing a potential risk of future landslides if not addressed promptly. Moreover, the earthquake has induced rock fall issues in regions such as Jajarkot and Rukum, which were previously unaffected by such occurrences. Furthermore, there have been reports of disappearing spring water in certain locations, raising concerns about potential water shortages in the near future.

In Total 211 no of landslides and about 300 cracks were located from the field visit. The main objective of this study is to prioritize areas of risk for settlement considering subsidence and landslide factors. Among the identified location of landslides 27 critical sites (as Listed in Table 2) need detail study for the resettlement and relocation work.

Upon assessment, significant cracks were identified on the uphill side of the road slope, signaling potential instability. To address the threat of landslides, particularly in road vulnerable regions like the Chinabagar area of Sani Bheri Municipality along the Midhill highway, our study emphasized the importance of proactive mitigation strategies, especially considering the impending rainy season.

As an immediate measure, it's imperative to seal these cracks using appropriate clay fill materials to prevent further erosion and potential collapse. All the cracks located in red and yellow zones must be sealed and monitored.

Furthermore, it's crucial to promote effective water management practices, locally known as "bhal katne kaam," to minimize water infiltration and reduce the likelihood of aggravating landslide-prone areas. By implementing such measures, we can mitigate the risk and safeguard critical infrastructure.

For areas identified as high-risk and requiring extensive intervention, a comprehensive study must be undertaken to assess the feasibility of rehabilitation measures. In cases where relocation is deemed necessary, adherence to government regulations and protocols is paramount.

In essence, a multifaceted approach combining immediate action, sustainable practices, and adherence to regulatory guidelines is essential to mitigate the threat of landslides and safeguard both lives and infrastructure in vulnerable regions.

9 References

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10 ANNEX

ANNEX 1: Survey sample format

पहिरो प्रभावित क्षेत्रको अध्ययन

जिल्ला	
गाउँ/नगरपालिका	
वडा	
गाउँको नाम	
GPS Location	

स्थलगत/भौगर्भिक विवरण

Dimension of Landslide (पहिरोको नाप)	Length (लम्बाई)	Breadth (चौडाई)	Depth (गहिराई)	Slope Angle (जमिनको भिरालोपन)	
Presence of Rock (चट्टान भएको/नभएको)				छ	छैन
If Yes, are the rocks fractured? (चट्टान भएमा चिरा परेको/नपरेको)				छ	छैन
Soil Type (माटोको प्रकार)	Clayey (कालो/रातो माटो)	Sandy (बालुवा जस्तो)	Gravel (गिट्टी)	Mixed (मिश्रित)	
Moisture Content (माटोको चिस्यान)	Dry (सुक्खा)		Moist (हल्का ओसिलो)	Very Moist (निकै ओसिलो)	

स्थलगत/भौगर्भिक विवरण

Type of Landslide (पहिरोको प्रकार)	Rockfall (चट्टान खस्ने)	Debris Flow (गेग्रान बहन)	Slide (स्खलन)	River Erosion (नदी कटान)

Presence of cracks (चिराहरु भएको/नभएको)	छ	छैन

Seepage in Landslide (मुल फुटेको छ/छैन)	छ	छैन
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Is there Seepage within the landslide? (पहिरो मा मुल फुटेको छ/छैन?)	1. Stream flow within the landslide (पहिरोको बीच बाट पानी बग्छ)
	2. Stream is at the side of the landslide (पहिरोको छेउ बाट पानी बग्छ)
	3. Seepage from one point (कुनै एउटा स्थानमा मुल फुटेको छ)
	4. Seepage from multiple points (धेरै ठाउँ बाट मुल फुटेको छ)



पहिरोले गरेको क्षतिको विवरण:

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Type of Landslide (Earthquake induced or reactivated old landslide) भूकम्पले निम्त्याएको पहिरो/पुरानो पहिरो reactivate भएको हो?		
मानविय क्षति	छ	छैन
खतरामा रहेको घरधुरी		

Cracks in land (जमिनमा चिरा परेको)	No cracks (चिराहरु नरहेको)		
	Few cracks (स-साना चिराहरु रहेको)		
	Large cracks (पहिरो वरपर गहिरो, लामो चिराहरु रहेको)		
Dimension of the largest crack (सबैभन्दा ठूलोचिराको नाप)	Length (लम्बाई)	Breadth (चौडाई)	Depth (गहिराई)
Crack 1			

Crack 2			
Crack 3			
Crack 4			
Crack 5			
Crack 6			
चिराको अवस्था	बढेर गएको	उस्तै रहेको	आफै पुरिएको

जोखिममा रहेका बस्तीको विवरण:

Damages to houses (क्षति भएका घरहरू)	
Completely destroyed houses (पूर्ण क्षति भएका घरहरू)	

पहिरोले गरेको क्षतिको प्रकार

पूर्ण रूपले क्षति	
आंशिक क्षति	
क्षति भएको घरमा बस्न मिल्ने/नमिल्ने	

रोकथामका कस्ता उपायहरू अपनाए बस्न मिल्ने हुनेछ?



Anchor गरे	Retaining वाल लगाए	Bioengineering गरे
अन्य उपाय:		
स्थानान्तरण गर्न पर्ने	छ	छैन
गर्न परेमा कति घर?		
निकै खतरामा रहेका घरधुरी		

जोखिममा विवरण:

Any visible signs of instability (जोखिमका संकेतहरू)	1. Cracks on land (जमिनमा चिरा)
	2. New spring (नयाँ पानीको मुल)
	3. Drying up of old spring (पानीको मुल सुक्ने)

	4. Muddy water from spring (मुलबाट लेदो पानी आएको)
	5. Disturbed terraces (धाजा फाटेको खेतका गाह)
	6. Overhanging rocks (झुन्डिएका चट्टान)
	7. Tilted poles/trees (ढल्केको रुख र बत्तीको पोल)
	8. others (अन्य)

चिराहरु घरबाट कति टाडा छन्?			
पानीको व्यवस्थापन कस्तो छ?			
जोखिम न्यूनीकरणको कुनै उपाय अपनाइएको छ/छैन	छ	छैन	

Recommendation (सिफारिस)

Final Impression (निष्कर्ष)	१. बस्ती जोखिम मा छ/छैन
	२. रोकथामका उपाय अपनाएर बस्न मिल्ने/नमिल्ने
	३. निष्कर्ष मा जान नसकिने
	४. बस्न नसकिने, स्थानान्तरण गर्न पर्ने
Does the settlement require simple means of slope protection measures? (के-कस्ता सामान्य रोकथामका उपायहरु अपनाउन सकिन्छ?) 	1. Surface water drainage (सतही पानीको व्यवस्थापन)
	2. Ground water drainage (जमिन मुनिको पानीको व्यवस्थापन)
	3. Sealing of cracks (चिराहरु पुर्ने)
	4. Bioengineering
	5. Slope trimming (भीरको भिरलोपन घटाउने)
	6. Scaling of rocks (अस्थिर चट्टानहरु हटाउने)
	7. Retaining walls (टेवा पर्खाल)
	8. Anchoring the slope
	9. Others

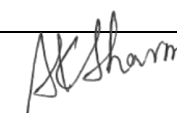
ANNEX II: List of contact person from the Jajarkot and Rukum selected municipalities

Municipality	Ward No.	Effects	Location	Contact information
Triveni Rural Municipality	1	No major cracks		Ganesh Kumar K.C. (R.M chairperson) (9829878360) 9868949539
	2	No problems	Jumli Pokhari, Barkhe takura, Simruti Bazar, Paandera	9768538470
	3			Unreachable 9813457706
	4			Unreachable 9810961060
	5	Cracks(C)	Saudera Shyamdhari(C)	9810181949
	6	No problems	421 houses damaged	9815601326
	7	Cracks	Ringthan Ghorekot (C), Khani Khola-Pitibang road, houses damaged in Melpokhari	9809717149
	8	Subsidence(S),	Aarmalidera, Garena, Purnagaun, Dhaireni	9819599250
	9	No problems		9822980081
	10	Cracks	Mulpani rojkhel, Salghari, Khandada, Damar, Trishule, Khatgara, Jiurekhola, Bhumke khola	9822847482

Municipality	Ward No.	Effects	Location	Contact information
Musikot Municipality	1	Landslide (L)	Chisapani (forest area)	Mahendra (Mayor) (9847852025) Gyanu Chaad/ Chairperson (9866904526)
	2	Subsidence, dried spring(DS), crack	Sangro Ooralo, jugaena spring, simkhola spring	Dinesh Kadka/Chairperson (9849968989)
	3	Landslide, subsidence	Mid-hill highway, kalahasingh, Pipalbot, Dhairinichaur	(9841570510)
	4	No landslides or cracks present /144 houses under reconstruction after earthquake		(9857848665)
	5			Unreachable (9841207418)
	6			Unreachable
	7	Subsidence	Janajagriti Ma. Bi.	Ek Bahadur Pandey (9847852636)
	8			Unreachable
	9	No problems		Sashiram Pun (9863679913)
	10			Unreachable
	11	Landslide, crack	Rithabot(6 houses at risk), Jaankalyan Ma. Vi Rampacha(above)	Govinda Bahadur Buda (9866907136)
	12	Landslides	Kurinichaur, Harkhola, Sungur Khola	Dhiren Shreepali (9847852283)
	13-Sadanchaur	Subsidence, Dried Spring	Lamodada tallo takura(S), Bhatke khola(DS+S)	Khal Bahadur Khadka(9844125525)

Municipality	Ward No.	Effects	Location	Contact information
	14	1.5 feet land subsidence	Lamadada(S), Chewon khola, Khalchepakha tole, wukkal dado, Lolingwang, Trishule, Whang khola	Chairperson 9866907140
Banphikot Rural Municipality	1	few cracks in slope of Kalatal, old landslide at Dhiklana has some new cracks that is 50m from settelment		Janak Kumar Batha Magar (R.M chairperson) (9844941181) Dhan Bdr. Khadka (Ward Chariperson) 9868900722
	2	No slide, subsidence or crakcs	Pipal	Prakash Khatri 9851097851
	3			9869982603
	4	Slide activated below Irrigation Canal at 10m subsidence, 60m from settelment	Simtaru	Ser Bdr Oli 9809509484
	5	No significant slide, cracks. Few spring dried up only		Manu Sahi 9844949199
	6	15'' subsidence at Jipsing Basti, Cracks at Pariban, Irrigation Canal subsidence & landslide at Pariban, Mangma Health post land subsidence and upliftment, Slide along Biekhola-Jipsing-Dhampu Road		Manoj Kumar Adhikari 9863198647
	7			9822954236
	8	Cracks near singhala basti, small slides along Turi dhara & Majha dhara, slide between Halbadhe & Sihala Road section		Dhan Bdr Khatri 9822982310

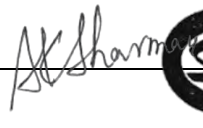
Municipality	Ward No.	Effects	Location	Contact information
	9			Jhog Bahadur Chandra 9810958848
	10	No significant slides, cracks		Hari Bahadur Chandra 9809770337
Sanibheri Rural Municipality	1	Landslides, cracks	Mid-hill highway(L)	Er. Sangam B. K. (9848009636) 9864360617
	2			9866923207
	3	Crack, landslide	Chaukhe tole road	9844915543
	4			9868907522
	5			9847478353
	6			9851142824
	7	Landslide	Mid-hill, Khahare dada rockfall, increase in spring discharge	9847953581
	8			9848795925
	9	Cracks, Landslide, Subsidence	Syallbot(C), Suliale, Kartike dhik(S)	9868052648
	10	Cracks	Dadaarna	9848106951
	11	Landslide	Nale Khanepani, Pathialna, 3-4 spring covered(rockfall)	9860905529




Municipality	Ward No.	Effects	Location	Contact information
Aathbiskot Municipality	1	No significant impact in land or cracks observed, spring dried	Thunu & Chaina Khola (S.D)	Dhan kumari Shahi (Vice-Mayor) (9843602687)/ Er. Rajesh Gautam (Spokesperson) (9860890574) 9864360617
	2			9866923207
	3			9844915543
	4			9868907522
	5			9847478353
	6	No cracks on land & landslides , few spring dried	Sisne, Joldi Khola	9851142824
	7	Small landslide above Dhumila Sadak, no significant slide & cracks	Dhumila	9847953581
	8			9843276478
	9			9848106951
	10	No significant slide		9847836977
	11	Cracks along the way to Danda Gaun , Cracks on road at Bojhabari section,Lanslide above road on way towards Goeri	Chisapani, Rithaa, Godire, Ruji gaun	
	12		Thuli Barkhu, Sani Barkhu	

Municipality	Ward No.	Effects	Location	Contact information
	13	Landslides near Jarjale, Jan Yuva School, Bhalna & Jan Jagaran School, Gaunbari near landslide, Cracks above the settlement of Puri Dhara, Cracks and Subsidence in Pakhaban, Bhalna Area, In previous ward no-5 area 1.5m cracks/subsidence	Bhalna, Gaunbati, Hampa / 8+ death severely affected zone	9864353590 Chakra Bdr BC
	14		Chepare, Takura basti, Jyamire kot /17+ death, severely affected zone	
Aathbiskot Municipality	1	Subsidence	Gharigaun Irrigation	Indra Bahadur Giri 9800595509
	2	No problems		Bir Bahadur Rana 9849659754
	3	Subsidence, Cracks	Khauchaurni-aamachour Irrigation Canal(S), Hattisaar-lamabagar(S), Karkitole(C)	Lal Bahadur Buda 9866123333
	4			Unreachable Gopal Bahadur Bhandari 9868019409
	5	No problems		Khub Bahadur Khadka 9858033506
	6	Subsidence	Mulpani Aadharbhut School, Chisapani-pakeykheth Irrigation	Jaya Prakash Bohara 9844983727
	7	No problems		Bhim Prasad Jaisi

Municipality	Ward No.	Effects	Location	Contact information
				9847849099
	8			Unreachable Sashiram B. K. 9867877312
	9	Subsidence, Cracks, Dried spring	Increased discharge in Panekhola, Badakhola agricultural land(S+C), Bhatkhola-Kairabesi(S), Kamigaun(C) may lead to landslide	Mahesh Sworcharya 9868946061
	10	Subsidence, Dried spring	Maankot-Kholagaun-Neta Gabion wall(S),	Birendra Rokaya 9745530560
	11	Cracks, Dried spring	Sarkina gaun(C), Belsiga(C+DS)	Tulse Oli 9866916814
	12	Cracks, Subsidence, Dried spring	Jaminpokhari gaun(C), Tallo ulleri gaun and road(C), Pateri gaun(C), Janakalyan Ma. Vi. Sanokhola(S), Pannekhola and Shisne Khola(S+DS)	Tika Ram Oli 9864935914
	13	Dried spring, Landslides, Subsidence, Cracks	Ratamata tole(DS+L), Bhuwati tole(DS+L), Sakhta khola irrigation and Dhursu bagar(S), Jhaffre khola(C), Janakalyan Aadharbhut School(L). Bhuwatijaula taap irrigation kulo(S)	Shree Krishna Sharma 9868606699
	14	No problems		Thal Bahadur Kunwar 9822805113




Annex III Damage details Jajarkot

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 45 ' 60"	82° 16 ' 52"	Kalpat Road	Nalgad M	Subsidence of 19 cm and crack having length of 36.6 m, width 15 cm, depth 50 cm.
1	28° 45 ' 60"	82° 16 ' 51"	Kalpat Road	Nalgad M	Extension of previous crack
1	28° 46 ' 0"	82° 16 ' 50"	Kalpat Road	Nalgad M	Extension of previous crack
1	28° 45 ' 59"	82° 16 ' 32"	Kalpat Road	Nalgad M	Cracks on road having length 15 m, subsidence 12 cm, depth 54 cm, width 10 cm
1	28° 45 ' 58"	82° 16 ' 32"	Kalpat Road	Nalgad M	Extension of previous crack
1	28° 45 ' 58"	82° 16 ' 32"	Kalpat Road	Nalgad M	Extension of previous crack; cracks are mostly covered with sediments manually for road access.
1	28° 45 ' 49"	82° 16 ' 19"	Kalpat Road	Nalgad M	Crack having length of 16.2 m, width 20 cm, depth 20 cm and subsidence of 10 cm
1	28° 45 ' 49"	82° 16 ' 18"	Kalpat Road	Nalgad M	Extension of previous location of crack
1	28° 45 ' 49"	82° 16 ' 19"	Kalpat Road	Nalgad M	Extension of previous location of crack
1	28° 45 ' 41"	82° 15 ' 54"	Kalpat Road	Nalgad M	Landslide induced by earthquake
1	28° 45 ' 40"	82° 15 ' 54"	Kalpat Road	Nalgad M	Another GPS of previous landslide
1	28° 45 ' 40"	82° 15 ' 54"	Kalpat Road	Nalgad M	Another GPS of previous landslide
1	28° 45 ' 41"	82° 15 ' 51"	Kalpat Road	Nalgad M	Landslide crown part

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 45 ' 41"	82° 15 ' 51"	Kalpat Village	Nalgad M	35 cm subsidence; crack start point gps which is filled now the crack of 11.4 m lies near the crown of landslide.this crack may later form the new crown of the landslide.
1	28° 45 ' 41"	82° 15 ' 51"	Kalpat Village	Nalgad M	Extension of previous crack
1	28° 45 ' 40"	82° 15 ' 50"	Kalpat Village	Nalgad M	Crack of length which is filled now; starting point
1	28° 45 ' 40"	82° 15 ' 51"	Kalpat Village	Nalgad M	Previous crack end point having 7.2 m length
1	28° 45 ' 41"	82° 15 ' 50"	Kalpat Village	Nalgad M	Landslide crown part; rock boulders at road; dry, highly fractured rocks and gravel is present on the landslide.
1	28° 45 ' 40"	82° 15 ' 48"	Kalpat Village	Nalgad M	Crack having length of 11.2 m , width of 16 cm and depth of 20 cm.subsidence of 18 cm is also observed in the site.
1	28° 45 ' 41"	82° 15 ' 47"	Kalpat Village	Nalgad M	Extension of previous location of crack
1	28° 45 ' 41"	82° 15 ' 47"	Kalpat Village	Nalgad M	Extension of previous location of crack
1	28° 45 ' 41"	82° 15 ' 47"	Kalpat Village	Nalgad M	Crack having width of 9 cm, depth of 10 cm and length of 5.4 m.
1	28° 45 ' 41"	82° 15 ' 47"	Kalpat Village	Nalgad M	Cracks with width of 16 cm, depth of 30 cm, subsidence of 11cm and length of 4.8 m
1	28° 45 ' 44"	82° 15 ' 49"	Kalpat, Timichaur	Nalgad M	Crack with length of 10.8 m, width of 6 cm.depth of 25 cm ;starting point of crack
1	28° 45 ' 45"	82° 15 ' 49"	Kalpat Village	Nalgad M	End point of previous crack
1	28° 45 ' 45"	82° 15 ' 49"	Kalpat Village	Nalgad M	Crack with length 8.4 m, depth 24 cm, width 5 cm, and subsidence of 7 cm

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 45 ' 46"	82° 15 ' 49"	Kalpat Village	Nalgad M	Extension of previous location of crack
1	28° 45 ' 59"	82° 16 ' 16"	Kalpat Village	Nalgad M	Crack which is filled now. 15m length according to the local people.
1	28° 46 ' 30"	82° 17 ' 18"	Chiure	Nalgad M	Crack observed.
1	28° 46 ' 33"	82° 17 ' 20"	Chiure	Nalgad M	Crack which is filled now.
1	28° 46 ' 36"	82° 17 ' 21"	Chiure	Nalgad M	Crack of 6cm length, 100 cm depth, and breadth 10 cm
1	28° 46 ' 39"	82° 17 ' 22"	Chiure	Nalgad M	Crack of length 7 m, 15 cm breadth, and 23 cm depth
1	28° 46 ' 37"	82° 17 ' 21"	Chiure	Nalgad M	Crack observed.
1	28° 46 ' 24"	82° 17 ' 6"	Chiure	Nalgad M	Landslide (rockfall) toe part GPS
1	28° 46 ' 24"	82° 17 ' 6"	Chiure	Nalgad M	Landslide (rockfall) crown part GPS
1	28° 46 ' 45"	82° 17 ' 6"	Chiure	Nalgad M	School area is affected; cracks inside and outside the school premises which are filled now
1	28° 45 ' 16"	82° 17 ' 4"	Chiure	Nalgad M	Slide /rockfall
1	28° 46 ' 7"	82° 17 ' 6"	Pali	Nalgad M	Deep seated landslide; boulder fall; toe meets the bheri river; not feasible to visit; only located the location from the crown part; reactivated; previously small in size now landslide is large in size.

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 46 ' 29"	82° 17 ' 19"	Chiure (Zero Point)	Nalgad M	Alluvial deposits, 30 m height; had done road blockage after earthquake; the deposits is extended to downside of river; contains of huge number of boulders, earthquake induced
1	28° 44 ' 60"	82° 16 ' 19"	Pauwadadha	Nalgad M	Crack observed.
1	28° 44 ' 59"	82° 16 ' 23"	Pauwadadha	Nalgad M	Crack of 2 m length, 7 cm width, 30 cm depth
1	28° 44 ' 55"	82° 16 ' 21"	Pauwadadha	Nalgad M	Crack observed.
1	28° 44 ' 53"	82° 16 ' 20"	Chaurila	Nalgad M	Crack of 5.8 m length, 10 cm width, and 1 m depth
1	28° 44 ' 53"	82° 16 ' 20"	Chaurila	Nalgad M	Crack observed.
1	28° 44 ' 53"	82° 16 ' 20"	Chaurila	Nalgad M	Extension of previous crack
1	28° 44 ' 52"	82° 16 ' 20"	Chaurila	Nalgad M	Crack observed.
1	28° 44 ' 52"	82° 16 ' 19"	Chaurila	Nalgad M	Crack; settlement below this gps
1	28° 44 ' 52"	82° 16 ' 19"	Chaurila	Nalgad M	Crack observed.
1	28° 44 ' 51"	82° 16 ' 19"	Chaurila	Nalgad M	Crack observed.
1	28° 44 ' 51"	82° 16 ' 19"	Chaurila	Nalgad M	Crack observed.
1	28° 44 ' 51"	82° 16 ' 18"	Chaurila	Nalgad M	Crack of length 4 m, breadth 20 cm and depth 0.5 m
1	28° 44 ' 51"	82° 16 ' 18"	Chaurila	Nalgad M	Extension of previous crack

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 44 ' 51"	82° 16 ' 18"	Chaurila	Nalgad M	Crack observed
1	28° 44 ' 50"	82° 16 ' 17"	Chaurila	Nalgad M	Crack observed
1	28° 44 ' 48"	82° 16 ' 16"	Chaurila	Nalgad M	Crack of length 2 m, depth 0.60 m and breadth 15 cm
1	28° 44 ' 48"	82° 16 ' 16"	Chaurila	Nalgad M	Extension of previous crack
1	28° 44 ' 48"	82° 16 ' 16"	Chaurila	Nalgad M	Crack observed
1	28° 44 ' 48"	82° 16 ' 16"	Chaurila	Nalgad M	Crack observed
1	28° 44 ' 47"	82° 16 ' 15"	Chaurila	Nalgad M	Crack observed
1	28° 45 ' 8"	82° 16 ' 31"	Pauwadada	Nalgad M	Crack of 6 cm subsidence and 1m length
2	28° 47 ' 47"	82° 17 ' 0"	Chaite Kafal	Nalgad M	Crack 6 m length 2-3 cm breadth
2	28° 47 ' 47"	82° 17 ' 5"	Chaite Kafal	Nalgad M	Crack assumed to be 10-15 m before, now filled
2	28° 47 ' 48"	82° 17 ' 7"	Chaitekafal	Nalgad M	Crack 16m length 17 cm breadth depth 37 cm
2	28° 47 ' 49"	82° 17 ' 7"	Chaitekafal	Nalgad M	Extension of previous crack
2	28° 47 ' 49"	82° 17 ' 7"	Chaitekafal	Nalgad M	Extension of previous crack
2	28° 47 ' 48"	82° 17 ' 7"	Rumali	Nalgad M	Crack of length 4.5 m, breadth 5 cm, height 18 cm
2	28° 47 ' 48"	82° 17 ' 7"	Rumali	Nalgad M	Extension of previous crack
2	28° 47 ' 54"	82° 16 ' 57"	Rumali	Nalgad M	Crack

Ward	Latitude	Longitude	Village	Municipality	Description
2	28° 47 ' 54"	82° 16 ' 57"	Rumali	Nalgad M	Crack of 21cm breadth, 1.2 m depth,45 m length
2	28° 47 ' 55"	82° 16 ' 57"	Rumali Tol	Nalgad M	Extension but depth more than 2 m
2	28° 47 ' 55"	82° 16 ' 57"	Rumali Tol	Nalgad M	Extension of previous crack
2	28° 47 ' 55"	82° 16 ' 57"	Rumali Tol	Nalgad M	Extension of previous crack
2	28° 48 ' 2"	82° 16 ' 45"	Ayardadha	Nalgad M	Crack of 4 m length, 20 cm depth, 10 cm breadth
2	28° 48 ' 2"	82° 16 ' 45"	Ayardadha	Nalgad M	Crack which is filled now
2	28° 48 ' 2"	82° 16 ' 44"	Ayardadha	Nalgad M	Crack observed
2	28° 48 ' 4"	82° 16 ' 43"	Ayardadha	Nalgad M	Subsidence and crack, 20 cm subsidence; depth 47 cm breadth 7 cm
2	28° 48 ' 4"	82° 16 ' 44"	Ayardadha	Nalgad M	Extension of previous crack
2	28° 48 ' 4"	82° 16 ' 44"	Ayardadha	Nalgad M	Extension of previous crack
2	28° 48 ' 3"	82° 16 ' 45"	Ayardadha	Nalgad M	Extension of previous crack
2	28° 48 ' 3"	82° 16 ' 46"	Ayardadha	Nalgad M	Subsidence
2	28° 48 ' 4"	82° 16 ' 46"	Ayardadha	Nalgad M	Crack observed
2	28° 48 ' 4"	82° 16 ' 46"	Aaidari	Nalgad M	Crack observed
2	28° 48 ' 5"	82° 16 ' 43"	Aaidari	Nalgad M	Crack observed
2	28° 48 ' 5"	82° 16 ' 45"	Aaidari	Nalgad M	Crack observed

Ward	Latitude	Longitude	Village	Municipality	Description
2	28° 48 ' 5"	82° 16 ' 46"	Aaidari	Nalgad M	Crack observed
2	28° 48 ' 7"	82° 16 ' 44"	Aaidari	Nalgad M	Huge boulder displacement due to subsidence
3	28° 49 ' 22"	82° 14 ' 16"	Baskoti	Nalgad M	Crack and subsidence
3	28° 49 ' 22"	82° 14 ' 16"	Baskoti	Nalgad M	Extension of previous crack which is filled now ;length that could be observed now is 15 m depth 87 cm subsidence 50 cm and width 25 cm
3	28° 49 ' 22"	82° 14 ' 17"	Baskoti	Nalgad M	Crack observed
3	28° 49 ' 22"	82° 14 ' 17"	Baskoti	Nalgad M	Crack of length 13.3 m breadth 23 cm depth 55 cm and 22 cm subsidence
3	28° 49 ' 23"	82° 14 ' 17"	Baskoti	Nalgad M	Extension of previous crack
3	28° 49 ' 23"	82° 14 ' 17"	Baskoti	Nalgad M	Extension of crack and here length is 13.5 m, 12 cm width and breadth 16 cm
3	28° 49 ' 23"	82° 14 ' 20"	Baskoti	Nalgad M	Extension 30 cm breadth depth 90 cm
3	28° 49 ' 23"	82° 14 ' 21"	Baskoti	Nalgad M	Extension of previous crack below the road .
3	28° 48 ' 9"	82° 14 ' 60"	Bhadale	Nalgad M	Crack of 80 m length width 10 cm and 1m depth
3	28° 48 ' 19"	82° 14 ' 38"	Chawale	Nalgad M	Crack observed
3	28° 48 ' 19"	82° 14 ' 37"	Chawale	Nalgad M	Susidence in school premises
3	28° 48 ' 19"	82° 14 ' 37"	Chawale	Nalgad M	Crack which is filled now

Ward	Latitude	Longitude	Village	Municipality	Description
3	28° 48 ' 20"	82° 14 ' 37"	Chawale	Nalgad M	Crack observed
3	28° 48 ' 32"	82° 13 ' 46"	Sanaban	Nalgad M	Landslide crown one end gps
3	28° 48 ' 32"	82° 13 ' 47"	Sanaban	Nalgad M	Landslide crown middle part gps
3	28° 48 ' 32"	82° 13 ' 47"	Sanaban	Nalgad M	Landslide crown another end gps
3	28° 48 ' 29"	82° 13 ' 45"	Sanaban	Nalgad M	Crack which is filled
3	28° 48 ' 29"	82° 13 ' 45"	Sanaban	Nalgad M	Crack which is assumed to be 15 m and is now filled
3	28° 48 ' 31"	82° 13 ' 9"	Karuwa	Nalgad M	Crack observed
3	28° 48 ' 31"	82° 13 ' 9"	Karuwa	Nalgad M	Extension of previous crack which is 11 cm long, breadth 45 cm depth 8.8 cm
3	28° 48 ' 32"	82° 13 ' 13"	Karuwa	Nalgad M	Crack which has 24 cm subsidence and 17.8 m length
3	28° 48 ' 32"	82° 13 ' 16"	Karuwa	Nalgad M	Crack which is filled now
3	28° 48 ' 26"	82° 13 ' 9"	Karuwa	Nalgad M	Crack of 23 m length
3	28° 48 ' 26"	82° 13 ' 8"	Karuwa	Nalgad M	Extension of previous crack having subsidence of 40 cm
3	28° 48 ' 27"	82° 13 ' 9"	Karuwa	Nalgad M	Crack ; length 2 m,breadth 4 cm ,depth 32 cm at school
3	28° 48 ' 26"	82° 13 ' 9"	Karuwa	Nalgad M	Crack inside classroom;20 cm width, 2.5 m length and 35m depth
3	28° 48 ' 40"	82° 12 ' 46"	Parikaruwa	Nalgad M	Crack observed

Ward	Latitude	Longitude	Village	Municipality	Description
3	28° 48 ' 40"	82° 12 ' 46"	Parikaruwa	Nalgad M	Extension of previous crack ;10 m length, 1.5 m depth, 25 cm breadth; high risk to settlement
3	28° 48 ' 40"	82° 12 ' 47"	Parikaruwa	Nalgad M	Crack; length 12 m breadth 32 cm subsidence
3	28° 48 ' 40"	82° 12 ' 48"	Parikaruwa	Nalgad M	Extension of previous crack
3	28° 48 ' 40"	82° 12 ' 47"	Parikaruwa	Nalgad M	Extension of previous crack
3	28° 48 ' 41"	82° 12 ' 49"	Parikaruwa	Nalgad M	Crack observed
3	28° 48 ' 41"	82° 12 ' 49"	Parikaruwa	Nalgad M	Crack observed
3	28° 48 ' 40"	82° 12 ' 51"	Parikaruwa	Nalgad M	Crack now filled
3	28° 48 ' 40"	82° 12 ' 51"	Parikaruwa	Nalgad M	Crack now filled
3	28° 48 ' 40"	82° 12 ' 51"	Parikaruwa	Nalgad M	Subsidence in landslide
3	28° 48 ' 43"	82° 12 ' 48"	Parikaruwa	Nalgad M	Landslide in parikaruwa one end gps at toe
3	28° 48 ' 43"	82° 12 ' 48"	Parikaruwa	Nalgad M	Landslide another end gps at toe
3	28° 49 ' 5"	82° 13 ' 29"	Parikaruwa	Nalgad M	Crack observed
3	28° 49 ' 4"	82° 13 ' 29"	Parikaruwa	Nalgad M	Crack observed
3	28° 49 ' 5"	82° 13 ' 30"	Katke	Nalgad M	Crack observed
3	28° 48 ' 60"	82° 13 ' 52"	Katke	Nalgad M	Crack observed

Ward	Latitude	Longitude	Village	Municipality	Description
4	28° 48 ' 46"	82° 16 ' 35"	Bagmarey	Nalgad M	Landslide
4	28° 48 ' 46"	82° 16 ' 34"	Bagmarey	Nalgad M	Cracks with 16m length, front bearing of 60°, width of 1.2 m, depth 80 cm.
4	28° 48 ' 47"	82° 16 ' 37"	Darpakha	Nalgad M	Crack of 13 cm width, depth of 30 cm, length of 11.4 m
4	28° 48 ' 47"	82° 16 ' 36"	Darpakha	Nalgad M	Another gps of crack
4	28° 49 ' 5"	82° 15 ' 44"	Sarkibara Gau	Nalgad M	Crack having length of 21 m and depth of 3-4 m (according to the local people; now its filled) ;cultivation land covered; downhill from this GPS point more crack
4	28° 48 ' 45"	82° 15 ' 18"	Aule	Nalgad M	Dry landslide
4	28° 48 ' 14"	82° 15 ' 35"	Aule	Nalgad M	Rock fall
4	28° 48 ' 31"	82° 15 ' 17"	Aule	Nalgad M	Crack of 26m length now filled
4	28° 48 ' 32"	82° 15 ' 16"	Aule	Nalgad M	Extension of previous crack not filled
4	28° 48 ' 28"	82° 15 ' 32"	Aule	Nalgad M	Crack which is filled now
4	28° 48 ' 25"	82° 15 ' 36"	Aule	Nalgad M	Crack of total 28 m length, depth 15 cm
4	28° 48 ' 25"	82° 15 ' 36"	Aule	Nalgad M	Crack observed
4	28° 48 ' 24"	82° 15 ' 36"	Aule	Nalgad M	Crack observed
4	28° 48 ' 24"	82° 15 ' 37"	Aule	Nalgad M	Crack on temple area

Ward	Latitude	Longitude	Village	Municipality	Description
4	28° 48 ' 23"	82° 15 ' 37"	Aule	Nalgad M	Crack observed
4	28° 48 ' 22"	82° 15 ' 37"	Aule	Nalgad M	Crack observed
4	28° 48 ' 23"	82° 15 ' 29"	Aule	Nalgad M	Crack observed
4	28° 48 ' 23"	82° 15 ' 31"	Aule	Nalgad M	Crack observed
4	28° 48 ' 18"	82° 15 ' 29"	Aule	Nalgad M	Crack observed
4	28° 48 ' 21"	82° 15 ' 32"	Aule	Nalgad M	Crack observed
5	28° 50 ' 19"	82° 16 ' 57"	Ranigau	Nalgad M	Crack observed
5	28° 50 ' 19"	82° 16 ' 56"	Ranigau	Nalgad M	Crack observed
5	28° 50 ' 19"	82° 16 ' 56"	Ranigau	Nalgad M	Crack observed
5	28° 50 ' 18"	82° 16 ' 54"	Ranigau	Nalgad M	Crack observed
5	28° 50 ' 17"	82° 16 ' 54"	Ranigau	Nalgad M	Crack observed
5	28° 50 ' 30"	82° 16 ' 40"	Pari Gau	Nalgad M	Crack observed
5	28° 50 ' 30"	82° 16 ' 40"	Parigau	Nalgad M	Extension of previous crack
5	28° 50 ' 30"	82° 16 ' 40"	Parigau	Nalgad M	Extension of previous crack
5	28° 51 ' 27"	82° 15 ' 52"	Kafalpata	Nalgad M	Crack which is filled now
5	28° 51 ' 28"	82° 15 ' 52"	Kafalpata	Nalgad M	Crack which is filled now

Ward	Latitude	Longitude	Village	Municipality	Description
5	28° 51 ' 29"	82° 15 ' 46"	Kafalpata	Nalgad M	Crack of length 30 m, breath 6 cm, depth 30 cm at road
5	28° 49 ' 06"	82° 17 ' 43"	Syali	Nalgad M	Earthquake induced landslide
5	28° 48 ' 44"	82° 17 ' 36"	Syali	Nalgad M	Earthquake induced landslide
6	28° 49 ' 38"	82° 14 ' 56"	Farsa Gau	Nalgad M	Crack of 6.6 m length
6	28° 49 ' 42"	82° 14 ' 50"	Farsa Gau	Nalgad M	Crack of 2 cm width ,4.2 m length
6	28° 49 ' 46"	82° 14 ' 46"	Farse Gau	Nalgad M	Crack and damaged house
6	28° 49 ' 39"	82° 14 ' 40"	Garagli	Nalgad M	Crack observed; start point
6	28° 49 ' 39"	82° 14 ' 39"	Garagli	Nalgad M	Crack with width of 20 cm, depth of 50 cm, length of 48 cm
6	28° 49 ' 37"	82° 14 ' 40"	Garagli	Nalgad M	Crack with length of 16 cm, depth 70 cm, subsidence of 20 cm
7	28° 48' 44"	82° 18' 40"	Latachaur	Nalgad M	Reactivated landslide (Rockfall)
8	28° 48' 13"	82° 21' 47"	Halchaur	Nalgad M	Crack
8	28° 49' 03"	82° 21' 47"	Kabra	Nalgad M	Crack
8	28° 49' 13"	82° 21' 48"	Kabra	Nalgad M	Crack
8	28° 48' 15"	82° 21' 57"	Gamrakha	Nalgad M	Crack
1	28° 48 ' 28"	82° 6 ' 5"	Shree Tapodhenu Mabi School	Kushe RM	Land subsidence and school wall damaged not more harmful in near area. Crack length is about 10 m.
1	28° 48 ' 2"	82° 5 ' 36"	Dhaun gau	Kushe RM	One house can be affected by stream erosion; not earthquake induced.

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 48 ' 19"	82° 5 ' 13"	Dhakhola	Kushe RM	Crack during earthquake but filled later. (according to the local people).
1	28° 48 ' 36"	82° 5 ' 35"	Hamalgau	Kushe RM	10-12 settlement; Small cracked width 3cm and length was about 70-80 m; currently crack filled up by mud.
1	28° 48 ' 33"	82° 6 ' 39"	Sagri gau	Kushe RM	Reactivated landslide (debris flow) ,1 house mainly affected and should be migrated; below landslide school is vulnerable of debris that flow from upside of landslide. Crown part to school is 200m.
2	28° 50 ' 18"	82° 6 ' 38"	Talegau	Kushe RM	Reactivated landslide
2	28° 49 ' 46"	82° 6 ' 12"	Mannedhunga ma vi	Kushe RM	Landslide crown section
2	28° 49 ' 45"	82° 5 ' 57"	Bagge khola	Kushe RM	Land subsidence
2	28° 49 ' 53"	82° 5 ' 30"	Kodal dunga	Kushe RM	Land subsidence
2	28° 50 ' 22"	82° 5 ' 14"	Jhargaun	Kushe RM	Land subsidence after earthquake in paddy field; large amount of water flow from spring in terrace; subsidence of land is about 12 m length and depth 15 to 20 cm. N344W fore bearing from this coordinate.
2	28° 50 ' 39"	82° 4 ' 57"	Dogri	Kushe RM	Crack developed after earthquake but filled later (according to local people).
2	28° 50 ' 39"	82° 4 ' 57"	Dogri	Kushe RM	Crack developed after earthquake but filled later (according to local people).

Ward	Latitude	Longitude	Village	Municipality	Description
2	28° 50 ' 39"	82° 4 ' 56"	Dogri	Kushe RM	Crack developed after earthquake but filled later (according to local people).
2	28° 50 ' 43"	82° 4 ' 48"	Dogri	Kushe RM	Reactivated landslide before a year but subsidence is about one meter per year (according to local people).
2	28° 50 ' 6"	82° 4 ' 37"	Bhera	Kushe RM	Crack and land subsidence
2	28° 50 ' 4"	82° 4 ' 35"	Bhera	Kushe RM	Cracks 20-22 cm width and 25-30 cm depth
2	28° 50 ' 4"	82° 4 ' 34"	Bhera	Kushe RM	Cracks line first coordinate
2	28° 50 ' 4"	82° 4 ' 34"	Bhera	Kushe RM	Crack line mid coordinate
2	28° 50 ' 6"	82° 4 ' 38"	Bhera	Kushe RM	Crack line last coordinate
2	28° 50 ' 6"	82° 4 ' 39"	Bhera	Kushe RM	Land subsidence and crack
3	28° 52 ' 10"	82° 7 ' 7"	Sallosare	Kushe RM	Land subsidence
3	28° 52 ' 10"	82° 7 ' 7"	Datior	Kushe RM	Crack line affect the two houses
3	28° 52 ' 3"	82° 7 ' 8"	Datior	Kushe RM	Land subsidence problem consist clay and silt
3	28° 50 ' 27"	82° 6 ' 49"	Tapchaur	Kushe RM	Reactivated landslide
3	28° 49 ' 6"	82° 7 ' 52"	Maltakura	Kushe RM	Reactivated Landslide and Subsidence
3	28° 52 ' 0"	82° 7 ' 12"	Matekhara	Kushe RM	Emerge new spring

Ward	Latitude	Longitude	Village	Municipality	Description
4	28° 49 ' 21"	82° 7 ' 55"	Tumti	Kushe RM	Affected to orange farm
4	28° 49 ' 21"	82° 7 ' 55"	Tumti	Kushe RM	Orange farm affected due to earthquake
4	28° 50 ' 1"	82° 7 ' 37"	Patbanne thati	Kushe RM	Road crack and Subsidence
4	28° 50 ' 0"	82° 7 ' 37"	Patbane thati	Kushe RM	Road subsidence
4	28° 49 ' 59"	82° 7 ' 38"	Patbanne thati	Kushe RM	Road subsidence
5	28° 50 ' 26"	82° 10 ' 38"	Khurpa	Kushe RM	Crown part 60-70m; zone of accumulation to crown part length is about 200m; Debris flow; schist fragments GW-GM; 1house destroyed and 2 household near toe part of landslide were replaced; 10-15 house below toe part i.e. about 500m and further below 20 houses
5	28° 48 ' 3"	82° 10 ' 16"	Damdala	Kushe RM	Rockfall after earthquake; wedge failure ; strike N25E dip direction S260W
7	28° 53 ' 49"	82° 10 ' 48"	Samayala gau	Kushe RM	Shallow old landslide
7	28° 51 ' 57"	82° 12 ' 1"	Dhanrasi	Kushe RM	Shallow old landslide
8	28° 55 ' 14"	82° 8 ' 47"	Pahada gau	Kushe RM	Reactivated old landslide
8	28° 55 ' 31"	82° 8 ' 52"	Pahada gau	Kushe RM	Reactivated old landslide
8	28° 55 ' 46"	82° 8 ' 57"	Pahadgau	Kushe RM	New spring after earthquake

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 38 ' 29"	81° 57 ' 47"	Aale, Sisne	Shivalaya RM	Large scale landslide
1	28° 38 ' 50"	81° 58 ' 16"	Dingaghar tol, Sisne	Shivalaya RM	Landslide
1	28° 38 ' 51"	81° 58 ' 12"	Dingaghar tol,Sisne	Shivalaya RM	Landslide (around 30 houses at risk)
1	28° 38 ' 49"	81° 58 ' 13"	Dingaghar tol, Sisne	Shivalaya RM	Reactivated landslide
1	28° 38 ' 46"	81° 58 ' 14"	Dingaghar tol,Sisne	Shivalaya RM	Reactivated landslide
1	28° 38 ' 45"	81° 58 ' 14"	Dingaghar tol, Sisne	Shivalaya RM	Reactivated landslide
1	28° 38 ' 26"	81° 58 ' 13"	Aale, Sisne	Shivalaya RM	Reactivated landslide
1	28° 38 ' 53"	81° 58 ' 58"	Kafal gaire ,Sisne	Shivalaya RM	Drying spring after earthquake
1	28° 38 ' 47"	81° 58 ' 58"	Kafal gaire, Sisne	Shivalaya RM	Multiple cracks observed
1	28° 38 ' 46"	81° 58 ' 58"	Kafal gaire, Sisne	Shivalaya RM	Crack observed
1	28° 38 ' 45"	81° 58 ' 57"	Kafal gaire,Sisne	Shivalaya RM	Crack observed
1	28° 38 ' 53"	81° 59 ' 11"	Babiya chaur,Sisne	Shivalaya RM	Large scale landslide
1	28° 38 ' 43"	81° 58 ' 24"	Sisne(Shree Nepal Rastriya Adharbhut Bidhyalaya)	Shivalaya RM	Subsidence up to 1m in classroom
1	28° 38 ' 49"	81° 58 ' 37"	Ritharukh nira	Shivalaya RM	Crack observed
1	28° 38 ' 49"	81° 58 ' 38"	Ritharukh nira	Shivalaya RM	Crack observed
1	28° 38 ' 51"	81° 58 ' 28"	Ratamata tol	Shivalaya RM	Reactivated landslide

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 39 ' 53"	81° 59 ' 46"	Ghogabuya	Shivalaya RM	Crack
1	28° 39 ' 52"	81° 59 ' 50"	Ghogabuya	Shivalaya RM	Reactivated landslide (blocked road)
1	28° 39 ' 56"	82° 0 ' 4"	Biurekuna raha	Shivalaya RM	Crack seen in road section (Morkha-Kachali road)
1	28° 39 ' 54"	82° 0 ' 7"	Purano tip	Shivalaya RM	Crack seen in road section (Morkha-Kachali road)
1	28° 39 ' 52"	82° 0 ' 11"	Purano tip	Shivalaya RM	Crack observed in road section (Morkha - Kachali road)
2	28° 39 ' 9"	81° 56 ' 43"	Dungra	Shivalaya RM	Subsidence
2	28° 38 ' 36"	81° 56 ' 27"	Dungra	Shivalaya RM	Reactivated landslide
2	28° 38 ' 34"	81° 56 ' 29"	Dungra	Shivalaya RM	Reactivated landslide
2	28° 38 ' 24"	81° 56 ' 33"	Dungra	Shivalaya RM	Reactivated landslide
2	28° 38 ' 20"	81° 56 ' 39"	Dungra	Shivalaya RM	Dry spring
2	28° 38 ' 37"	81° 56 ' 50"	Dungra	Shivalaya RM	Multiple crack observed
2	28° 38 ' 42"	81° 56 ' 48"	Dungra	Shivalaya RM	Reactivated landslide
2	28° 38 ' 45"	81° 56 ' 50"	Dungra	Shivalaya RM	Reactivated landslide
2	28° 38 ' 46"	81° 56 ' 54"	Dungra	Shivalaya RM	Reactivated landslide
2	28° 38 ' 56"	81° 57 ' 2"	Dungra	Shivalaya RM	Large scale landslide (earthquake induced)
4	28° 41 ' 6"	81° 53 ' 54"	Jamna gaira	Shivalaya RM	Subsidence

Ward	Latitude	Longitude	Village	Municipality	Description
4	28° 41 ' 9"	81° 53 ' 54"	Jamna gaira	Shivalaya RM	Subsidence
4	28° 41 ' 11"	81° 53 ' 55"	Jamna gaira	Shivalaya RM	Multiple Crack Observed
4	28° 41 ' 9"	81° 54 ' 3"	Betada gaira	Shivalaya RM	Reactivated large Scale landslide
4	28° 41 ' 36"	81° 54 ' 27"	Pahire pakha	Shivalaya RM	Large Scale Landslide; reactivated
4	28° 41 ' 31"	81° 54 ' 48"	Kolpari tol	Shivalaya RM	Reactivated Large Scale landslide
5	28° 41 ' 54"	81° 53 ' 19"	Dalit tol	Shivalaya RM	Reactivated landslide
5	28° 41 ' 46"	81° 53 ' 18"	Dalit tol	Shivalaya RM	Subsidence (1m depth)
5	28° 41 ' 40"	81° 53 ' 16"	Irja naula khola	Shivalaya RM	Crack observed
5	28° 41 ' 23"	81° 53 ' 13"	Sauka mandera	Shivalaya RM	Reactivated landslide
5	28° 41 ' 35"	81° 53 ' 5"	Kuseni	Shivalaya RM	Subsidence
5	28° 41 ' 29"	81° 52 ' 55"	Salleri	Shivalaya RM	Reactivated landslide
5	28° 41 ' 21"	81° 52 ' 50"	Dhoyediya galpa	Shivalaya RM	Crack observed
5	28° 41 ' 18"	81° 52 ' 41"	Salleri	Shivalaya RM	Reactivated landslide
5	28° 41 ' 0"	81° 52 ' 40"	Kaula pani mula	Shivalaya RM	Reactivated landslide
5	28° 41 ' 34"	81° 53 ' 32"	Biral sim	Shivalaya RM	Reactivated landslide
5	28° 41 ' 58"	81° 53 ' 43"	Salleri(dhaura khola pahiro)	Shivalaya RM	Reactivated landslide; (25-houses are at risk),large scale landslide

Ward	Latitude	Longitude	Village	Municipality	Description
6	28° 43 ' 2"	81° 52 ' 22"	Lothpakha	Shivalaya RM	Large scale landslide (Reactivated)
6	28° 43 ' 5"	81° 52 ' 16"	Lothpakha	Shivalaya RM	Small scale landslide
6	28° 43 ' 7"	81° 52 ' 13"	Lothpakha	Shivalaya RM	Small scale landslide
6	28° 43 ' 8"	81° 51 ' 50"	Rokaya jaat tol	Shivalaya RM	Crack observed
6	28° 43 ' 8"	81° 51 ' 50"	Rokaya jaat tol	Shivalaya RM	Crack observed
6	28° 42 ' 33"	81° 52 ' 8"	Virpana	Shivalaya RM	Reactivated landslide
6	28° 42 ' 20"	81° 52 ' 0"	Pokhare	Shivalaya RM	Reactivated landslide (effect on road)
7	28° 42 ' 16"	81° 55 ' 18"	Syala gaun	Shivalaya RM	Subsidence (Behind Ward Office)
7	28° 42 ' 32"	81° 55 ' 0"	Viusa	Shivalaya RM	Landslide; 19 houses are in risk
7	28° 42 ' 43"	81° 55 ' 6"	Thanbada, Viusa	Shivalaya RM	Large scale landslide
7	28° 42 ' 41"	81° 54 ' 59"	Thanbada, Viusa	Shivalaya RM	Multiple crack observed
7	28° 42 ' 35"	81° 54 ' 54"	Thanbada, Viusa	Shivalaya RM	Large scale landslide
7	28° 42 ' 34"	81° 54 ' 37"	Gosai tol, Viusa	Shivalaya RM	Large scale landslide
7	28° 42 ' 20"	81° 54 ' 41"	Malubela chaur	Shivalaya RM	Landslide; 1 house destroy; 7 houses are in danger
7	28° 42 ' 4"	81° 54 ' 34"	Khirra	Shivalaya RM	Road destruction; large scale landslide
7	28° 42 ' 34"	81° 53 ' 15"		Shivalaya RM	Reactivated landslide

Ward	Latitude	Longitude	Village	Municipality	Description
8	28° 40 ' 46"	81° 56 ' 15"	Daha	Shivalaya RM	Large Scale landslide
8	28° 40 ' 48"	81° 56 ' 14"	Daha	Shivalaya RM	Large Scale landslide
8	28° 41 ' 20"	81° 57 ' 46"	Nigale kholi	Shivalaya RM	Reactivated landslide; Road destruction, two houses affected on the area.
8	28° 41 ' 24"	81° 57 ' 20"	Talla mela odar	Shivalaya RM	Reactivated landslide
8	28° 41 ' 43"	81° 56 ' 36"	Kathaya tol, Lispani	Shivalaya RM	Multiple cracks observed
8	28° 41 ' 41"	81° 56 ' 34"	Kathaya tol, Lispani	Shivalaya RM	Crack in landslide
8	28° 41 ' 31"	81° 56 ' 29"	Kathaya tol, Lispani	Shivalaya RM	Reactivated landslide
8	28° 41 ' 40"	81° 56 ' 32"	Kathaya tol, Lispani	Shivalaya RM	Large scale landslide (Reactivated)
1	28° 49 ' 5"	81° 57 ' 8"	Karkigau(Saraswati Adharbhut School)	Junichande RM	Reactivated landslide
1	28° 48 ' 4"	81° 56 ' 23"	Karkigau	Junichande RM	Reactivated landslide
1	28° 48 ' 4"	81° 56 ' 24"	Karkigau	Junichande RM	Reactivated landslide
1	28° 48 ' 5"	81° 56 ' 23"	Karkigau	Junichande RM	Reactivated landslide
1	28° 48 ' 4"	81° 56 ' 23"	Karkigau	Junichande RM	Reactivated landslide
1	28° 48 ' 8"	81° 55 ' 59"	Karkigau	Junichande RM	Old landslide
1	28° 48 ' 37"	81° 56 ' 6"	Karkigau	Junichande RM	Old landslide, creeping observed; Reactivated; affect village downward
1	28° 48 ' 2"	81° 55 ' 57"	Karkigau	Junichande RM	Reactivated landslide

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 48 ' 3"	81° 55 ' 59"	Karkigau	Junichande RM	Reactivated landslide
1	28° 48 ' 24"	81° 56 ' 33"	Batkule	Junichande RM	Reactivated landslide
1	28° 48 ' 21"	81° 56 ' 50"	Karkigau	Junichande RM	Reactivated landslide
2	28° 50 ' 1"	81° 57 ' 43"	Mujkot	Junichande RM	Reactivated landslide
2	28° 49 ' 7"	81° 56 ' 4"	Mujkot	Junichande RM	Reactivated landslide
3	28° 50 ' 2"	81° 56 ' 43"	Sanokholagau	Junichande RM	Excavation induced landslide
3	28° 49 ' 52"	81° 56 ' 42"	Near Sanokholagau	Junichande RM	Excavation induced landslide
3	28° 49 ' 47"	81° 56 ' 40"	Near Majhdanda	Junichande RM	Excavation induced landslide
3	28° 49 ' 18"	81° 56 ' 38"	Majhdanda	Junichande RM	Excavation induced landslide
3	28° 49 ' 9"	81° 57 ' 45"	Badibati Tole	Junichande RM	Reactivated landslide
3	28° 50 ' 11"	81° 57 ' 28"	Sanokholagau	Junichande RM	Excavation induced landslides/slope failure
3	28° 48 ' 11"	81° 56 ' 35"	Karkigau	Junichande RM	Reactivated landslide
4	28° 49 ' 4"	81° 57 ' 50"	Badibati Tole	Junichande RM	Rockfall
4	28° 49 ' 4"	81° 57 ' 49"	Badibati Tole	Junichande RM	Rockfall
4	28° 49 ' 6"	81° 57 ' 50"	Badibati Tole	Junichande RM	Decrease in water discharge
4	28° 49 ' 2"	81° 57 ' 41"	Badibati Tole	Junichande RM	Landslide reactivated

Ward	Latitude	Longitude	Village	Municipality	Description
4	28° 49 ' 2"	81° 57 ' 41"	Badibati Tole	Junichande RM	Reactivated landslide
4	28° 49 ' 7"	81° 57 ' 45"	Badibati Tole	Junichande RM	Reactivated landslide
4	28° 49 ' 8"	81° 57 ' 47"	Badibati Tole	Junichande RM	Excavation induced landslide, seepage problem
4	28° 49 ' 18"	81° 58 ' 3"	Patalpani,Khattekhola	Junichande RM	Reactivated landslide
4	28° 49 ' 20"	81° 58 ' 3"	Patalpani,Khattekhola	Junichande RM	Reactivated landslide
4	28° 49 ' 20"	81° 58 ' 3"	Patalpani,Khattekhola	Junichande RM	Reactivated landslide
9	28° 50 ' 24"	81° 56 ' 24"	Chotrakot	Junichande RM	Reactivated landslide
11	28° 49 ' 44"	81° 57 ' 30"	Kortan	Junichande RM	Excavation induced landslide, road cut slope failure
11	28° 49 ' 44"	81° 57 ' 29"	Kortan	Junichande RM	Excavation induced landslide, road cut slope failure
11	28° 49 ' 60"	81° 57 ' 44"	Kortan	Junichande RM	Reactivated landslide
11	28° 49 ' 60"	81° 57 ' 44"	Kortan	Junichande RM	Reactivated landslide
11	28° 50 ' 1"	81° 57 ' 43"	Kortan	Junichande RM	Reactivated landslide
1	28° 43 ' 11"	82° 14 ' 9"	Maide	Bheri M	Reactivated landslide and crack; settlement need to migrate
1	28° 43 ' 14"	82° 14 ' 8"	Red cross tent (Between Maide and Chaupani)	Bheri M	Reactivated landslide and crack
1	28° 43 ' 16"	82° 14 ' 8"	Chaupani	Bheri M	Reactivated landslide and crack
1	28° 43 ' 35"	82° 14 ' 8"	Chaur	Bheri M	Reactivated landslide and crack

Ward	Latitude	Longitude	Village	Municipality	Description
2	28° 44 ' 23"	82° 11 ' 31"	Bauthana	Bheri M	Crack line is observed
2	28° 43 ' 56"	82° 43 ' 56"	Dulley	Bheri M	Reactivated landslide and crack settlement
2	28° 43 ' 57"	82° 12 ' 57"	Puranrisang, Dulley	Bheri M	Reactivated landslide and crack
2	28° 42 ' 51"	82° 11 ' 43"	Puranrisang, Saldada	Bheri M	Reactivated landslide and crack
3	28° 41 ' 50"	82° 12 ' 36"	Thaple	Bheri M	Landslide, one building is at risk
3	28° 41 ' 39"	82° 12 ' 53"	Thaplechaur	Bheri M	Crack line is observed.
3	28° 41 ' 58"	82° 12 ' 27"	Thaple	Bheri M	Landslide; may affect village which lies at downhill slope
3	28° 41 ' 54"	82° 12 ' 14"	Thati Bazar	Bheri M	Khalanga-Rimna road is affected by crack and shallow landslide
3	28° 41 ' 56"	82° 11 ' 57"	Jajarkot Durbar	Bheri M	Landslide and Creeping
4	28° 41 ' 52"	82° 11 ' 24"	Ranagaun	Bheri M	Landslide and Creeping
4	28° 42 ' 23"	82° 10 ' 41"	Ranagaun	Bheri M	Landslide and Creeping
4	28° 41 ' 50"	82° 11 ' 33"	Ranagaun	Bheri M	Landslide and Creeping
4	28° 40 ' 60"	82° 11 ' 12"	Along Chinchu-Jajarkot Road	Bheri M	Landslide and Creeping
5	28° 41 ' 6"	82° 10 ' 6"	Eber, Ata basti	Bheri M	Shallow landslide which has affect retaining wall
5	28° 40 ' 46"	82° 10 ' 42"	Rokayagaun	Bheri M	Reactivated landslide and creeping

Ward	Latitude	Longitude	Village	Municipality	Description
5	28° 40 ' 51"	82° 10 ' 33"	Sukeysarna, Bamnikamal	Bheri M	Creep
6	28° 44 ' 6"	82° 8 ' 42"	Naumuli dada	Bheri M	Creep
6	28° 43 ' 37"	82° 9 ' 30"	Paurana	Bheri M	Creep
6	28° 41 ' 41"	82° 7 ' 50"	Dadakhet, Gangatiya	Bheri M	Land subsidence
6	28° 42 ' 11"	82° 7 ' 42"	Yagud, Gangatiya	Bheri M	Crack about 15 m long from bank of Bungre Khola
7	28° 42 ' 52"	82° 7 ' 58"	Rupakot	Bheri M	Reactivated landslide
7	28° 42 ' 8"	82° 6 ' 46"	Punma, Swahara	Bheri M	Land subsidence which might cause disturbance to farming and settlement land.
7	28° 42 ' 4"	82° 6 ' 48"	Darim, Okharatol	Bheri M	Land subsidence which might cause disturbance to farming and settlement land.
7	28° 41 ' 15"	82° 5 ' 30"	Dokana tol	Bheri M	Landslide
8	28° 40 ' 2"	82° 6 ' 18"	Harla tole	Bheri M	Old landslide; no settlement
8	28° 40 ' 5"	82° 6 ' 32"	Harla tole	Bheri M	Shallow landslide
8	28° 39 ' 56"	82° 7 ' 11"	Harla tol	Bheri M	Landslide on river bank slope angle=41°; seven house are on high risk.
9	28° 39 ' 19"	82° 6 ' 38"	Simlayana	Bheri M	Land subsidence; per year subsidence is 0.5 m to 0.2 m (according to the local people).
9	28° 39 ' 41"	82° 6 ' 16"	Bhoor	Bheri M	Land subsidence and erosion due to rainfall
9	28° 39 ' 43"	82° 6 ' 10"	Bhoor	Bheri M	Land subsidence
9	28° 39 ' 45"	82° 6 ' 2"	Playa, Kopacha Ghosa Khola	Bheri M	Reactivated landslide; affects multiple houses and the farming terrace.
9	28° 39 ' 53"	82° 5 ' 58"	Jarja Khola	Bheri M	Erosion of agricultural land on large area.

Ward	Latitude	Longitude	Village	Municipality	Description
9	28° 39 ' 54"	82° 5 ' 55"	Jarja Khola	Bheri M	Old shallow landslide and subsidence; no earthquake induced
9	28° 39 ' 51"	82° 5 ' 49"	Jarja Khola	Bheri M	Old landslide
9	28° 39 ' 57"	82° 6 ' 12"	Jarja Khola	Bheri M	Landslide has affected spring water and creates water problems
9	28° 39 ' 58"	82° 6 ' 58"	Lamakatiya	Bheri M	Old landslide
9	28° 39 ' 31"	82° 7 ' 13"	Lamakatiya	Bheri M	Landslide caused due to erosion by river (length=100 m breadth=150 m, height=20 m)
9	28° 39 ' 26"	82° 7 ' 3"	Lamakatiya	Bheri M	Erosion; new spring emerge after earthquake; about 19 houses are in risk due to erosion.
10	28° 37 ' 22"	82° 6 ' 2"	Bhoor chaur	Bheri M	Land subsidence and landslide
10	28° 37 ' 46"	82° 4 ' 31"	Aale	Bheri M	Rockfall
10	28° 38 ' 56"	82° 7 ' 3"	Palsani	Bheri M	Landslide; Detail description is on survey sheet.
10	28° 38 ' 10"	82° 7 ' 36"	Balyatar, Damahabagar, Dhulpahiro	Bheri M	Reactivated landslide (length=50 m breadth=25 m height=10 m); slope angle=71° aspect=110°; per year erosion 4 m (according to the local people).
10	28° 38 ' 40"	82° 7 ' 28"	Burunge Khola	Bheri M	Large landslide on river bank aspect=81° slope angle=57°; Affects farming land.
1	28° 56 ' 17"	82° 14 ' 31"	Okhaldhunga	Barekot RM	Seepage creep
1	28° 55 ' 45"	82° 13 ' 56"	Dhaulakot	Barekot RM	Reactivated landslide
1	28° 56 ' 14"	82° 15 ' 11"	Koteli	Barekot RM	Reactivated landslide

Ward	Latitude	Longitude	Village	Municipality	Description
1	28° 56 ' 6"	82° 15 ' 14"	Koteli	Barekot RM	Reactivated landslide
1	28° 55 ' 55"	82° 15 ' 14"	Koteli	Barekot RM	Reactivated landslide
1	28° 55 ' 54"	82° 16 ' 2"	Koteli	Barekot RM	Reactivated landslide
1	28° 55 ' 24"	82° 16 ' 39"	Bayal	Barekot RM	Reactivated landslide
1	28° 55 ' 3"	82° 16 ' 34"	Bayal	Barekot RM	Reactivated landslide
2	28° 57 ' 3"	82° 14 ' 42"		Barekot RM	Creep
3	28° 59 ' 42"	82° 12 ' 18"	Pahada Mulcham	Barekot RM	Reactivated landslide
4	28° 58 ' 21"	82° 15 ' 19"	Jaltari	Barekot RM	Reactivated landslide
4	28° 58 ' 8"	82° 16 ' 0"	SarkiJuila	Barekot RM	Reactivated landslide
5	28° 57 ' 42"	82° 15 ' 34"	Bhailokhan	Barekot RM	landslide road induced
5	28° 56 ' 54"	82° 15 ' 30"	Tolkhan	Barekot RM	Reactivated landslide
5	28° 57 ' 10"	82° 15 ' 14"	Siligichaur	Barekot RM	Reactivated landslide
5	28° 56 ' 5"	82° 16 ' 22"	Karuka	Barekot RM	Reactivated landslide
6	28° 55 ' 26"	82° 17 ' 54"		Barekot RM	Reactivated landslide
6	28° 55 ' 1"	82° 18 ' 3"	Nauli, Dimyari	Barekot RM	Jaha khola stream erosion
6	28° 55 ' 2"	82° 17 ' 9"		Barekot RM	Reactivated landslide induced
7	28° 56 ' 35"	82° 17 ' 10"	Shyala	Barekot RM	Reactivated landslide induced

Ward	Latitude	Longitude	Village	Municipality	Description
8	28° 53 ' 56"	82° 16 ' 7"	Mayabata	Barekot RM	Reactivated landslide
8	28° 53 ' 24"	82° 16 ' 12"	Bantaure	Barekot RM	Road slope erosion
8	28° 53 ' 11"	82° 16 ' 21"		Barekot RM	Road slope erosion
8	28° 53 ' 2"	82° 16 ' 28"	Bansura khola	Barekot RM	Stream
8	28° 53 ' 4"	82° 16 ' 37"	Gapichaur	Barekot RM	Road slope erosion
8	28° 53 ' 8"	82° 16 ' 48"	Paigapati	Barekot RM	Reactivated landslide
8	28° 53 ' 16"	82° 16 ' 50"	Tindhare	Barekot RM	Reactivated landslide
8	28° 53 ' 39"	82° 16 ' 48"	Sakala chaur	Barekot RM	Reactivated landslide
9	28° 53 ' 31"	82° 17 ' 51"	Rawaldhara	Barekot RM	Reactivated landslide
9	28° 53 ' 51"	82° 17 ' 34"	Bandala	Barekot RM	Reactivated landslide
9	28° 53 ' 44"	82° 17 ' 14"	Sini khola	Barekot RM	Reactivated landslide
1	28° 44 ' 11"	81° 55 ' 42"	Salma Indrajyoti School	Chhedagad M	Subsidence and cracks
2	28° 45 ' 20"	81° 52 ' 56"	Suyada	Chhedagad M	Crack first point at Suyada
2	28° 45 ' 20"	81° 52 ' 56"	Suyada	Chhedagad M	Suyada crack end point
2	28° 45 ' 20"	81° 52 ' 56"	Suyada	Chhedagad M	Suyada crack point
2	28° 45 ' 27"	81° 53 ' 31"	Bata chaur	Chhedagad M	Huge crack

Ward	Latitude	Longitude	Village	Municipality	Description
2	28° 45 ' 27"	81° 53 ' 31"	Bata chaur	Chhedagad M	Huge crack along road
2	28° 45 ' 27"	81° 53 ' 31"	Bata chaur	Chhedagad M	Crack end point
2	28° 45 ' 27"	81° 53 ' 31"	Bata chaur	Chhedagad M	Shallow landslide
2	28° 45 ' 31"	81° 53 ' 31"	Bata chaur	Chhedagad M	Shallow landslide
2	28° 45 ' 35"	81° 53 ' 31"	Jamli khola	Chhedagad M	Crack on road
2	28° 46 ' 1"	81° 54 ' 16"	Dedh kula	Chhedagad M	Shallow landslide (Eathquake induced)
2	28° 45 ' 26"	81° 55 ' 58"	Ghalgale	Chhedagad M	Dried spring
2	28° 45 ' 13"	81° 55 ' 20"	Darpakha jiban jyoti	Chhedagad M	School damage due to crack
3	28° 45 ' 39"	81° 57 ' 1"	Dalitbasti	Chhedagad M	Land crack, potential of slide
3	28° 45 ' 54"	81° 56 ' 59"	Thadi ukali	Chhedagad M	Land subsidence
3	28° 46 ' 9"	81° 56 ' 30"	Pannepanna	Chhedagad M	Cracks, land subsidence, risk to road
3	28° 45 ' 24"	81° 56 ' 14"	Kada khet	Chhedagad M	Land subsidence effect to lamara
3	28° 45 ' 13"	81° 56 ' 14"	Lamara	Chhedagad M	Land subsidence
3	28° 45 ' 46"	81° 58 ' 5"	Dankheri	Chhedagad M	Landslide risk to road
3	28° 45 ' 28"	81° 59 ' 20"	Dashera	Chhedagad M	Shallow landslide
4	28° 45 ' 28"	81° 59 ' 58"	Chaur jewa	Chhedagad M	Landslide extend up to kalyan khola

Ward	Latitude	Longitude	Village	Municipality	Description
4	28° 44 ' 55"	81° 59 ' 52"	Kalyan khola	Chhedagad M	Crack line
5	28° 45 ' 4"	82° 2 ' 48"	Phalihalna	Chhedagad M	Land subsidence, slide; risk to road ; just in front of ward 5 office
5	28° 45 ' 23"	82° 3 ' 12"	Phalihalna tole	Chhedagad M	Road subsidence
5	28° 45 ' 56"	82° 3 ' 4"	Dhanikhet	Chhedagad M	Road subsidence
5	28° 43 ' 33"	82° 1 ' 34"	Lewal bazar	Chhedagad M	Shallow landslide
6	28° 46 ' 23"	82° 1 ' 26"	Suwadeula	Chhedagad M	Shallow landslide
6	28° 47 ' 13"	81° 59 ' 33"	Sajhmela	Chhedagad M	Reactivated landslide
10	28° 46 ' 4"	82° 4 ' 50"	Khraya danda	Chhedagad M	Large crack line during quake but later filled (according to local people).
12	28° 43 ' 53"	82° 1 ' 16"	Sagachaur	Chhedagad M	Subsidence of huge land mass
12	28° 42 ' 58"	82° 1 ' 58"	Sagachaur	Chhedagad M	Subsidence landmass
12	28° 43 ' 11"	82° 2 ' 10"	Sagachaur	Chhedagad M	Subsidence
12	28° 43 ' 13"	82° 2 ' 35"	Janakalyan school	Chhedagad M	Cracks filled with dirt
13	28° 40 ' 34"	82° 4 ' 13"	Bagaha	Chhedagad M	Reactivated landslide ; No residents around it
13	28° 39 ' 52"	82° 3 ' 43"	Bhumpa	Chhedagad M	Reactivated landslide after earthquake
13	28° 40 ' 30"	82° 3 ' 44"	Bhumpa	Chhedagad M	Shallow landslide
13	28° 40 ' 55"	82° 4 ' 14"	Almullah	Chhedagad M	Shallow landslide

Annex IV Damage details of Rukum

Damage details Aathbiskot Municipality, West Rukum				
Ward	Latitude	Longitude	Village	Description
9	28° 45 ' 41"	82° 22 ' 54"	Tatapani	Stream erosion; risk to 2 houses
11	28° 47 ' 11"	82° 20 ' 52"	Gharapali	Reactivated landslide
11	28° 46 ' 29"	82° 20 ' 11"	Mungraha	Old creep; road is swept away
11	28° 46 ' 25"	82° 19 ' 60"	Mungraha	Road slope failure
11	28° 46 ' 0"	82° 17 ' 17"	Road	Old landslide which reactivated after the earthquake.
12	28° 45 ' 7"	82° 21 ' 45"	Kotdanda	Road slope failure
13	28° 45 ' 29"	82° 22 ' 06"	Maurighari	Reactivated landslide
13	28° 45 ' 34"	82° 18 ' 51"	Kalapahira	Old regenerated rockfall, crack is also observed.
13	28° 44 ' 45"	82° 20 ' 08"	Hampa	Reactivated landslide
13	28° 45 ' 27"	82° 19 ' 33"	Mathillo Kuridhara	Reactivated rockfall
13	28° 45 ' 12"	82° 17 ' 40"	Bhalma	Road slope failure
13	28° 44 ' 24"	82° 17 ' 46"	Ghartipakha	Reactivated landslide
14	28° 44 ' 5"	82° 17 ' 42"	Banke Gaira	Reactivated landslide
Damage details Banphikot Rural Municipality, West Rukum				
Ward	Latitude	Longitude	Village	Description
3	28° 41' 25"	82° 29' 12"	Syarphu lake	Crack is observed.
4	28° 41' 14.06"	82° 27' 34.28"	Sallarai	Tiltation of irrigation slope
4	28° 41' 13.84"	82° 27' 45.99"	Sallarai mul	Decrease in water discharge in spring
5	28° 41' 37.04"	82° 28' 06.56"	Talla kada	Crack of 5.4 m length, 14 cm breadth and 10 cm depth. Crack lies near the house.

5	28° 41' 41.31"	82° 28' 16.51"	Kurlengjur	Cracks on dolomite rock; it is not activated by earthquake but is increasing yearly according to the local people; sink holes also observed; have poor water management.
5	28° 41' 44.19"	82° 28' 16.14"	Kurlengjur	Cracks which started about 9-10 years ago and is increasing yearly (according to the local people). Its length is 3 m, width is 15 - 30 cm and depth is 1.5 m. Crack is observed in the boulder.
6	28° 42' 34.33"	82° 26' 55.72"	Jipsing	Small subsidence but no crack is observed. The slope is little tilted. No seepage
6	28° 42' 41.01"	82° 26' 43.44"	Jipsing	Reactivated landslide. Detailed description is on survey sheet.
6	28° 42' 44.19"	82° 26' 43.35"	Jipsing	Reactivated landslide. Detailed description is on survey sheet.
6	28° 42' 53.9"	82° 26' 34.74"	Jipsing	Slide initiated by road excavation.
7	28° 42' 56.35"	82° 25' 07.84"	Okhardhara	Crack in agricultural land with length of 17.7 m; depth of 70 cm and width about 6 -15 cm. Slope angle is 41°; dip direction 254°; crack locates on 40 m downhill from house.
7	28° 43' 10.87"	82° 25' 49.89"	Dulingga	Subsidence due to cracks; length of crack is 30 m, width 15 cm, and depth 60 cm. small portion of crack is still visible and some portions are filled.
7	28° 43' 34.88"	82° 25' 03.29"	Barsut	Crack of length 4 m and width of 6 m. Half house was covered by sediments. Crack was developed by earthquake and lead to slope failure. Landslide can be mitigated by building retaining wall. No threat to the human being but further failure can affect one house.
7	28° 44' 5.46"	82° 24' 21.21"	Pariban	Crack of length 25 m, width 60 cm and depth 80 cm; most parts of crack is filled by sediments, still small parts of crack is visible. Crack is also present in the road. Slope angle is 41°. Slope aspect 30°.



7	28° 44' 06.41"	82° 24' 17.21"	Sunchari	Landslide
7	28° 44' 15.27"	82° 24' 00.45"	Sunchari	Landslide reactivated by earthquake
7	28° 43' 38.57"	82° 25' 07.9"	Barsut	Crack with length of 11.7 m, 15 cm width, 60 cm depth. One house is located just 1 m above the crack and 2 houses are just below the slope, crack was sealed using cement and rock. Small crack is still visible on the ground.
7	28° 43' 38.77"	82° 25' 10.29"	Barsut	Subsidence of 0.5 m to 1 m. Crack of length 5 m, depth 2 m, width 60 cm; 2 houses near the subsidence. Crack was sealed using cement and rock.
7	28° 43' 40.75"	82° 25' 09.42"	Duwanath	Crack which is filled now.
7	28° 43' 42"	82° 25' 06.67"	Duwanath	Crack which is filled now. According to the local people crack had a 3m length, 1 cm width and depth 10 cm
7	28° 43' 42.45"	82° 25' 01.83"	Duwanath	Crack which is filled now.
7	28° 43' 43.24"	82° 25' 00.9"	Duwanath	Crack of about 19.7 m long, 60 cm width, depth of 1 m and front bearing of 247°; crack was sealed and now a wall is built in this place; no visible crack on recent time.
7	28° 43' 43.78"	82° 25' 42.62"	Gorekada	Small crack; gentle slope; no effect to the houses.
7	28° 43' 50.03"	82° 25' 23.49"	Gorekada	Crack which is filled now.
8	28° 42' 12.41"	82° 25' 10.81"	Moche	Crack is inside and outside the house but no photo evidence. Crack have about 8 m length, width of 2 cm and depth of 40 cm. 3 number of cracks within the area.



8	28° 42' 17.79"	82° 25' 02.04"	Turdhara	No visible crack but have obtained old photo. According to the local people, length of crack was 15-20 m, width 5 cm, and depth 0.5 m- 1 m.
8	28° 42' 20.46"	82° 25' 1.07"	Turdhara	Crack with length 4.5 m, width 3-6 cm depth 28 cm. No houses near the crack. Slope angle 28°, Slope dip direction 143°
8	28° 42' 18.62"	82° 25' 01.66"	Turdhara	Crack which is filled now.
8	28° 42' 23.59"	82° 24' 59.78"	Turdhara	Most of the crack was filled but some parts are still visible. Multiple cracks (3-4 in number) having length 4 m, width 10- 18 cm, depth 58 cm; Cracks on agriculture land which is dry. One house is about 55 m uphill from crack. Slope angle 35°; Slope direction 149°
8	28° 42' 45.52"	82° 25' 13.43"	Khohol khelna Dada	Crack of 6 m length, 2 cm depth and 1 cm width in road.
9	28° 41' 00.64"	82°26' 54.91"	Kalabang tol	Crack inside the house. About 1.5 m long, 3 cm wide and 5 cm in depth. Another crack was also seen about 7 m downhill of house.
9	28° 41' 01.86"	82° 26' 56.02"	Kalabang tol	Crack of length 10.5 m, width about 8 cm and depth 45 cm. Steep slope in front of house. Mitigation techniques are not used. Crack was filled using soil and rock. Slope failure is also seen in front of house. Slope dip direction and amount is 197° and 39° respectively. Only soils no rock is present.
9	28° 41' 05.46"	82°26' 51.99"	Kalabang tol	Slope failure due to cracks. 2 houses were covered by sediments. Crack is about 11.2 m long 15 cm wide and depth of 50 cm. 3 cracks which are not continuous can be observed in the area.
9	28° 41' 03.27"	82°26' 56.38"	Kalabang tol	Subsidence of 4 cm in front of house.
9	28° 41' 10.19"	82°26' 48.91"	Tiruwa tol	5 m slide due to steep slope behind the house.

9	28° 41' 38.63"	82° 26' 36.16"	Kalegau	Crack of length 8.4 m, width 6 cm depth 70 cm and bearing of 246° inside the house. Steep slope about 4 m in front of the house. Stone wall is built to make the slope stable. The slope also consists of black soil in it.
9	28° 41' 17.4"	82° 26' 45.36"	Kavra tol	Crack inside the house. 8 m in length, width 1.5 cm, and depth 70 cm. Crack was filled using soil. Crack was present outside the house also according to the local people. Crack bearing is 47°. Slope dip amount 41° and slope dip direction 174°.
9	28° 41' 18.17"	82° 26' 48.03"	Kavra tol	Crack which is filled now.
9	28° 41' 20.91"	82° 26' 36.08"	Pipal fewcha	Crack which is filled now.
9	28° 41' 20.26"	82° 26' 35.86"	Pipal fewcha	Cracks with length of 5.5 m, width 8 cm and depth 60 cm. Soil type: Gravel with silt. Slope dip direction 216° and slope dipping 24°.
9	28° 41' 20.69"	82° 26' 36.42"	Pipal fewcha	Crack with length 12 m, width 8 cm, and depth 12 cm.
9	28° 41' 23.08"	82° 26' 38.77"	Pipal fewcha	Crack which is filled now.
9	28° 41' 23.43"	82° 26' 37.58"	Pipal fewcha	Crack which is filled now.
9	28° 41' 23.39"	82° 26' 36.91"	Pipal fewcha	Multiple cracks. Length 13 m, width 8-13 cm and depth 28 cm. Most of the crack was filled using rocks, soil. Some portion is still visible.
10	28° 40' 01.98"	82° 26' 13.07"	Sima gaira	Old landslide
10	28° 40' 1.67"	82° 26' 12.61"	Purano thati, Karyan pakha	Dry old spring. Length 25 m, width 90 cm and depth 80 cm. About 2.7 m crown height. Crack in rock is also observed. Dip direction 259° dip amount 16°. Slope direction 142° and slope dip 47°.

10	28° 40' 02.01"	82° 26' 15.08"	Sima gaira	Cracks; Crack have length of 7.8 m, width 20 cm and depth 60 cm. Slope angle 44°, dip direction 139°. Presence of slate in the slope.
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Damage details Triveni Rural Municipality, West Rukum

Ward	Latitude	Longitude	Village	Description
1	28° 32 ' 55"	82° 27 ' 39"	Near Khare khola	Landslide with minor cracks
1	28° 32 ' 59"	82° 27 ' 59"	Near Khare khola	Crack is present.
1	28° 32 ' 50"	82° 28 ' 28"	Balle	The cracks developed on the wall of houses.
1	28° 32 ' 48"	82° 28 ' 34"	Balle	Landslide is observed and two house are in danger.
1	28° 32 ' 48"	82° 28 ' 35"	Balle	
1	28° 32 ' 45"	82° 28 ' 41"	Waski Thala	The whole village slide about 1 m to 2 m per year (according to the local people).
2	28° 34 ' 26"	82° 26 ' 13"	Barkhe Takura	Crown part of the landslide; old and triggered by the roadcut. Detailed description is in the survey sheet.
2	28° 34 ' 28"	82° 26 ' 10"	Barkhe Takura	
2	28° 34 ' 29"	82° 26 ' 11"	Barkhe Takura	
3	28° 32 ' 50"	82° 23 ' 11"	Haiban	Landslide which affects 20 houses; road is also disturbed.
3	28° 32 ' 51"	82° 23 ' 13"	Haiban	Another GPS of previous landslide
3	28° 32 ' 50"	82° 23 ' 23"	Haiban	Another GPS of previous landslide
3	28° 32 ' 45"	82° 23 ' 30"	Haiban	Another GPS of previous landslide
3	28° 32 ' 55"	82° 23 ' 50"	Haiban	The landslide is reactivated after the earthquake; gravel clayey sand present; the trees are tilted; no water management; the landslide consists of 100 m length, 80 m breadth, and depth of 1m and is on dry condition; has 32° slope angle; no cracks and subsidence occur; has damage the road and there are 6 houses which are in danger.
3	28° 32 ' 52"	82° 24 ' 14"	Hampal	Deep seated landslide
3	28° 32 ' 47"	82° 24 ' 40"	Hampal/ Khara	Landslide



4	28° 32 ' 22"	82° 25 ' 25"	Damju	Shallow landslide
4	28° 32 ' 19"	82° 25 ' 27"	Damju	Major to minor cracks is observed.
4	28° 32 ' 17"	82° 25 ' 32"	Damju	
4	28° 32 ' 17"	82° 25 ' 31"	Damju	
4	28° 33 ' 39"	82° 25 ' 52"	Jyalan	Landslide is observed and two school buildings are in danger. Detailed description is on survey sheet.
4	28° 33 ' 41"	82° 25 ' 51"	Jyalan	
4	28° 33 ' 41"	82° 25 ' 50"	Jyalan	
4	28° 33 ' 41"	82° 25 ' 50"	Jyalan	
5	28° 33 ' 38"	82° 26 ' 31"	Chaukidhunga	School is in danger due to landslide
5	28° 33 ' 37"	82° 26 ' 29"	Chaukidhunga	Landslide triggered by excavation
5	28° 32 ' 23"	82° 26 ' 29"	Lalit	Landslide
5	28° 32 ' 26"	82° 26 ' 25"	Lalit	Another GPS of previous landslide
5	28° 32 ' 25"	82° 26 ' 24"	Lalit	Another GPS of previous landslide
5	28° 32 ' 24"	82° 26 ' 22"	Lalit	Another GPS of previous landslide
5	28° 32 ' 23"	82° 26 ' 22"	Lalit	Another GPS of previous landslide
6	28° 33 ' 53"	82° 24 ' 43"	Ghoredhunga	GPS of landslide at crown part
6	28° 33 ' 49"	82° 24 ' 10"	Ghoredhunga	GPS of landslide at crown part
7	28° 34 ' 47"	82° 25 ' 32"	Rithyang	Deep seated old landslide.
7	28° 34 ' 47"	82° 25 ' 33"	Rithyang	Another GPS of previous landslide
7	28° 34 ' 49"	82° 24 ' 16"	Melpokhari	Crack; have bearing of 220°N, length of 20m, breadth of 10 cm to 20 cm and thickness up to 80 cm.
7	28° 34 ' 10"	82° 24 ' 1"	Near Durpati Khola	Old landslide and large cracks are found.
7	28° 34 ' 13"	82° 23 ' 57"	Muru	Minor crack on road.
7	28° 35 ' 4"	82° 24 ' 17"	Muru	Cracks; are filled according to local people; only trace is found.
7	28° 35 ' 11"	82° 24 ' 13"	Muru	
7	28° 35 ' 15"	82° 23 ' 59"	Malamedada	About 5 m crack found.
7	28° 35 ' 16"	82° 23 ' 56"	Gyanna	
8	28° 35 ' 3"	82° 23 ' 13"	Paeugha	Crack observed in Shree Janapriya Madhyamik Bidhyalaya.
8	28° 35 ' 0"	82° 22 ' 55"	Kurepani	Crack observed



8	28° 34 ' 53"	82° 22 ' 58"	Panera	About 15 m long crack; front bearing of 20°.
8	28° 34 ' 55"	82° 22 ' 39"	Thulogaira	Rock fall and road blockage
8	28° 34 ' 33"	82° 22 ' 41"	Mathigauon	Crack; front bearing 330°; 5 m length
8	28° 34 ' 42"	82° 22 ' 48"	Thulogaira, Kanjia	Crack with 15 m length, 13 cm breadth and depth of 2 m. The crack was filled by local due to danger of slide. 4 houses are just below about 10m the crack which are in danger.
8	28° 34 ' 42"	82° 22 ' 46"	Thulogaira, Kanjia	Crack having front bearing of 20° and about 5 m length.
10	28° 35 ' 24"	82° 23 ' 3"	Bairaban	Crack having front bearing of 145° and about 5 m long.
10	28° 35 ' 30"	82° 22 ' 55"	Bairaban	Minor crack (length of 2 m, breadth of 20 cm and thickness of 10 to 30 cm)
3 and 6 junctions	28° 33 ' 50"	82° 24 ' 8"	Dharegaira	Landslide (Rock types: phyllite and siliceous dolomite)
3 and 6 junctions	28° 32 ' 56"	82° 23 ' 1"	Dharegaira	Landslide
3 and 6 junctions	28° 32 ' 50"	82° 22 ' 59"	Dharegaira	Landslide with the crack having fore bearing of about 350°
3 and 6 junctions	28° 32 ' 46"	82° 22 ' 55"	Dharegaira	Another GPS of previous landslide
3 and 6 junctions	28° 32 ' 56"	82° 22 ' 55"	Dharegaira	Another GPS of previous landslide
3and 6 junctions	28° 32 ' 55"	82° 22 ' 56"	Dharegaira	GPS at the crown part of the landslide; crack and subsidence occurs within the landslide

Damage details Musikot Municipality, West Rukum

Ward	Latitude	Longitude	Village	Description
1	28°39'56.24"	82°29'47.31"	Bhalkhola gau	Earth material slide, triggered by earthquake in slow creeping old landslide mass.
1	28°39'53.44"	82°29'23.63"	Charikhola gau	Approximately 0.5 m land subsidence after earthquake.
1	28°40'09.62"	82°29'07.08"	Sagindada	Land subsidence after earthquake.



5	28°37'12"	82°28'26"	Ghualpani	Reactivated old landslide by earthquake.
6	28°36'52"	82°29'01"	Jharigau, Dabang	Earthquake induced landslide and crack of length 70 m and bearing of 170°.
6	28°37'00"	82°29'02"	Mathi Gaira	Landslide triggered by earthquake in old deep seated gravitational slope deformation mass.
6	28° 36 ' 39"	82° 29 ' 4"	Bhandari kanda	Reactivated landslide
6	28°36'37"	82°28'54"	Jharigau	Shallow failure after earthquake.
7	28°36'24"	82°30'25"	Pandey gau, Basina Khola	Landslide triggered by earthquake in old deep seated gravitational slope deformation mass.
7	28°36'04"	82°30'14"	Basina Khola gau, Retaban	Shallow failure after earthquake in toe part of landslide mass
7	28°36'16.44"	82°30'40.32"	Kauleni gau	Shallow failure after earthquake in middle part of landslide mass
7	28°36'20.11"	82°30'40.59"	Gyamigaira	Crack of length 20 m and 1m subsidence in bearing of 90°.
7	28°36'39.23"	82°30'41.82"	Sakh	Land subsidence of 2 m in 150m length
7	28°36'43.39"	82°31'06.72"	Sakh	Mid hill highway subsidence in length of 50 m after earthquake and shallow failure in slope in slow creeping landslide mass
7	28°36'50.45"	82°31'36.88"	Sagine gau	0.5 m Subsidence of Mid hill Highway in 60m length after earthquake
7	28°36'34.49"	82°31'15.35"	Gyammakhola gau	Rock fall hazard after earthquake
7	28°36'07.03"	82°32'00.13"	Gyammakhola gau	Shallow failure initiated after earthquake
7	28°36'25.81"	82°32'20.6"	Okhreni gau	Crack of length 5 m and subsidence of 0.5 m
8	28°35'55.44"	82°32'03.54"	Chelabang	Shallow failure initiated by earthquake
8	28°35'22.16"	82°33'55.31"	Masinapakha / Ghatingkhola	Cracks on land and slow-moving landslide
8	28°35'22.29"	82°34'02.18"	Thuligaira	Cracks of 20 m length and 1.2 m depth with bearing of 200°.

8	28°35'39.73"	82°31'58.67"	Kyamma (Okhartharabhir)	Plane failure initiated after earthquake.
8	28°35'13.15"	82°32'21.19"	Khanidada	Shallow failure after earthquake.
8	28°35'30.43"	82°34'14.44"	Ghorneti	5 m subsidence at Mid hill Highway after Earthquake.
8	28°35'36.85"	82°34'26.08"	Ghorkhani	Series of cracks after earthquake.
8	28°35'22.16"	82°33'55.3"	Masinpakha	Failure initiated after earthquake.
9	28°35'46.68"	82°31'01.1"	Dhaule pahi	Deep seated landslide triggered by earthquake; 3 houses are directly at risk.
9	28°35'06.59"	82°30'15.93"	Hol gau	crack of length of 5 m and subsidence of 1 m; high risk in Suryadaya School and Hol village.
9	28°35'26.06"	82°29'51.91"	Jhulkhet	Rock slide and fall after earthquake.
10	28°33'37.67"	82°30'47.31"	Galampati	Crack and drying of the spring after earthquake
10	28°34'23.45"	82°29'58.86"	Thulopahi	Multiple shallow failure in huge massive landslide mass after earthquake
10	28°34'08.17"	82°29'51.41"	Jhyang Gau	Shallow failure in old landslide mass after earthquake; 7 houses are at risk
11	28°36'00.89"	82°27'42.52"	Rithabot	Land subsidence of 15 cm, rock slide and plane failure after earthquake.
11	28°36'03.24"	82°27'54.09"	Rithabot	Plane failure problem in slow creeping landslide mass.
12	28°35'56.35"	82°26'21.52"	Kanshpokhari	Land subsidence of 0.6 m and crack formed in length of 12 m in width of 45 cm with bearing of 70°.
12	28°36'15.82"	82°26'11.62"	Naya gau, Dhamdhame	Earthquake induced deep seated landslide in which 4 houses are in risk.
12	28°36'09"	82°25'37.92"	Harale basti	Deep seated landslide triggered by earthquake; 11 houses are directly at risk.
13	28° 38 ' 14"	82° 25 ' 3"	Kasera Pakha	Crack after Earthquake
13	28° 37 ' 11"	82° 24 ' 59"	Bhaludula	Old landslide
13	28° 37 ' 14"	82° 24 ' 41"	Khopikhelne	Rockfall triggered by landslide
13	28° 37 ' 23"	82° 25 ' 29"	Kalupatal	Landslide induced by roadcut
14	28° 37 ' 7"	82° 22 ' 13"	Ukaledadha	Landslide induced by roadcut

Damage Details Sanibheri Rural Municipality, West Rukum

Ward	Latitude	Longitude	Village	Description
1	28° 42 ' 1"	82° 15 ' 33"	Chinabagar	Crack along the Midhill highway
1	28° 42 ' 1"	82° 15 ' 35"	Chinabagar	Crack which is 50 m long along the uphill slope of Midhill highway with 44° slope angle; has highly affected water tank
1	28° 42 ' 2"	82° 15 ' 35"	Chinabagar	Shallow landslide (45° slope angle) which lies between colluvium and alluvium deposit.
1	28° 42 ' 3"	82° 15 ' 33"	Chinabagar	Long crack along Midhill highway with front bearing of 95°
1	28° 41 ' 60"	82° 15 ' 39"	Chinabagar	Landslide
1	28° 41 ' 55"	82° 15 ' 46"	Cinabagar	Landslide with slope angle of 66°
1	28° 41 ' 51"	82° 15 ' 56"	Chainabagar	Crack of about 55 m length along Midhill highway with bearing of 320°
1	28° 41 ' 51"	82° 15 ' 57"	Chinabagar	Crack extension of previous location
1	28° 41 ' 51"	82° 15 ' 57"	Chinabagar	Crack extension of previous location
1	28° 42 ' 12"	82° 14 ' 13"	Tarika	Cracks, shallow landslide and subsidence around the forest area observed; Slope angle is 72°. The slope consists of large boulder mix with clay and gravel. Detailed description is on survey sheet.
1	28° 42 ' 11"	82° 14 ' 12"	Tarika	Crack with length of 9 m
1	28° 42 ' 11"	82° 14 ' 12"	Tarika	Crack with length of 16 m, breadth of 0.1 m and depth of 0.5 m
1	28° 42 ' 17"	82° 14 ' 9"	Swota Village	Rock fall (topple) due to earthquake with bearing of 305°/56°(slope).
1	28° 42 ' 30"	82° 14 ' 19"	Swota	Rockfall; affect the Midhill Highway; have bearing 335°/54°(slope)
2	28° 40 ' 50"	82° 17 ' 59"	Githakot	Crack having depth of 5 cm, breadth of 5 cm and length of 5.5 m and with bearing of 150°



2	28° 41 ' 21"	82° 17 ' 10"	Githakot	Crack on Midhill highway having length of 45 m, breadth of 4 cm, depth of 5 cm and front bearing of 280°.
2	28° 41 ' 14"	82° 17 ' 8"	Githakot	Land subsidence of 150 m with 55 cm length in settlement area; the most affected area; 7 houses are completely affected in the area. Thus, the settlement area should be migrated.
2	28° 41 ' 25"	82° 16 ' 24"	Basandanda	Crack observed
2	28° 41 ' 24"	82° 16 ' 26"	Basandanda	Subsidence of 35 cm; have 40% moisture content
2	28° 41 ' 23"	82° 16 ' 27"	Basandanda	Consists of multiple shallow landslide
3	28° 39 ' 41"	82° 21 ' 34"	Gajetakuri	Crack on the uphill slope having breadth of 8 cm, depth of 50 cm and 200 m length, the crack looks like arc shape.
3	28° 39 ' 37"	82° 21 ' 33"	Gajetakuri	GPS of toe of landslide
3	28° 40 ' 28"	82° 21 ' 24"	Sirchaur	Land subsidence of 1.9 m; The crack extend towards two direction from the GPS point; front bearing of 80° at length of 50 m and front bearing of 265° at length of 150 m
3	28° 40 ' 25"	82° 21 ' 28"	Sirchaur	Water tank leakage due to crack
4	28° 41 ' 29"	82° 17 ' 21"	Near Rimana bazar	Crack with front bearing of 130°, length of 22 m, breadth of 10 cm and depth of 0.5 m.
4	28° 41 ' 15"	82° 18 ' 28"	Balebang	Crack of about 90 m length near settlement area; multiple crack lines present. The crack decreases towards bearing of 240° with width of 15 cm, depth of 30 cm, and slope of 145°/69°.
4	28° 41 ' 8"	82° 18 ' 22"	Balebang	Crack with length of 40 m, breadth of 20 cm, and front bearing of 195°.
6	28° 41 ' 46"	82° 22 ' 17"	Kotgau Bijayapar	Crack with length of 100 m, breadth of 0.5 m; multiple parallel crack lines are present having perpendicular distance of 0.7 m, depth greater than 10 m, front bearing towards slope is 110° and front bearing towards settlement is towards 280°.

8	28° 39 ' 4"	82° 20 ' 58"	Tallo aamdanda	Crack line having length of 70 m, breadth of 10 cm and depth of 0.5 m; destruction of 4 houses after earthquake; have front bearing of 65°
8	28° 38 ' 13"	82° 20 ' 40"	Sarkemare	Dry spring after earthquake
8	28° 38 ' 45"	82° 20 ' 12"	Biphaya	Dry spring after earthquake
8	28° 38 ' 34"	82° 20 ' 18"	Biphaya	Dry spring after earthquake. Before earthquake, discharge of water was 32 cubic meter per 12 hours but now the spring is dry. The slope angle is 64° and the slope aspect is 130°. The sandstone bed has dip direction and dip amount of 45° and 60° respectively.
9	28° 39 ' 20"	82° 21 ' 48"	Kurmidhik	Crack of 2 m length
9	28° 39 ' 20"	82° 21 ' 49"	Kurmidhik	Crack on Midhill highway; front bearing of 281° and 255°, length of 15 m, breadth of 4 cm.
9	28° 38 ' 52"	82° 22 ' 59"	Kurmidhik	Crack on Midhill highway having length of 5 m and breadth of 5 cm
9	28° 39 ' 23"	82° 22 ' 10"	Kurmidhik	Landslide
9	28° 39 ' 20"	82° 21 ' 36"	Ranjapokharadhik	Crack having length 500 m, breadth 5 cm depth 0.5 m and front bearing of 306°
11	28° 36 ' 34"	82° 21 ' 26"	Jharpokhara	Dry spring after earthquake
11	28° 37 ' 3"	82° 22 ' 6"	Sadanakhosi	Crack having length of 20 m, breadth of 5 cm, and depth of 0.5 m.
11	28° 37 ' 28"	82° 21 ' 4"	Kafalbot	Crack having length of 20 m
11	28° 37 ' 28"	82° 21 ' 4"	Kafalbot	Crack having length of 10 m

Damage details Chaujhahari Rural Municipality, West Rukum

1	28° 38 ' 15"	82° 12 ' 2"	Chaurjahari	Reactivated landslide
6	28° 39 ' 13"	82° 16 ' 48"	Mulpani	Shree Jyanjyoti Adharbhut School is highly affected by earthquake; Multiple crack lines develop on the walls of school.
9	28° 39 ' 41"	82° 14 ' 43"	Majhmela	Landslide observed along road cut slope.
9	28° 39 ' 39"	82° 14 ' 38"	Majhmela	Reactivated landslide



9	28° 39 ' 36"	82° 14 ' 42"	Majhmela	Landslide induced by road excavation
9	28° 36 ' 56"	82° 14 ' 40"	Majhmela	Old reactivated landslide with large debris flow
11	28° 38 ' 4"	82° 17 ' 13"	Sharkena	Crack has affected Shree Gyan Jyoti School
11	28° 38 ' 2"	82° 17 ' 11"	Sharkena	Reactivated landslide; slightly affected the area
11	28° 39 ' 40"	82° 17 ' 11"	Sharkena	Reactivated landslide
11	28° 38 ' 21"	82° 18 ' 26"	Jogineta	Highly affects for Janakalyan School; crack is not visible due to filling
12	28° 38 ' 22"	82° 18 ' 16"	Jumlipokhari	Most of the houses are damaged by recent earthquake. Detail description is on survey sheet.
12	28° 38 ' 17"	82° 17 ' 58"	Pathihale	Slightly affected by landslide and crack
12	28° 37 ' 51"	82° 18 ' 20"	Pathihalne	Landslide
12	28° 38 ' 21"	82° 18 ' 54"	Ulleri Dharelidanda	Reactivated landslide
12	28° 38 ' 29"	82° 18 ' 53"	Tallo ulleri	Reactivated landslide
12	28° 38 ' 35"	82° 19 ' 2"	Tallo ulleri	Reactivated landslide
12	28° 38 ' 19"	82° 19 ' 8"	Tallo ulleri	Reactivated landslide
12	28° 37 ' 57"	82° 18 ' 52"	Tallo ulleri	Reactivated landslide
12	28° 38 ' 33"	82° 18 ' 46"	Ulleri	Reactivated landslide
13	28° 35 ' 60"	82° 17 ' 8"	Ranapakha	Reactivated landslide
13	28° 35 ' 59"	82° 18 ' 5"	Ranapakha	Reactivated landslide
13	28° 35 ' 59"	82° 18 ' 37"	Ratamata	Multiple crack lines developed at Janakalyan Adharbhut School
13	28° 35 ' 59"	82° 18 ' 37"	Ratamata	Crack extension coordinate of previous landslide
13	28° 36 ' 0"	82° 18 ' 37"	Ratamata	Crown of landslide
13	28° 36 ' 2"	82° 18 ' 38"	Ratamata	Reactivated landslide
13	28° 36 ' 1"	82° 18 ' 38"	Ratamata	Reactivated landslide and crack is observed
13	28° 36 ' 1"	82° 18 ' 39"	Ratamata	Recent shallow landslide on the school area
13	28° 36 ' 1"	82° 18 ' 39"	Ratamata	GPS of previous landslide
13	28° 36 ' 1"	82° 18 ' 38"	Ratamata	GPS of previous landslide.
13	28° 36 ' 0"	82° 18 ' 38"	Ratamata	GPS of previous landslide.
13	28° 36 ' 1"	82° 18 ' 38"	Ratamata	GPS of previous landslide.



13	28° 36 ' 1"	82° 18 ' 39"	Ratamata	Landslide that has affect the school
13	28° 35 ' 60"	82° 18 ' 39"	Ratamata	GPS of previous landslide
13	28° 35 ' 58"	82° 18 ' 38"	Ratamata	GPS of previous landslide
13	28° 35 ' 59"	82° 18 ' 38"	Ratamata	GPS of landslide toe part
13	28° 35 ' 58"	82° 18 ' 39"	Ratamata	GPS of landslide toe part
13	28° 35 ' 58"	82° 18 ' 38"	Ratamata	GPS of landslide toe part
13	28° 36 ' 0"	82° 18 ' 38"	Ratamata	GPS coordinate landslide toe part
13	28° 36 ' 0"	82° 18 ' 34"	Ratamata	Reactivated landslide
13	28° 36 ' 0"	82° 18 ' 35"	Ratamata	Reactivated landslide
13	28° 36 ' 1"	82° 18 ' 33"	Ratamata	Reactivated landslide; Gabion wall protection
13	28° 36 ' 1"	82° 18 ' 31"	Ratamata	Reactivated landslide
13	28° 36 ' 3"	82° 18 ' 30"	Ratamata	Reactivated landslide
13	28° 36 ' 0"	82° 18 ' 27"	Ratamata	Hanging block of limestones about diameter of 20 m.
13	28° 36 ' 1"	82° 18 ' 20"	Ratamata	Possible landslides observed near the road section, creeping that could affect house of downslope.
13	28° 35 ' 60"	82° 18 ' 20"	Ratamata	Reactivated landslide
13	28° 36 ' 5"	82° 18 ' 20"	Ratamata	Reactivated landslide
13	28° 36 ' 5"	82° 18 ' 11"	Jhapekhola	Subsidence and crack.
13	28° 36 ' 1"	82° 18 ' 11"	Jhapekhola	Crack observed
13	28° 35 ' 59"	82° 18 ' 0"	Near Jhapekholagau	Reactivated and excavation induced landslide
13	28° 35 ' 59"	82° 17 ' 57"	Near Jhapekholagau	Reactivated landslide
13	28° 36 ' 1"	82° 17 ' 51"	Near Jhapekholagau	Reactivated landslide
13	28° 36 ' 55"	82° 17 ' 33"	Near Jhapekholagau	Old landslide activated due to road excavation
13	28° 36 ' 54"	82° 17 ' 4"	Near Jhapekholagau	Landslide on road
13	28° 37 ' 43"	82° 17 ' 3"	Near Jhapekholagau	Landslides on road